

Colored tracked changes

Blue underlined text shows additions. ~~Red struck text shows deletions.~~ The baseline is the original pre-Codex manuscript preserved on September 10, 2026. Main Tables 1–2 are shown in full before (red) and after (blue); new appendix tables are blue. Other tables retain current layout and unchanged numerical values. Current typography and bibliography are used. Relocations can appear as both deletions and additions.

Locally Robust Commensurate BART for Borrowing External Controls

Abstract

Real-world data (RWD) can ~~supply control observations that supplement the control arm of~~ a randomized controlled trial (RCT) ~~cannot afford to collect, but whether an external patient is a fair stand-in for a trial control depends on who that patient is.~~ ~~Meta-analytic, but the comparability of external and trial controls may vary with patient characteristics.~~ ~~Study-level meta-analytic predictive, commensurate, and power priors decide how much to borrow through one study-level parameter, so the same discount applies to every patient.~~ ~~apply a common borrowing parameter across patients.~~ We propose locally robust commensurate Bayesian additive regression trees (LRC-BART) ~~, which write to model the RCT control surface as a pooled BART surface fitted to both sources plus a sparse discrepancy ensemble, and place a~~. A spike-and-slab commensurate prior on each ~~leaf of the discrepancy ensemble~~ ~~discrepancy leaf~~ allows borrowing to vary across covariate regions. The spike ~~is the commensurate component, the slab represents commensurability and the slab serves as~~ the vague component of ~~the a~~ robust meta-analytic predictive prior, ~~and the tree prior decides region by region which.~~ ~~Tree partitions determine the regions over which each component applies.~~ All leaf marginals are exact, so the sampler targets the exact posterior. ~~The A variance-based prior effective sample size is defined at the estimand, the RCT-standardized control mean, and calibrated against the RWD posterior with an explicit ceiling.~~ We prove that the RWD-induced shift of a leaf estimate is bounded and vanishes at a Gaussian rate under conflict, that the posterior probability of a small local discrepancy is consistent ~~for the true state of each region under fixed partitions and representability of the discrepancy,~~ and that the calibration is monotone and well posed. A covariate-resolved ~~borrowing map reports, for map reports posterior compatibility at~~ each patient profile, ~~how much external information the analysis used; realized regional effective sizes summarize borrowing.~~ Simulations with Gaussian and survival outcomes and a single-arm myeloma trial with a registry control illustrate the method.

1 Introduction

Real-world data can ~~supply control observations that supplement~~ a randomized trial ~~cannot afford to collect.~~ ~~Whether an external patient is a fair stand-in for a trial control, however, depends on who that patient is: selection into the trial when collecting a sufficiently large control arm is difficult.~~ ~~Their usefulness depends on whether external outcomes represent the outcomes trial patients would have experienced under control.~~ Trial selection, treatment practice ~~outside it,~~ and outcome ascertainment ~~all may~~ vary with age, disease stage, ~~and prior therapy, so external controls that match the trial well for one kind of patient can be badly off for another.~~ A single borrowing

~~parameter, applied to every patient alike, must then choose between the bias of over-borrowing where the sources disagree and the inefficiency of under-borrowing. External and trial controls may therefore be comparable for some patient profiles and differ substantially for others.~~

A common borrowing parameter may overborrow where sources differ or discard useful information where they agree. We ~~take the view that borrowing is a decision to be made region by region in covariate space, with the regions learned from the data, and that the analysis should report, for each patient profile, how much external information it used. We build a Bayesian tree model that does both.~~ propose a Bayesian tree model with borrowing that varies over learned covariate regions. We accompany the treatment-effect estimate with profile-specific summaries of compatibility and external information.

~~The established priors for historical borrowing~~ Established borrowing priors ~~act on a study-level parameter. The meta-analytic predictive (MAP) prior of Neuenschwander et al. [17] and Schmidli et al. [23] treats the control parameters of the historical studies and of the new study as exchangeable draws from a common population and takes the historical and current control parameters as exchangeable and uses the current parameter's posterior predictive distribution of the new study's parameter as its prior; the robust MAP prior mixes that predictive component. Robust MAP mixes it~~ with a vague one so that a current study in conflict with the historical data can switch borrowing off component to accommodate conflict. ~~The commensurate prior of Hobbs et al. [9, 10] conditions the current parameter on the historical one, $\theta | \theta_0 \sim \mathcal{N}(\theta_0, 1/\tau)$, and learns the commensurability~~ learns the ~~precision τ from the data, with~~ under ~~a spike-and-slab hyperprior so that the posterior can move between near-complete borrowing and near-none. The power prior of Ibrahim and Chen [11] raises discounts~~ the historical likelihood to a power by a_0 , normalized by; the normalizations of ~~Duan et al. [7] and Neuenschwander et al. [16] so that a_0 can carry its own prior. show that with~~ allow a prior on a_0 . With ~~a single historical study, the ease we treat here, the normal MAP, commensurate, and power-prior forms nearly coincide. The three families share the location of the decision: one~~ [21, 22]. Their common ~~heterogeneity or discount scale, with no dependence on the covariates of the patient does not depend on patient covariates.~~

Locally robust commensurate BART (LRC-BART) models ~~the two control sources jointly as both control sources with~~ a pooled Bayesian additive regression tree (BART) surface f fitted to RCT and RWD controls alike, plus and ~~a sparse discrepancy ensemble g that only the RCT controls~~

~~inform, so that the~~. The RWD follow f and ~~the~~ RCT controls follow $f + g$. Each leaf of g carries a spike-and-slab prior whose spike is a commensurate component of small scale and whose slab is a vague component on the outcome scale, so the tree prior of g decides, region by region, whether the RCT control surface is the RWD surface up to a calibrated discrepancy or a free local offset has a small-scale commensurate spike and an outcome-scale vague slab. Its posterior state allows local agreement or a discrepancy over regions learned by the tree partitions.

~~Three components of the method come from the literature. The decomposition into a pooled surface and a discrepancy ensemble multiplied by an indicator is the Bayesian causal forest of~~, with the source indicator in the place of the treatment indicator The decomposition follows Bayesian causal forests [8], using source membership in place of treatment. The leaf prior ~~is the commensurate prior of in spike-and-slab form, with the slab in the role of the vague component of~~; conditioning on the pooled surface is the commensurate conditioning on the historical parameter, and since with one external source ~~this is close to a~~ combines commensurate shrinkage [9] with a vague component [23]; its relation to single-study MAP prior [22], ~~we rest no claim on the distinction. The calibration uses the expected local information ratio effective sample size is shared with those established priors. Calibration averages variance-based precision ratios, motivated by the normal-prior ESS of Neuenschwander et al. [18]. What we add is the combination and what it makes possible. Placing the spike-and-slab commensurate prior on the leaves of a discrepancy ensemble makes the choice between borrowing and not borrowing per region of covariate space, with the regions learned by the same tree prior that fits the surface, which a single ensemble with two means per leaf cannot do~~

The leaf mixture lets borrowing vary over learned regions, with separate priors for the external mean and discrepancy (Web Appendix G.6). ~~Every leaf marginal is in closed form, so the tree moves target the exact posterior and no surrogate likelihood is needed. The prior effective sample size is defined at the estimand, the~~ Closed-form leaf marginals permit exact tree updates. We calibrate a variance-based ESS at the RCT-standardized control mean, ~~and calibrated against the RWD posterior; it has a ceiling equal to the information the RWD carries about that estimand, and a target can be stated as a fraction of the ceiling. We prove that the RWD-induced shift of a leaf estimate is bounded and~~ with targets stated relative to its RWD-only ceiling. The conditional theory establishes bounded leaf influence that vanishes at a Gaussian rate ~~as the local conflict grows;~~

that the posterior probability of a small local discrepancy is consistent for the true state of each region, and that the calibration is monotone and well-posed under conflict, map consistency under fixed partitions and representability, and monotone calibration. The posterior of g ~~then gives a~~ covariate-resolved borrowing map, which reports at each patient profile the posterior probability that the local discrepancy is below a pre-specified threshold, together with a pointwise effective sample size. The model extends to right-censored outcomes through an accelerated supplies a profile-specific compatibility probability alongside information summaries. Accelerated failure time (AFT) ~~likelihood with data augmentation, and to the data augmentation accommodates right censoring; a single-arm setting in which no RCT control exists~~ variant represents uncertainty about an unobserved trial control surface.

1.1 Related work

A second group of methods lets the amount of borrowing vary with covariates, through propensity-score strata [4, 25], a commensurate precision that depends on the covariates through kernel local regression [13], exchangeability over sources or over latent patient classes [1, 14, 15], a global congruence statistic [12], or clustering of the patients on the covariates before the outcome model is fitted [2, 3]; Web Appendix G.0 reviews them, and the closest ~~parametric~~ kernel-local relative of our borrowing map is the covariate-adaptive ~~hybrid~~ historical control borrowing of Jin et al. [13].

Among tree methods, Zhou and Ji [27] is the ~~direct ancestor~~ most directly related approach. They fit one BART ensemble [5] to RCT and external controls with the data source as a splitting covariate, so that trees pool the sources wherever they do not split on the source; borrowing is implicit and has no parameter, and their discussion proposes as future work a prior discouraging source splits together with a source offset whose scale controls borrowing. ~~LRC-BART is that offset made nonparametric and given a prior that can be calibrated~~ We model the proposed offset nonparametrically and give it a calibratable prior. Pan et al. [19] apply BART to patient-level borrowing across subgroups. HE-BART [26] places group-specific random effects inside each terminal node with a shared precision, the structural precedent for a hierarchical parameter inside a leaf, ~~but it is built for~~ with the aim of mixed-effects prediction ~~and has no current study to protect~~ rather than controlling historical borrowing for a current study. flexBART [6] lets a tree assign several levels of a categorical predictor to one branch, which with the source as predictor gives the implicit

borrowing of Zhou and Ji [27] in a richer split space. None of these places a commensurate prior on a leaf, and among the covariate-adaptive methods only Jin et al. [13] and Alt et al. [1] report borrowing at the resolution of a patient profile, ~~neither on a nonparametric outcome surface, with CAHB using nonparametric kernel estimates. LRC-BART instead uses joint BART surfaces and spike-and-slab discrepancy leaves.~~

Section 2 ~~sets out the model, the sampler, the effective sample size and its calibration, the borrowing map with its guarantees, and the single-arm variant.~~ develops the method. Section 3 reports simulation studies with Gaussian and survival outcomes, including scenarios in which the external data are compatible on part of the covariate space ~~only~~ evaluates it in simulations, Section 4 ~~applies the method to~~ illustrates a single-arm myeloma ~~trial with a registry control~~ comparison, and Section 5 ~~concludes~~ discusses limitations. Proofs are in the Web Appendix.

2 Methods

2.1 Setting and estimands

We observe an RCT with N subjects, of whom n_1 are controls and $N - n_1$ are treated, and an external source of n_2 RWD controls. For subject i we record the outcome y_i , the covariate vector $x_i \in \mathbb{R}^p$, and the source $g_i \in \{1, 2\}$, with $g_i = 1$ for an RCT control and $g_i = 2$ for an RWD control; $D_i = \mathbf{1}\{g_i = 1\}$ indicates an RCT control. Continuous outcomes are modelled as Gaussian given the covariates. Survival outcomes are modelled on the log scale under a log-normal AFT model with right censoring, and y_i then denotes the log survival time, observed or imputed as described in Section 2.3.

The estimands are standardized to the RCT covariate distribution. Let $f_T(x)$ be the mean outcome under treatment and $f_1(x)$ the mean outcome of an RCT control at covariate x . For Gaussian outcomes the average treatment effect is $\text{ATE} = N^{-1} \sum_{i \leq N} \{f_T(x_i) - f_1(x_i)\}$, the average over all N RCT profiles, treated and control. For survival outcomes we report the ratio of population median survival times $q_{0.5}(\bar{S}_T)/q_{0.5}(\bar{S}_1)$, where $\bar{S}_a(t) = N^{-1} \sum_i S_a(t|x_i)$ is the arm- a survival function standardized to the RCT profiles, and the ratio of restricted mean survival times $\int_0^{t^*} \bar{S}_T(t) dt / \int_0^{t^*} \bar{S}_1(t) dt$ at a horizon t^* inside the follow-up. The treatment surface f_T is estimated from the RCT treated subjects alone with standard BART or AFT-BART. ~~The~~ External

~~observations inform the control surface f_1 is where the external data enter, and the~~. We define effective sample size ~~is defined~~ through the RCT-standardized control mean $\mu = N^{-1} \sum_{i \leq N} f_1(x_i)$.

Both ensembles below are BART models [5]. A BART surface is a sum of H regression trees, $\sum_{h=1}^H m(x; T_h, \theta_h)$, where T_h is a binary tree whose internal nodes split on one covariate at one cutpoint and whose terminal nodes (leaves) $\ell = 1, \dots, L_h$ carry means $\theta_{h\ell}$, and $m(x; T_h, \theta_h)$ returns the mean of the leaf to which x is routed. The tree prior splits a node at depth d with probability $\alpha(1+d)^{-\beta}$, chooses the splitting variable and cutpoint uniformly, and the leaf means are independent $\mathcal{N}(0, \lambda^2)$ with $\lambda = (\max y - \min y)/(2k\sqrt{H})$ and $k = 2$, the range rule, so that the sum of H trees spans the outcome range with high prior probability.

2.2 The model

The two control sources are modelled jointly as

$$y_i = f(x_i) + D_i g(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_{g_i}^2), \quad (1)$$

~~so that the RWD controls follow f , the RCT controls follow $f_1 = f + g$, and the sources keep separate residual variances.~~ Here $f(x)$ is the external-control response mean and $f_1(x) = f(x) + g(x)$ is the RCT-control response mean, with a separate residual variance for each source. Both surfaces are estimated jointly.

The pooled surface f is a standard BART ensemble of $H_f = 50$ trees with the default split prior $(\alpha_f, \beta_f) = (0.95, 2)$ and leaf prior $\mathcal{N}(0, \lambda_f^2)$ with λ_f from the range rule. The discrepancy surface g is a small, shallow ensemble of $H_g = 5$ trees with split prior $(\alpha_g, \beta_g) = (0.5, 3)$, so that a g tree is a stump a priori with probability one half and rarely deeper than two levels, and its leaves ~~carry~~ represent differences from the external-control response mean. The commensurate component centers $f_1(x)$ at $f(x)$. In particular, conditional on f , the discrepancy-tree partitions T_g and the spike variance, if all H_g leaves reached by x are in the spike, then

$$\underbrace{f_1(x) | f, T_g, \{z_{h, \ell_h(x)} = 1\}_{h=1}^{H_g}, \tau_0^2}_{\sim \mathcal{N}(f(x), H_g \tau_0^2)}.$$

Thus agreement means that the RCT-control response mean is close to the external-control response

mean. We implement this prior with zero-centered leaves for $g(x) = f_1(x) - f(x)$:

$$\theta_{h\ell} | z_{h\ell} \sim z_{h\ell} \mathcal{N}(0, \tau_0^2) + (1 - z_{h\ell}) \mathcal{N}(0, \tau_1^2), \quad z_{h\ell} \stackrel{\text{iid}}{\sim} \text{Bernoulli}(w), \quad (2)$$

with $w \sim \text{Beta}(1, 1)$, $\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)$ with $\nu_0 = 3$, and $\tau_1^2 = (\max y - \min y)^2 / (4k^2 H_g)$ fixed by the range rule for an H_g -tree ensemble. We require $\tau_0^2 < \tau_1^2$, Assumption A1 of the Web Appendix, and the sampler draws τ_0^2 from its conditional truncated to $(0, \tau_1^2)$. The residual variances carry independent inverse-gamma priors. The spike, $z = 1$, ~~is the commensurate component: the RCT control surface is locally the RWD surface up to a discrepancy of~~ shrinks an individual discrepancy contribution toward zero and thereby shrinks the RCT response mean toward the external response mean. Section 2.4 calibrates its leaf scale τ_0 ~~, the scale the effective sample size calibration of Section 2.4 sets~~ through the effective sample size of the full ensemble. The slab, $z = 0$, is the vague component of the robust MAP prior placed in a leaf of g : a free local discrepancy on the outcome scale, sized by the range rule. ~~Integrating If the scale prior is left untruncated, integrating z and τ_0^2 , a g -leaf has the marginal prior gives the leaf marginal~~ $w t_{\nu_0}(\theta; 0, s_0^2) + (1 - w) \mathcal{N}(\theta; 0, \tau_1^2)$, and the. The sampler uses the truncated scale prior, for which the spike marginal is not exactly Student- t . ~~The~~ The local discrepancy $g(x) = f_1(x) - f(x)$ ~~is a sum of sums H_g such draws~~ leaf contributions, one per tree. More generally, conditional on f, T_g , the reached-leaf indicators and both scales, $f_1(x)$ is normal with mean $f(x)$ and variance $\sum_{h=1}^{H_g} \{z_{h, \ell_h(x)} \tau_0^2 + (1 - z_{h, \ell_h(x)}) \tau_1^2\}$. The slab increases the permitted departure from $f(x)$ while retaining the same prior center. These distributions describe the latent control mean; observation-level residual variance enters through (1).

Three special cases place the model among its comparators. With $w = 1$ every leaf is in the spike and the model is a commensurate BART (C-BART) with a single commensurability scale τ_0^2 ; with $w = 1$ and $\tau_0^2 \rightarrow 0$ the discrepancy vanishes and the model is BART fitted to the pooled controls (BART-CP); with $w = 0$ every leaf is in the slab and g is a free nonparametric source offset regularized by the range rule, the tree analogue of BART with the source indicator as a covariate (BART-PP). LRC-BART with w learned from the data moves between these per leaf.

~~The decomposition is what lets a regional shift be detected: in the alternative design, one ensemble with two means per leaf coupled through a common variance, the cheapest fit spreads a shift across all H trees under the spike, and a prior-cost calculation (Separate ensembles allow the~~

complexity of the external mean and that of the source discrepancy to be controlled independently. A small g ensemble with shallow trees expresses our prior preference for discrepancies that are simpler than the outcome surface. The fixed-partition calculation in Web Appendix G.6 shows that at $H = 50$ no shift of clinical size flips an indicator, whereas in (1) the shift lands in one or two leaves of a small g ensemble after a single split, where the spike cannot hide it by spreading it. Identification is by support: where the RCT has no support g is a prior draw and describes how the number of discrepancy contributions affects their prior cost; it does not establish a detection threshold for the fitted model or rule out regional borrowing with shared leaves. At the population level, f is the RWD surface, and where both sources have support f and $f + g$ are identified on their respective source supports, and their difference is identified on the intersection. Outside that intersection, decomposition or prediction requires extrapolation. A g separate because f is pinned by the RWD and g is shrunk by the spike leaf with no RCT controls has its prior conditional on the partition and current hyperparameters, but shared leaves and learned hyperparameters can transmit trial information beyond trial support (Web Appendix E).

2.3 Posterior computation

Inference is by We use Markov chain Monte Carlo in the backfitting form with the backfitting scheme of Chipman et al. [5], one tree at a time with the others held fixed. Every factor entering a tree move is in closed form, because conditionally The tree-move factors are closed form: conditional on (τ_0^2, τ_1^2, w) the leaf prior (2) is a finite Gaussian mixture. For a leaf of an f tree the partial residuals $r_i = y_i - f_{-h}(x_i) - D_i g(x_i)$ come from both sources with their own variances; with $A = n_1/\sigma_1^2 + n_2/\sigma_2^2$ and $B = S_1/\sigma_1^2 + S_2/\sigma_2^2$, where n_g and S_g are the count and sum of the source- g residuals in the leaf, the observation-level factors $\prod_i \phi(r_i; 0, \sigma_{g_i}^2)$ cancel from every tree-move ratio and the leaf factor is

$$M_f = (1 + \lambda_f^2 A)^{-1/2} \exp\left\{\frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)}\right\}, \quad (3)$$

the heteroscedastic form of the standard BART marginal. For a leaf of a g tree the partial leaf uses residuals $r_i = y_i - f(x_i) - g_{-h}(x_i)$ run over the from RCT controls in the leaf only that leaf,

with count n_1 and mean $\bar{y}_1$, and writing $v_1 = \sigma_1^2/n_1$,

$$M_g = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}{\phi(\bar{y}_1; 0, v_1)}, \quad (4)$$

~~a two-component Gaussian mixture in the leaf mean, with $M_g = 1$ in a leaf holding for a leaf with~~
no RCT controls. Grow, prune ~~;~~ and change moves ~~for both ensembles use the standard multiply~~
~~the tree-prior and proposal ratios times the ratio of products of M_f or M_g over the affected leaves,~~
~~by the affected leaf factors;~~ a grow move ~~contributing contributes~~ $M_L M_R / M_P$. ~~Since these are the~~
~~exact collapsed ratios, the tree kernels leave the exact posterior invariant with the leaf parameters~~
~~and indicators integrated out, and the full sweep is a partially collapsed Gibbs sampler with the~~
~~joint posterior as its invariant law~~ Each collapsed tree move is followed immediately by conditional
leaf and indicator draws. This ordering preserves the joint posterior (Lemma A1 and Proposition
A1 of Web Appendix A).

~~Given~~ Conditional on the trees, ~~the leaf draws are conjugate~~ we update the leaf parameters
conjugately. An f leaf draws $\theta \sim \mathcal{N}(\lambda_f^2 B / (1 + \lambda_f^2 A), \lambda_f^2 / (1 + \lambda_f^2 A))$. A g leaf is updated as a
block: first

$$P(z = 1 | \bar{y}_1) = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1)}{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}, \quad (5)$$

then $\theta | z \sim \mathcal{N}(\tau_z^2 \bar{y}_1 / (\tau_z^2 + v_1), \tau_z^2 v_1 / (\tau_z^2 + v_1))$ with τ_z^2 equal to τ_0^2 in the spike and τ_1^2 in the
slab. With K_0 spike leaves among the L_g leaves of g and S_0 the sum of their squared means, the
hyperparameters update as $\tau_0^2 | \cdot \sim \text{scaled-Inv-}\chi^2(\nu_0 + K_0, (\nu_0 s_0^2 + S_0) / (\nu_0 + K_0))$ truncated to $(0, \tau_1^2)$
and $w | \cdot \sim \text{Beta}(1 + K_0, 1 + L_g - K_0)$, and the residual variances update from their inverse-gamma
conditionals using the residuals $y_i - f(x_i) - D_i g(x_i)$ of each source. ~~We fix~~ The range rule fixes τ_1^2
~~by the range rule because slab leaves are few and the slab's job is to be wide on the outcome scale.~~
An f leaf ~~holding only with~~ one source is an ordinary BART leaf, ~~and the~~ The minimum leaf size
~~of five is enforced on each leaf's own data, pooled is five, using pooled data~~ for f and ~~RCT-only~~
RCT controls for g . A sweep costs a standard BART on $n_1 + n_2$ observations plus a small ensemble
on n_1 .

For ~~right-censored survival outcomes~~ we use the AFT model $\log y_i = f(x_i) + D_i g(x_i) + \varepsilon_i$ with
~~data augmentations~~ survival outcomes, $y_i = \log T_i$ denotes log time. At each iteration ~~a censored~~

subject's log survival time is censored values are imputed from $\mathcal{N}(f(x_i) + D_i g(x_i), \sigma_{g_i}^2)$ truncated to $(\log C_i, \infty)$, where C_i is the censoring time, and treated as observed in the used in the subsequent tree and parameter updates that follow; the. The treatment-arm AFT-BART is imputed the same way. Nothing else in the sampler changes. uses the same augmentation.

2.4 Prior effective sample size and calibration

The spike scale s_0^2 is hard to set directly, because a given value means different amounts of borrowing depending on the residual scale, the leaf sizes, and H_g . We calibrate it through a prior. We calibrate the spike scale through a variance-based effective sample size (ESS) defined at the estimand. Conditionally on the RWD, the prior for the RCT control surface is $f_1 = f + g$ with $f | \text{RWD}$ the posterior of a standard BART fitted for the standardized control mean. Stage 1 fits f to the RWD alone and g drawn from its prior estimates $V_\mu^f = \text{Var}(N^{-1} \sum_i f(x_i) | \text{RWD})$. For a scalar parameter with prior normal prior of variance V and unit Fisher information $1/\sigma_1^2$, a reference observation with distribution $\mathcal{N}(\mu, \sigma_1^2)$, σ_1^2/V equals the expected local-information-ratio (ELIR) ESS of Neuen-schwander et al. [18] is σ_1^2/V , the number of RCT control observations carrying the same information as the prior, and we apply this to μ . Write $V_\mu^f = \text{Var}(N^{-1} \sum_{i \leq N} f(x_i) | \text{RWD})$, which is read directly from the posterior draws of the RWD-only BART evaluated at the RCT profiles, and $c_g = \mathbb{E}[N^{-2} \sum_{i,j} \mathbf{1}\{x_i, x_j \text{ share a leaf}\}]$ under the g -tree prior, the leaf-sharing constant, which equals one for stumps and is computed once. The marginal BART and scale-mixture priors here are generally nonnormal. We therefore use the precision ratios below as working information summaries; they are not exact marginal ELIR values. For survival outcomes, σ_1^2 sets uncensored log-time reference units, not the information in one censored observation.

For fixed discrepancy partitions $\mathcal{T}$, let $a_{h\ell}$ be the proportion of the N standardization profiles in leaf ℓ of tree h . Independence of the leaf priors gives

$$\mu_g = \sum_{h,\ell} a_{h\ell} \theta_{h\ell}, \quad V_g(\mathcal{T}, z, \tau_0^2) = \sum_{h,\ell} a_{h\ell}^2 \{z_{h\ell} \tau_0^2 + (1 - z_{h\ell}) \tau_1^2\}.$$

A standardized mean can involve several leaves in each tree. Its prior average must therefore integrate over all their indicators; a binomial count of H_g indicators suffices only when one leaf per

tree has positive standardization weight, including a pointwise profile. Web Appendix D gives the general leaf-state average.

For calibration we set every discrepancy leaf to the spike as an informative reference. Define $c_g = \mathbb{E}_{T_h}[\sum_{\ell} a_{h\ell}^2]$, estimated by Monte Carlo. ~~Given the spike and slab states, g adds $c_g \sum_h \tau_{z_h}^2$ to the prior variance of μ . The ESS of the commensurate component, with every g leaf in the spike, is~~

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)} \left[\frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right],$$

under the tree prior; $c_g = 1$ for stumps. We replace the random sum $\sum_{h,\ell} a_{h\ell}^2$ by $c_g H_g$ and define

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)} \left[\frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right]. \quad (6)$$

~~a one-dimensional integral, and the prior-averaged ESS is~~

$$\text{ESS}(w, s_0^2) = \sum_{k=0}^{H_g} \binom{H_g}{k} w^k (1-w)^{H_g-k} \sigma_1^2 \mathbb{E}[\{V_\mu^f + c_g(k\tau_0^2 + (H_g - k)\tau_1^2)\}^{-1}].$$

~~We calibrate ESS_0 . This averages inverse conditional variances over the untruncated scale law, with the mean partition coefficient substituted before inversion. The fitting prior truncates the scale at τ_1^2 ; equation (6) is the calibration convention used for all reported results. It differs from averaging over the full partition prior or computing the information of the marginal mixture. We calibrate this informative reference and report w separately; as robust MAP reports the ESS of its informative component and the robust weight separately: with per-leaf indicators the prior probability that every leaf is commensurate is. Conditional on w , the probability that all leaves reached by a profile are in the spike is w^{H_g} , so the averaged ESS is dominated by w and cannot reach a target of the order of n_2 whatever s_0^2 is; for a standardized mean involving L_+ distinct leaves it is w^{L_+} .~~

The definition has a ceiling: as $s_0^2 \rightarrow 0$, $\text{ESS}_0 \rightarrow \sigma_1^2 / V_\mu^f$, the information the RWD posterior carries about μ before any discrepancy is added. ceiling of (6) is σ_1^2 / V_μ^f . It includes the regularization of the RWD-only BART. For a single tree with RCT leaf proportions equal to the RWD ones, $V_\mu^f = \sigma_2^2 / n_2$ and the ceiling is fixed tree with a flat leaf prior and RWD support in every target leaf, it equals $n_2 \sigma_1^2 / \sigma_2^2$ exactly, the RWD sample size in RCT units; when source leaf proportions

match and decreases by the factor $\sum_\ell a_\ell^2/\pi_\ell$ under covariate shift ~~it is lower, since RWD in regions without RCT patients carries no information about μ (Theorem 3(d)). The regularization of f itself contributes a term of order $\sigma_1^2/(H_f\lambda_f^2)$ that is negligible at $H_f=50$ and is not subtracted. A target N_{target} may be given as an absolute number of RCT controls, feasible only below the ceiling, or as a fraction of the ceiling, which is always feasible and which we recommend as the primary sensitivity ladder when the~~ This equality and direction of change need not hold under finite leaf-prior regularization. We recommend a sensitivity ladder of fractions of the fitted ceiling when overlap is poor.

The calibration has two stages, of which Stage 1 fits standard BART to the RWD alone and records, for each posterior draw, the standardized surface $N^{-1}\sum_{i \leq N} f^{(b)}(x_i)$ at the RCT profiles and the residual variance; the draws are cut ~~We divide the Stage-1 draws into B consecutive blocks, and each block gives one value~~ blocks and rescale their variances $V^{(b)}$ of the estimand variance, ~~resealed so that the block mean equals the whole-chain variance to have mean V_μ^f .~~ The RCT residual variance σ_1^2 is the posterior mean from a standard BART fitted to the RCT controls alone, which puts ESS_0 in RCT units; before RCT outcomes exist, σ_2^2 or a design value is substituted. Stage 2 evaluates (6) on a ~~On a~~ grid $\mathcal{S}$ of candidate scales for each block and selects by the Pr-rule is

$$\hat{s}_{0\star}^2 = \inf \{s_0^2 \in \mathcal{S} : \hat{G}(s_0^2) \geq q\}, \quad \hat{G}(s_0^2) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\text{ESS}_0(V^{(b)}; s_0^2) \leq N_{\text{target}}\}, \quad (7)$$

with $q = 0.95$: ~~the most informative spike whose prior ESS respects the target in at least a fraction q of the Stage-1 blocks. Stage 1 does not depend on s_0^2 , so the same draws serve any target, grid, or level q . When N_{target}~~ If the requested target is at or above the ceiling the rule whole-chain ceiling, the implementation instead returns the smallest grid ~~value, the point-mass limit that is BART-CP, and the target is flagged as capped. In the two-arm analysis τ_0^2 is then updated from the spike leaves, so the calibrated prior is informative but not fixed, and we report both the prior ESS_0 and the realized ESS, $\mathbb{E}_{\text{post}}[\sigma_1^2/\{V_\mu^f + e_g \sum_h \tau_{z_h}^2\}]$ with the indicators and τ_0^2 drawn from the posterior. A pointwise version, $\text{ESS}(x) = \sigma_1^2/\{V_f(x) + H_g \tau_0^2\}$ scale and flags the fit as capped; this is a separate override of (7). If no grid value satisfies the rule, it returns and flags the largest scale.~~

The reported realized ESS is $\mathbb{E}_{\text{post}}[\sigma_1^2/\{V_\mu^f + V_g(\mathcal{T}, z, \tau_0^2)\}]$, with $V_f(x)$ the RWD exact squared

leaf proportions at each posterior state. Here V_g is a prior variance evaluated at learned states, not the posterior variance of f at x , gives the local ESS map of Section 4 μ_g or a measured reduction in posterior uncertainty. Regional summaries standardize separately within each region and are not additive. The application uses the pointwise summary $\text{ESS}(x) = \hat{\sigma}_1^2 / \{V_f(x) + H_g \bar{\tau}_0^2\}$, where $V_f(x)$ is the Stage-1 variance and $\bar{\tau}_0^2$ is the fitted mean spike variance. Appendix D distinguishes this plug-in from averaging pointwise precision over the scale prior.

2.5 The borrowing map and its guarantees

The output that distinguishes the method is a map $\mathcal{B}_c$ over covariate space of how much external information was used: by the posterior probability that the local discrepancy is smaller than a threshold $c > 0$ fixed in advance on the outcome scale,

$$\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data}), \quad (8)$$

computed as the fraction of posterior draws, together with the posterior mean and interval of $g(x)$ and the pointwise ESS of Section 2.4. The $\mathcal{B}_c$ also retains the all-spike probability $B(x)$, the posterior probability that every leaf covering x is in the spike given the data, is kept belongs to the spike, as a diagnostic; its prior scale is w^{H_g} and, as Theorem 2 explains, it does not settle as the RCT grows its limiting value need not identify the true compatibility state, whereas $\mathcal{B}_c$ does is consistent under the stated conditions. In the simulation study $c = 0.5$, one third of the RCT residual standard deviation for Gaussian outcomes and 0.5 on the log-time scale for survival outcomes.

Three results follow. We give three theoretical results. The first two are stated at the leaf level concern leaf-level inference and condition on the pooled surface f , on the partition of the g trees, and on $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$, and consider a g leaf with n RCT controls, residual mean $\bar{y}$, and $v = \sigma_1^2/n$. Write $\rho_j = \tau_j^2 / (\tau_j^2 + v)$ for $j \in \{0, 1\}$, so that $\hat{\theta}_1 = \rho_1 \bar{y}$ is the slab estimate, close to the RCT-only residual mean, and $\hat{\theta}_0 = \rho_0 \bar{y}$ the spike-shrunk estimate, close to zero, that is, to the pooled surface. Let $p = P(z = 0 \mid \bar{y})$ be the posterior slab probability and

$$\kappa = \frac{1}{2} \left(\frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right), \quad a = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2 + v}{\tau_1^2 + v}, \quad C = e^{-a}. \quad (9)$$

Theorem 1 (Bounded local influence). *Under Assumption A1, $0 < w < 1$, and the conditioning above, $\theta | \bar{y}$ is the mixture $p\mathcal{N}(\hat{\theta}_1, \rho_1 v) + (1-p)\mathcal{N}(\hat{\theta}_0, \rho_0 v)$ with $\text{logit} p = a + \kappa \bar{y}^2$, and the shift of the posterior mean relative to the slab estimate, $I(\bar{y}) = \mathbb{E}[\theta | \bar{y}] - \hat{\theta}_1 = -(1-p)(\rho_1 - \rho_0)\bar{y}$, satisfies*

$$\sup_{\bar{y}} |I(\bar{y})| \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}, \quad 1-p \leq Ce^{-\kappa \bar{y}^2},$$

so $I(\bar{y}) \rightarrow 0$ at a Gaussian rate as $|\bar{y}| \rightarrow \infty$. With $w = 1$ (C-BART) the shift is $-(\rho_1 - \rho_0)\bar{y}$, linear and unbounded, and with $w = 1$, $\tau_0^2 \rightarrow 0$ (BART-CP) it is $-\rho_1 \bar{y}$.

Theorem 1 is Theorem A1 of Web Appendix B. It ~~says that the commensurate component can move a leaf estimate away from what the RCT controls alone would give by at most a bounded amount, of the order of one RCT standard error of the leaf, and that once the local conflict is large the spike is switched off and the influence disappears. The bound is on the borrowing-specific part; the slab estimate itself is shrunk~~ bounds the spike-induced shift relative to the slab estimate, on the scale of an RCT leaf standard error. The slab itself shrinks toward the pooled surface by ~~the factor ρ_1 , the range-rule regularization any BART leaf applies, which is slight at ordinary leaf sizes~~ applied to a BART leaf. C-BART and BART-CP have ~~no such bound, and their bias grows linearly with the conflict~~ unbounded conditional shifts. Corollary A1 ~~carries~~ extends the bound to the ~~estimand: at fixed partitions of the g ensemble the RWD-induced shift of the standardized control mean relative to the free-offset model is bounded uniformly in the RCT data, whereas under~~ standardized control mean at fixed g partitions; the corresponding C-BART and BART-CP ~~it is a nonzero linear functional of the data and hence unbounded~~ shifts are linear and unbounded unless the linear functional vanishes.

Theorem 2 (Detection and the borrowing map). *Under Assumption A1, $0 < w < 1$, and the conditioning above, let the RCT residuals in the leaf be $d + \epsilon_i$ with $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_1^2)$, where d is the true local discrepancy.*

(a) *The spike probability $P(z = 1 | \bar{y}) = \{1 + \exp(a + \kappa \bar{y}^2)\}^{-1}$ is decreasing in $|\bar{y}|$, and when*

$\log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} > 0$ it equals one half at

$$d_{1/2} = \left[\frac{2(\tau_0^2 + v)(\tau_1^2 + v)}{\tau_1^2 - \tau_0^2} \left\{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} \right\} \right]^{1/2},$$

~~a few multiples of $(\tau_0^2 + \sigma_1^2/n)^{1/2}$, so the probability that the leaf abstains at true shift d is $\bar{\Phi}\{(d_{1/2} - d)/\sqrt{v}\} + \bar{\Phi}\{(d_{1/2} + d)/\sqrt{v}\}$, with $\bar{\Phi}$ the standard normal survival function.~~

(b) ~~For a single g tree, $\mathcal{B}_c(x) = p \Psi_1(c) + (1 - p) \Psi_0(c)$ with $\Psi_j(c) = P\{|\mathcal{N}(\rho_j \bar{y}, \rho_j v)| < c\}$, and as $n \rightarrow \infty$ with $|d| \neq c$, $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|d| < c\}$ almost surely.~~

(c) ~~As $n \rightarrow \infty$ with τ_0^2 fixed, the spike probability converges to a limit in $(0, 1)$ that is at most $\frac{w\tau_1}{(1-w)\tau_0} \exp\{-\frac{d^2}{2}(\tau_0^{-2} - \tau_1^{-2})\}$ for $d \neq 0$ and exceeds w for $d = 0$; it tends to zero only as $|d|/\tau_0 \rightarrow \infty$, or for every fixed $d \neq 0$ if $\tau_0^2 \rightarrow 0$ with n .~~

Theorem 2 ~~collects Theorem~~ combines Theorems A2 ~~(a),(b) and Theorem and~~ A3 of Web Appendix C. ~~Part (a) is the resolution of a leaf: a shift is detected only beyond a few multiples of the spike scale plus the RCT standard error of the leaf, which is why the simulation study includes shifts at and below this scale. Part (b) is the consistency of the map we display: as the RCT controls in a region accumulate, $\mathcal{B}_c$ converges to the truth about whether the local discrepancy is below c , whatever the spike indicators do. Part (c) is the reason we display $\mathcal{B}_c$ and not the all-spike probability: the spike is a Gaussian component of scale τ_0 , not a point mass, so a discrepancy of that order is consistent with it and the indicator's question stays uncertain in the limit, whereas $\mathcal{B}_c$ asks a question the RCT controls settle. In the two-arm sampler τ_0^2 is learned under a prior calibrated to a small scale, so the exponentially small bound in (c) is the operative statement and abstention under a shift of several spike scales is near certain; Section 3 shows what happens at a shift near $d_{1/2}$. Proposition A2 extends (b) to the $H_{\bar{g}}$ -tree ensemble with fixed partitions, provided the true discrepancy is representable on those partitions, with the map converging cell by cell as the RCT count in a cell grows. These results are conditional on the partitions of the g trees: we have no statement about the posterior over the partitions themselves, hence none about $\mathcal{B}_c$ under the full posterior, since the two-ensemble decomposition with a spike-and-slab discrepancy prior lies outside the available posterior-concentration theory for BART. The leaf-level results also treat~~ The ensemble extension requires fixed partitions and a discrepancy representable on their additive span. The leaf results also condition on f as fixed, whereas with f estimated the residual in a g leaf carries the RWD posterior error of f at the RCT profiles as an additive perturbation of d . The separation of shifted from compatible regions by $\mathcal{B}_c$; they do not establish consistency under the full posterior ~~is shown empirically in Section 3~~ over both ensembles. Further interpretation is given

in Web Appendix G.3.

Theorem 3 (Monotone, well-posed calibration). *Fix $\sigma_1^2 > 0$, $V > 0$, ~~and~~ $c_g H_g > 0$, ~~and~~ $N_{\text{target}} > 0$ and $q \in (0, 1)$, and write $\text{ESS}_0(V; s)$ for (6) at $V_\mu^f = V$ and $s = s_0^2$.*

- (a) *$s \mapsto \text{ESS}_0(V; s)$ is continuous, convex, and strictly decreasing on $(0, \infty)$, with limits σ_1^2/V as $s \downarrow 0$ and 0 as $s \uparrow \infty$, ~~whether or not the prior on τ_0^2 is truncated under the untruncated calibration prior in (6). Under truncation to $(0, \tau_1^2)$; $\text{ESS}(w, s)$ is in addition strictly decreasing in τ_1^2 and strictly increasing in w , continuity and strict decrease remain, but the limit as $s \uparrow \infty$ is $\sigma_1^2/(V + c_g H_g \tau_1^2) > 0$; convexity is asserted only for the untruncated law.~~*
- (b) ~~For~~ *Under the untruncated calibration law, for each V the sublevel set $\{s : \text{ESS}_0(V; s) \leq N_{\text{target}}\}$ is a half-line $[s^\dagger(V), \infty)$, so the population rule $G(s) = P\{s^\dagger(V) \leq s\}$ is a distribution function and $\{s : G(s) \geq q\} = [s_\star, \infty)$ has a unique attained infimum; $s_\star > 0$ if and only if $P(N_{\text{target}} < \sigma_1^2/V) > 1 - q$, and otherwise $s_\star = 0$, the ~~point-mass limit zero-scale limit of the informative reference.~~*
- (c) ~~If the Stage-1 draws form a stationary ergodic chain, $\hat{G}(s) \rightarrow G(s)$ almost surely for each s , uniformly in s when G is continuous, and on a finite grid with $G \neq q$ at every grid point the empirical selection equals $s_\star$ for all~~ *For a stationary, aperiodic Harris-ergodic full Stage-1 chain and a square-integrable standardized surface, fixed B and increasing block length select the smallest grid value satisfying the whole-chain rule, on a grid separated from its root. For fixed block length and increasing B large enough, almost surely; under geometric ergodicity a central limit theorem and an $O_p(B^{-1/2})$ rate hold, the rule converges to the selection based on the limiting rescaled block law, under the positivity, continuity and grid-separation conditions in Web Appendix D. These statements concern (7) with a nonempty admissible grid, not the ceiling override.*
- (d) *For a single f tree with fixed partition, RCT profile proportions a_ℓ and RWD proportions ~~π_ℓ over its leaves~~, $\lim_{s \downarrow 0} \text{ESS}_0(s) \geq n_2 \sigma_1^2 / \sigma_2^2 / \sum_\ell a_\ell^2 / \pi_\ell$, with equality ~~$\pi_\ell > 0$ on every target leaf, the ceiling is at least $n_2 \sigma_1^2 / \sigma_2^2 / \sum_\ell a_\ell^2 / \pi_\ell$. Equality holds in the flat-prior limit; the denominator is at least one, with equality if and only if $a = \pi$ (zero-weight 0/0 terms contribute zero).~~*

Theorem 3 ~~is Theorem A4 of~~, proved in Web Appendix D. ~~Parts (a) and (b) say that the Pr-rule is well defined : the ESS moves in one direction as the spike scale moves, so there is exactly one most informative admissible scale, and it is strictly positive exactly when the target is~~, concerns the defined calibration summary. Under the untruncated law, each positive target below the ceiling in enough Stage-1 blocks. ~~A leaf-level ESS diverges as the scale shrinks and is feasible for any target; here the ceiling is finite and feasibility depends on the data and the target, which is why the fractional target exists. Part (c) says that the Monte Carlo rule recovers the population selection. Part (d) ties the ceiling to the sample sizes: with matched covariate distributions the ceiling is the RWD sample size in RCT units, and covariate shift between the sources lowers it by the factor $\sum_{\ell} a_{\ell}^2 / \pi_{\ell}$, which measures how unevenly the RCT profiles are spread over the leaves relative to the RWD. has a unique positive root; truncation adds a positive lower limit. The block rule, finite grid and ceiling override are distinct parts of the implementation. These results establish neither predictive consistency of marginal ELIR nor a guarantee on frequentist operating characteristics.~~

2.6 The single-arm variant

When the trial has no control arm, $n_1 = 0$, every g leaf holds no RCT controls, $M_g = 1$, and the f moves are exactly those of standard BART on the RWD. The control surface is then $f_1 = f + g$ with f the RWD surface and g a prior draw, a zero-mean discrepancy process ~~of~~ with pointwise variance $H_g\{w\tau_0^2 + (1-w)\tau_1^2\}$ ~~at a profile covered by H_g distinct leaves~~ conditional on the scales and w , and τ_0^2 and w stay at their priors. ~~Nothing in the data informs the indicators, and slab draws only add prior noise of variance~~ The indicators receive no likelihood information, and the slab contributes $(1-w)H_g\tau_1^2$ to the counterfactual controls conditional pointwise variance, so we report the $w = 1$ model, the commensurate prior with the discrepancy carried by g at the calibrated scale, and use $w < 1$ as a sensitivity analysis. ~~The calibration is unchanged: Stage 1 is~~ Calibration uses the RWD-only fit, σ_1^2 is set to σ_2^2 in the absence of RCT outcomes, the ceiling is at most about n_2 , and ~~the target is stated as an absolute number of controls or a fraction~~ Stage-1 fit. We assume $\sigma_1^2 = \sigma_2^2$ for calibration and counterfactual survival prediction; single-arm data cannot check this equality. Targets are absolute control counts or fractions of the ceiling. With $w = 1$ the single-arm model is a hierarchically centered commensurate AFT-BART, $f_1(x) = f(x) + \sum_h e_h$ with $e_h \sim \mathcal{N}(0, \tau_0^2)$, ~~the discrepancy riding with the discrepancy defined~~ on the partition of g instead of that of f . ~~Because~~

~~the borrowing map is flat by construction here~~ The prior mean of the Gaussian control surface is preserved, but nonlinear survival-ratio means can change when discrepancy uncertainty is added. Because the compatibility map is constant across profiles in this setting, the covariate-resolved display for a single-arm analysis is the local ESS map $\text{ESS}(x)$, which ~~shows where the RWD support at the trial profiles is thin~~ describes variation in RWD support across trial profiles.

3 Simulation studies

3.1 Design

Each replicate consists of an RCT of $N = 300$ subjects and an external source of $n_2 = 300$ RWD controls, with $p = 10$ covariates $X_j \sim \mathcal{N}(\mu_j, 1)$, $\boldsymbol{\mu} = (1, -1, 0.5, -1, 2, -2, 2, -3, 1, 0)$, and X_1 replaced with probability 0.15 by a draw from $\mathcal{N}(\pm 3, 0.5^2)$ so that the sources overlap imperfectly. The RCT randomizes 2:1 for Gaussian outcomes, giving about $n_1 = 100$ controls, and 3:1 for survival outcomes, giving about 75, ~~the smaller control arm of~~ as in a hybrid design with a smaller control arm. The RCT control and RWD outcome surfaces are the same nonlinear function

$$f_0(\mathbf{W}) = \alpha + \beta_1 W_1^2 + \beta_2 e^{-(W_2+1)^2/2} + \beta_3 |W_3 - 0.5| + \beta_4 |W_4 + 1| + \beta_5 \tanh(W_5 - 2) + \beta_6 \tanh(W_6 + 2) + 2\{b(W_5 - 2) + b(W_6 + 2)\} + \sum_{j=7}^{10} \beta_j W_j, \quad (10)$$

with $b(u) = u^2 / \{1 + (u/1.5)^2\}$ and

$$\boldsymbol{\beta} = (-0.50, -0.75, -0.50, -0.50, -1.35, -0.80, -0.01, -0.01, -0.01, -0.01),$$

and $\mathbf{W} = \mathbf{X}$ except where stated; the treated RCT subjects add $\delta = 1$ for Gaussian outcomes and $\delta = \log 1.4$ on the log-time scale for survival outcomes. Residual standard deviations are $\sigma_1 = 1.5$ and $\sigma_2 = 1.8$ for Gaussian outcomes and 1.4 and 1.8 for survival outcomes, where the survival time is log-normal, dropout is Exponential(0.02), entry is Uniform(0, 1) and administrative censoring occurs at calendar time 3.

~~Five scenarios govern how the sources differ~~ We consider five patterns of agreement between sources. Sc1 assigns the source at random. Sc2 assigns it through $P(D = 1 | \mathbf{X}) = \text{logit}^{-1}\{(\ell - q_{0.5}(\ell))/0.5\}$ with $\ell = -1.20(X_5 + X_6)$, ~~a measured confounding of the source by~~ so that source

membership depends on the two covariates with the largest outcome effects. Sc3 drives the same assignment by unmeasured (U_1, U_2) that also replace (X_5, X_6) in the outcome, with the observed (X_5, X_6) correlated with $\mathbf{U}$ at $\rho \in \{-0.5, 0, 0.5\}$, so the RWD surface differs from the RCT surface everywhere in a way the covariates cannot remove; Web Appendix G lists the intercepts and the assignment constants.

Sc4 and Sc5 ~~are the scenarios a covariate-adaptive method exists for~~evaluate borrowing under regional and global outcome shifts. Sc4 keeps the assignment of Sc1 and raises the RWD surface by δ_2 outside a compatible region $\mathcal{R} = \{X_5 > 2\}$, so that the true discrepancy $g = f_1 - f$ is $-\delta_2$ there and 0 inside. The region holds half of the RCT population and lies on the covariate with the dominant nonlinear effect, so the shift has to be separated from a slope that f already models. We take $\delta_2 \in \{0.5, 1, 2\}$, one third to 1.3 RCT residual standard deviations, chosen around the leaf resolution $d_{1/2}$ of Theorem 2(a), which at $\sigma_1 = 1.5$, fifty RCT controls per leaf and the spike scale calibrated in Sc1 is about 0.5. Sc5 raises the RWD surface by $\delta_2 \in \{1, 2\}$ everywhere, so the compatible region is empty and ~~the right answer is to abstain~~borrowing should be suppressed throughout the covariate space. Survival outcomes use Sc4 with $\delta_2 \in \{0.5, 1\}$ on the log scale and Sc5 with $\delta_2 = 1$.

The estimands are those of Section 2.1, standardized to the replicate’s RCT profiles: the ATE for Gaussian outcomes, and for survival outcomes the ratio of standardized population medians, whose true value is 1.4, and the ratio of standardized restricted mean survival times at $t^* = 3$. Over $R = 500$ replicates we report the bias of the posterior mean, the mean posterior SD, the Monte Carlo RMSE of the posterior mean, the coverage of the central 95% credible interval, and power, the fraction of replicates whose 95% interval excludes the null value, 0 for the ATE and 1 for the ratios; the arm-specific means are summarized the same way. Monte Carlo standard errors are about 0.009 for a Gaussian ATE bias and 0.01 for a coverage or power, and the replicate data are paired across methods and scenarios, so paired standard errors of a contrast between two methods are much smaller. For Sc4 and Sc5 we add the RMSE of the posterior mean control surface over all N RCT profiles inside and outside $\mathcal{R}$, the mean of the borrowing map $\mathcal{B}_c(x)$ over the RCT control profiles in each region at $c = 0.5$ on the outcome scale, and the mean of the posterior mean of $g(x)$ in each region, whose true values are 0 in $\mathcal{R}$ and $-\delta_2$ outside it. ~~Every~~For every LRC-BART fit ~~also records~~we also record the prior ESS_0 , the realized ESS, the ceiling, and whether the target

was capped. Since ESS_0 averages $\sigma_1^2 / \{V^{(b)} + c_g H_g \tau_0^2\}$ over the Stage-1 blocks while the ceiling uses the whole-chain variance, Jensen’s inequality puts the printed ESS_0 of a capped fit a few percent above the printed ceiling.

~~The comparators are~~ We compare LRC-BART with Bayesian linear and log-normal AFT regressions fitted to the RCT controls alone (LM-NP, AFT-NP), to the pooled controls (LM-CP, AFT-CP) and to the pooled controls with the source indicator as a covariate (LM-PP, AFT-PP); standard BART and AFT-BART [5, 24] in the same three configurations (BART-NP, BART-CP, BART-PP), the last being the implicit borrowing of Zhou and Ji [27]; the propensity-score composite likelihood PSCL of Chen et al. [4]; the MAP prior for the control mean without covariate adjustment; hierarchical linear and AFT models (HierLM, HierAFT); and the two nested versions of the proposed model, C-BART ($w = 1$) and the free-offset LRC-BART ($w = 0$) of Section 2.2. Web Appendix G.2 gives their priors, samplers and implementation details. We also transfer the analysis component of CAHB [13] to Sc1, Sc4 and Sc5 in $R = 200$ Gaussian replicates paired with the first 200 full-study replicates. Our choices for the historical smoother, covariate selection, bandwidth, precision scales and joint draws are described in Web Appendix G.2, together with the ten-covariate and literal-constant variants.

LRC-BART is fitted as in Section 2.2, with $H_f = 50$, $H_g = 5$, $(\alpha_g, \beta_g) = (0.5, 3)$, the range-rule slab, $\nu_0 = 3$, $w \sim \text{Beta}(1, 1)$, and a treated-arm BART with 50 trees. ~~The calibration fits~~ In Stage 1, we fit the calibration model to the RWD with 1,000 retained draws cut into $B = 20$ blocks, ~~takes~~ take σ_1^2 from a plain BART fitted to the RCT controls, ~~computes~~ compute c_g once by Monte Carlo under the g -tree prior, and ~~searches~~ search a geometric grid of 141 spike scales from 10^{-6} to 10. We report an absolute ladder $N_{\text{target}} \in \{50, 75, 100\}$, ~~capped at the ceiling where infeasible~~ with the ceiling override applied when the requested target reaches it, and a fractional ladder at 0.25, 0.5 and 0.9 of the ceiling, with LRC-BART (100) as the reference configuration and LRC-BART (0.9) as the fractional reference; LRC-BART ($H_g = 10$) doubles the discrepancy ensemble. Averaged over scenarios and replicates in the full study, run on 16 forked workers, a full LRC-BART fit with calibration and treated arm took 2.3 seconds for Gaussian and 2.5 seconds for survival outcomes, against 1.6 and 2.1 seconds for BART-PP and 1.1 and 1.6 for BART-NP, and the whole study ran in 5.6 hours. The configuration of g was chosen on a development run of $R = 200$ paired with the first 200 replicates here (Web Appendix G.3).

3.2 Gaussian outcomes

Table 1 reports the two-arm Gaussian results for ~~the selected~~ scenarios and methods ~~that carry the comparison~~, and Web Table S1 the full set. ~~When the sources agree,~~

Under agreement between sources (Sc1), the RMSE of LRC-BART (100) is 0.172 against 0.205 for BART-PP and 0.220 for BART-NP, with coverage 0.94; the paired difference in mean squared error from BART-PP is -0.012 with standard error 0.001. ~~The four tree models order themselves by how much they borrow,~~ BART-CP ~~pooling without adjustment at~~ has RMSE 0.142, ~~then followed by~~ C-BART at 0.150, LRC-BART at 0.172 and BART-PP at 0.205. ~~The prior ESS₀ of LRC-BART (100) averages has mean prior ESS₀ of 82 RCT controls, below the target because the ceiling, ceiling 127 ~~on average, is exceeded by the target in 20% of the replicates and the calibration then returns the point-mass limit, and the realized ESS, computed with the~~ and realized ESS 53, using posterior indicators and τ_0^2 drawn from the posterior, ~~The target exceeds the ceiling in 20% of replicates, triggering the smallest-scale override. Tree-model power is 53. Power at $\delta = 1$ is 1.00 for every tree model and separates only the linear ones, so it carries no information here, and the same holds at~~ at $\delta = 1$, also with a 90% interval (Web Appendix G).~~

Under measured confounding, ~~(Sc2, every),~~ the covariate-adjusted ~~method stays near unbiased, methods have small biases:~~ LRC-BART (100) with bias 0.023 and RMSE 0.192 against 0.007 and 0.204 for BART-PP, whereas BART-CP, which pools the sources, and ~~C-BART, which cannot let the sources differ,~~ C-BART, which has a common discrepancy scale without a slab, carry biases of 0.058 and 0.042 with coverage 0.91. ~~Sc2 is also where the ceiling of The ESS ceiling is substantially lower in Sc2, consistent with the covariate-overlap calculation in~~ Theorem 3(d) ~~becomes visible:~~ the RCT profiles concentrate ~~where the RWD is thin in~~ regions with limited RWD support, the ceiling falls from 127 in Sc1 to 23, every absolute target is capped in every replicate and the three ~~rungs of the absolute ladder collapse onto one~~ absolute targets give the same fit (Web Table S3), whereas the fractional ladder still spans prior ESS 5, 10 and 16 with biases 0.012, 0.016 and 0.019. ~~When the~~ We therefore recommend reporting fractional targets when covariate overlap is poor, ~~the fractional target is the one to report.~~

Under unmeasured confounding, Sc3, the RWD surface differs from the RCT surface everywhere by an amount the covariates cannot explain, and the two ~~complete-pooling tree models fail as~~ tree

~~models with fixed commensurate components have substantial bias, as do~~ their linear counterparts ~~do~~: BART-CP has bias 0.98 at $\rho = 0$ with coverage 0, and C-BART, with one commensurability scale and no slab, has bias 0.23 with coverage 0.87, ~~the linear influence of~~ consistent with the unbounded influence under fixed components in Theorem 1 ~~with the roles of the components fixed.~~

LRC-BART (100) ~~switches borrowing off~~ largely suppresses borrowing. Its realized ESS is 1.6, its map averages 0.07, and its bias is 0.041 with coverage 0.94 and RMSE 0.280, against 0.029 and 0.273 for BART-PP and -0.002 and 0.289 for BART-NP, with ~~the same picture~~ similar results at the three values of ρ . The ~~residual bias of 0.04 lies 0.012 above 0.04~~ residual bias exceeds BART-PP's ~~with a by 0.012~~ (paired Monte Carlo standard error ~~of 0.001, so the gap is small but real, and it is the price of the spike~~). The spike provides a possible explanation: a leaf whose RCT residual mean is not far beyond $d_{1/2}$ keeps a small but nonzero posterior spike probability, and Theorem 1 bounds the resulting shift by about one RCT standard error of the leaf. Doubling the discrepancy ensemble raises the bias to 0.055, ~~because as~~ $H_g \tau_0^2$ doubles the shift the spike can absorb, which is why increases the variation permitted under the spike. This favors a small H_g ~~is kept small in this study.~~

~~Sc4 and Sc5 are the scenarios the method was built for, and~~ Table 2 and Figure 1 report ~~the map and the surface alongside the ATE~~ Sc4 and Sc5. At $\delta_2 = 2$ the map separates the regions: $\bar{B}_c$ is 0.64 over the RCT control profiles in $\mathcal{R}$ and 0.05 outside it, and the posterior mean of g averages -0.26 in $\mathcal{R}$ and -1.71 outside it against true values of 0 and -2 .

Figure 1 shows ~~where the shortfall lies~~ how discrepancy estimates vary across the boundary. Away from the boundary at $X_5 = 2$ the posterior mean of g is -1.8 outside $\mathcal{R}$ and -0.15 inside it, short of the truth by about a tenth of the shift on each side, and the map is 0.02 outside and 0.7 inside; the transition occupies the unit around the boundary, with g averaging -1.66 in the half-unit bin below it and -0.44 in the half-unit bin above it. The boundary is learned from the RCT controls alone, about four of them per 0.1 of X_5 , and the pooled surface f , which ~~sees~~ is estimated from both sources, smooths the RWD step across it, ~~so that leaving the shallow discrepancy ensemble~~ g has to undo a smoothed step with a shallow tree. The leak to estimate the remaining difference. The discrepancy underestimation persisted with f fixed at the truth and under every g prior we tried (Web Appendix G.3), so ~~it is a sample-size limit of the design and not a tuning problem. The ATE is nevertheless unbiased~~ the configurations examined did not resolve it.

The empirical ATE bias at $\delta_2 = 2$ ~~is~~ 0.000 ~~+~~ (coverage 0.92), because the errors on the two sides of the boundary have opposite signs and the estimand averages over both. ~~The control surface is where the leak costs~~ Control-surface estimation is less accurate: its RMSE over the RCT profiles in $\mathcal{R}$ is 0.824 against 0.785 for BART-PP, and outside $\mathcal{R}$ it is 0.857 against 0.822, so at $\delta_2 = 2$ LRC-BART does not beat the free-offset model on the surface, while C-BART, at 1.087 outside $\mathcal{R}$, ~~is far behind both~~ has a larger error than either method. A secondary version of Sc4 places the shift on X_7 , a covariate with negligible effect, and the map and ~~the ATE behave there as~~ ATE results are similar to those on X_5 at both $\delta_2 = 2$ and $\delta_2 = 1$ (Web Appendix G.3).

At $\delta_2 = 1$, a shift of about twice $d_{1/2}$, the map separates only partially (0.62 against 0.52), the posterior mean of g is -0.6 outside $\mathcal{R}$ and -0.25 inside it (Figure 1), and LRC-BART borrows too much outside $\mathcal{R}$: the ATE bias is -0.041 (paired difference from BART-PP -0.039 , standard error 0.003), the RMSE 0.223 against 0.207, and the coverage 0.89, whereas C-BART and BART-CP have biases of -0.21 and -0.34 .

At $\delta_2 = 0.5$, below the stated resolution, the map is flat at 0.84, g averages -0.15 in $\mathcal{R}$ and -0.18 outside it, and the bias is -0.043 with RMSE 0.200, below BART-PP's 0.203 because the ~~borrowing is worth more than the shift costs. The lower rungs of the ladder are less biased~~ variance reduction offsets the bias in this scenario. Lower fractional targets reduce bias under the moderate shift, -0.032 at a quarter of the ceiling against -0.041 , at a cost of 0.01 in RMSE in Sc1 (Web Table S3).

Under the global shift of Sc5 ~~the method abstains~~, LRC-BART suppresses borrowing: at $\delta_2 = 2$ the map averages 0.01, the realized ESS is 0.2, the bias is -0.001 and the RMSE 0.203, against BART-NP's 0.220 and BART-PP's 0.205, whereas C-BART and BART-CP have biases of -0.32 and -1.39 ; at $\delta_2 = 1$ the map is 0.10 and the bias -0.005 .

The ~~Gaussian study therefore shows three things. When the sources agree LRC-BART is more efficient than the free-offset model by about 16% in RMSE and by more on the control surface; when they disagree everywhere it abstains, at a bias of about 0.04; and when they disagree on a region it finds the region at a shift of 1.3 residual SDs, with a transition of about one unit of the splitting covariate around the boundary~~ realized regional ESS summarizes learned local precision (Table 2). Across the first 200 replicates, it is 32.7 and 31.9 on the two sides of the Sc4 partition under Sc1. Under Sc4 it is 15.1 in $\mathcal{R}$ and ~~a shortfall of a tenth of the shift on each side, but at a shift~~

of one residual SD it borrows across the boundary and pays a bias of 0.04 and a coverage of 0.89. The last is the resolution of Theorem 2(a) seen in the data, and it is the reason the borrowing map should be reported with the estimate 12.7 outside at $\delta_2 = 1$, and 9.5 and 0.44 at $\delta_2 = 2$. Under Sc5 with $\delta_2 = 2$, it is about 0.2 on both sides. These quantities use the control mean standardized within each region and are not additive. The prior pointwise ESS uses the RWD fit and cannot establish that trial outcomes support regional borrowing.

The surface loss persists away from the boundary. On RCT control profiles with $X_5 > 2.5$, the Sc4 $\delta_2 = 2$ RMSE is 0.747 for LRC-BART and 0.723 for BART-PP, a paired difference of 0.024 with standard error 0.005 over 200 replicates. Using all RCT profiles in $\mathcal{R}$ over those same replicates gives a difference of 0.039 with standard error 0.006. Web Appendix G.3 reports both profile sets. Thus the evidence for LRC-BART is its efficiency under Sc1 and its regional compatibility and ESS summaries, with a surface-estimation loss under Sc4.

The transferred CAHB analysis has Sc1 RMSE 0.258, compared with 0.175 for LRC-BART and 0.223 for BART-NP over the same 200 replicates. At Sc4 $\delta_2 = 2$, its bias is -0.333 and its historical precision shares average 0.49 and 0.41 in the two regions. At Sc5 $\delta_2 = 1$, its bias is -0.437 . These transfer-specific results do not establish a general ranking of CAHB and tree-based borrowing.

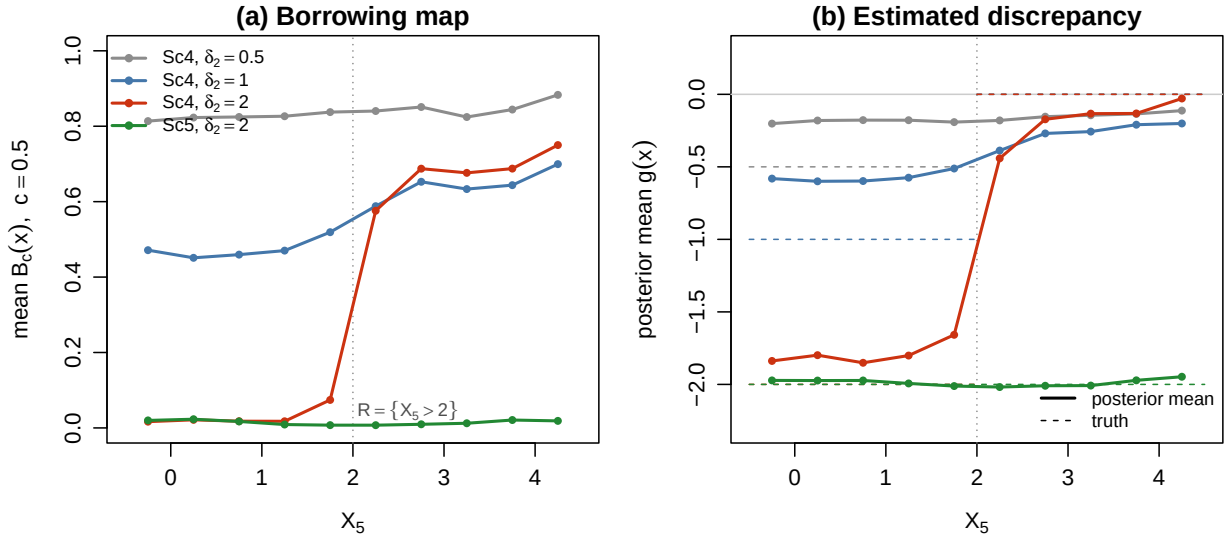


Figure 1: Regional and global shifts, Gaussian outcomes, LRC-BART (100): (a) the borrowing map $\mathcal{B}_c(x) = P(|g(x)| < 0.5 | \text{data})$ and (b) the posterior mean of $g(x)$, both averaged over the RCT control profiles in bins of width 0.5 on X_5 across the first 40 replicates. The RWD surface is raised by δ_2 where $X_5 \leq 2$ (Sc4) or everywhere (Sc5), so the true discrepancy, dashed in (b), is $-\delta_2$ there.

Before revision: Gaussian outcomes, two-arm design, $R = 500$ replicates: average treatment effect (ATE) bias, mean posterior SD, Monte Carlo RMSE, coverage of the 95% credible interval, and power (95% interval excludes zero at $\delta = 1$), with bias and RMSE of the RCT-standardized control mean. Sc4 raises the RWD control surface by δ_2 outside $\mathcal{R} = \{X_5 > 2\}$; Sc5 raises it everywhere. LRC-BART (100) targets a prior ESS of 100 RCT controls and LRC-BART (0.9) targets 0.9 of the ESS ceiling; C-BART is LRC-BART with $w = 1$. PSCL reports no arm-specific means. The full table with every scenario and method is Web Table S1.

Scenario	Method	ATE					Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE
Sc1	LM-PP	-0.002	0.347	0.337	0.94	0.85	0.001	0.243
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.001	0.185
	BART-CP	0.008	0.145	0.142	0.96	1.00	-0.003	0.096
	BART-PP	0.006	0.207	0.205	0.94	1.00	-0.001	0.170
	PSCL	0.005	0.269	0.261	0.95	0.97	-0.005	0.171
	MAP (100)	0.006	0.292	0.265	0.97	0.95	-0.005	0.174
	HierLM	-0.003	0.338	0.341	0.93	0.85	0.003	0.249
	C-BART	0.007	0.155	0.150	0.95	1.00	-0.002	0.106
	LRC-BART (100)	0.009	0.171	0.172	0.94	1.00	-0.004	0.134
	LRC-BART (0.9)	0.011	0.171	0.171	0.95	1.00	-0.005	0.132
Sc2	LM-PP	0.010	0.340	0.331	0.95	0.84	-0.010	0.248
	BART-NP	-0.008	0.201	0.219	0.93	1.00	0.009	0.180
	BART-CP	0.058	0.177	0.195	0.91	1.00	-0.056	0.151
	BART-PP	0.007	0.208	0.204	0.96	1.00	-0.005	0.163
	PSCL	0.024	0.321	0.292	0.97	0.91	-0.029	0.205
	MAP (100)	0.332	0.360	0.491	0.83	0.97	-0.337	0.424
	HierLM	-0.011	0.318	0.321	0.95	0.88	0.011	0.238
	C-BART	0.042	0.168	0.186	0.91	1.00	-0.041	0.141
	LRC-BART (100)	0.023	0.181	0.192	0.94	1.00	-0.021	0.148
	LRC-BART (0.9)	0.019	0.183	0.193	0.93	1.00	-0.017	0.151
Sc3, $\rho = 0$	LM-PP	0.043	0.356	0.346	0.95	0.85	-0.027	0.241
	BART-NP	-0.002	0.292	0.289	0.94	0.92	0.024	0.219
	BART-CP	0.984	0.202	1.003	0.00	1.00	-0.962	0.971
	BART-PP	0.029	0.275	0.273	0.94	0.96	-0.007	0.197
	PSCL	0.704	0.269	0.753	0.26	1.00	-0.693	0.713
	MAP (100)	0.363	0.360	0.504	0.82	0.97	-0.352	0.431
	HierLM	0.039	0.362	0.354	0.96	0.82	-0.023	0.253
	C-BART	0.226	0.300	0.380	0.87	0.98	-0.204	0.311
	LRC-BART (100)	0.041	0.284	0.280	0.94	0.96	-0.018	0.206
	LRC-BART (0.9)	0.037	0.282	0.277	0.94	0.96	-0.015	0.203
Sc4, $X_5, \delta_2 = 1$	LM-PP	-0.012	0.349	0.339	0.95	0.83	0.011	0.245
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.001	0.185
	BART-CP	-0.338	0.146	0.368	0.35	1.00	0.343	0.357
	BART-PP	-0.002	0.209	0.207	0.95	1.00	0.007	0.173
	PSCL	-0.246	0.271	0.358	0.87	0.80	0.247	0.300
	MAP (100)	-0.224	0.306	0.361	0.91	0.75	0.224	0.292
	HierLM	-0.025	0.338	0.343	0.94	0.83	0.024	0.250
	C-BART	-0.207	0.173	0.273	0.73	1.00	0.212	0.258
	LRC-BART (100)	-0.041	0.195	0.223	0.89	1.00	0.046	0.194
	LRC-BART (0.9)	-0.041	0.196	0.224	0.90	1.00	0.047	0.194
Sc4, $X_5, \delta_2 = 2$	LM-PP	-0.023	0.354	0.343	0.95	0.81	0.022	0.249
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.001	0.185
	BART-CP	-0.685	0.149	0.701	0.00	0.56	0.690	0.698
	BART-PP	-0.006	0.209	0.205	0.96	1.00	0.011	0.170
	PSCL	-0.498	0.273	0.562	0.54	0.44	0.498	0.528
	MAP (100)	-0.377	0.333	0.495	0.78	0.46	0.378	0.437
	HierLM	-0.044	0.339	0.345	0.94	0.81	0.043	0.253
	C-BART	-0.227	0.197	0.391	0.66	0.90	0.232	0.383
	LRC-BART (100)	0.000	0.189	0.202	0.92	1.00	0.005	0.168
	LRC-BART (0.9)	-0.000	0.189	0.202	0.93	1.00	0.006	0.170
Sc5, $\delta_2 = 2$	LM-PP	-0.042	0.347	0.339	0.94	0.80	0.042	0.246
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.001	0.185
	BART-CP	-1.390	0.152	1.398	0.00	0.72	1.395	1.399
	BART-PP	-0.012	0.208	0.205	0.95	1.00	0.017	0.171
	PSCL	-1.002	0.269	1.035	0.04	0.04	1.002	1.017
	MAP (100)	-0.395	0.348	0.525	0.78	0.41	0.395	0.465
	HierLM	-0.038	0.338	0.344	0.94	0.82	0.037	0.251
	C-BART	-0.316	0.194	0.650	0.76	0.93	0.322	0.648
	LRC-BART (100)	-0.001	0.190	0.203	0.92	1.00	0.005	0.171
	LRC-BART (0.9)	-0.001	0.190	0.205	0.92	1.00	0.006	0.171

Table 1: Gaussian outcomes, two-arm design, $R = 500$ replicates: average treatment effect (ATE) bias, mean posterior SD, Monte Carlo RMSE, coverage of the 95% credible interval, and power (95% interval excludes zero at $\delta = 1$), with bias and RMSE of the RCT-standardized control mean. Sc4 raises the RWD control surface by δ_2 outside $\mathcal{R} = \{X_5 > 2\}$; Sc5 raises it everywhere. LRC-BART (100) targets a prior ESS of 100 RCT controls and LRC-BART (0.9) targets 0.9 of the ESS ceiling; CAHB uses $R = 200$ paired replicates and our transfer described in Web Appendix G.2; other rows use $R = 500$. C-BART is LRC-BART with $w = 1$. PSCL reports no arm-specific means. The full table with every scenario and method is Web Table S1.

Scenario	Method	ATE					Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE
Sc1	LM-PP	-0.002	0.347	0.337	0.94	0.85	0.001	0.243
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.001	0.185
	BART-CP	0.008	0.145	0.142	0.96	1.00	-0.003	0.096
	BART-PP	0.006	0.207	0.205	0.94	1.00	-0.001	0.170
	PSCL	0.005	0.269	0.261	0.95	0.97	-0.005	0.171
	MAP (100)	0.006	0.292	0.265	0.97	0.95	-0.005	0.174
	HierLM	-0.003	0.338	0.341	0.93	0.85	0.003	0.249
	C-BART	0.007	0.155	0.150	0.95	1.00	-0.002	0.106
	LRC-BART (100)	0.009	0.171	0.172	0.94	1.00	-0.004	0.134
	LRC-BART (0.9)	0.011	0.171	0.171	0.95	1.00	-0.005	0.132
	CAHB	-0.003	0.410	0.258	1.00	0.78	-0.023	0.184
Sc2	LM-PP	0.010	0.340	0.331	0.95	0.84	-0.010	0.248
	BART-NP	-0.008	0.201	0.219	0.93	1.00	0.009	0.180
	BART-CP	0.058	0.177	0.195	0.91	1.00	-0.056	0.151
	BART-PP	0.007	0.208	0.204	0.96	1.00	-0.005	0.163
	PSCL	0.024	0.321	0.292	0.97	0.91	-0.029	0.205
	MAP (100)	0.332	0.360	0.491	0.83	0.97	-0.337	0.424
	HierLM	-0.011	0.318	0.321	0.95	0.88	0.011	0.238
	C-BART	0.042	0.168	0.186	0.91	1.00	-0.041	0.141
	LRC-BART (100)	0.023	0.181	0.192	0.94	1.00	-0.021	0.148
	LRC-BART (0.9)	0.019	0.183	0.193	0.93	1.00	-0.017	0.151
Sc3, $\rho = 0$	LM-PP	0.043	0.356	0.346	0.95	0.85	-0.027	0.241
	BART-NP	-0.002	0.292	0.289	0.94	0.92	0.024	0.219
	BART-CP	0.984	0.202	1.003	0.00	1.00	-0.962	0.971
	BART-PP	0.029	0.275	0.273	0.94	0.96	-0.007	0.197
	PSCL	0.704	0.269	0.753	0.26	1.00	-0.693	0.713
	MAP (100)	0.363	0.360	0.504	0.82	0.97	-0.352	0.431
	HierLM	0.039	0.362	0.354	0.96	0.82	-0.023	0.253
	C-BART	0.226	0.300	0.380	0.87	0.98	-0.204	0.311
	LRC-BART (100)	0.041	0.284	0.280	0.94	0.96	-0.018	0.206
	LRC-BART (0.9)	0.037	0.282	0.277	0.94	0.96	-0.015	0.203
Sc4, $X_5, \delta_2 = 1$	LM-PP	-0.012	0.349	0.339	0.95	0.83	0.011	0.245
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.001	0.185
	BART-CP	-0.338	0.146	0.368	0.35	1.00	0.343	0.357
	BART-PP	-0.002	0.209	0.207	0.95	1.00	0.007	0.173
	PSCL	-0.246	0.271	0.358	0.87	0.80	0.247	0.300
	MAP (100)	-0.224	0.306	0.361	0.91	0.75	0.224	0.292
	HierLM	-0.025	0.338	0.343	0.94	0.83	0.024	0.250
	C-BART	-0.207	0.173	0.273	0.73	1.00	0.212	0.258
	LRC-BART (100)	-0.041	0.195	0.223	0.89	1.00	0.046	0.194
	LRC-BART (0.9)	-0.041	0.196	0.224	0.90	1.00	0.047	0.194
	CAHB	-0.208	0.413	0.333	0.98	0.42	0.181	0.262

Table 1, continued

Scenario	Method	ATE					Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE
Sc4, X_5 , $\delta_2 = 2$	LM-PP	-0.023	0.354	0.343	0.95	0.81	0.022	0.249
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.001	0.185
	BART-CP	-0.685	0.149	0.701	0.00	0.56	0.690	0.698
	BART-PP	-0.006	0.209	0.205	0.96	1.00	0.011	0.170
	PSCL	-0.498	0.273	0.562	0.54	0.44	0.498	0.528
	MAP (100)	-0.377	0.333	0.495	0.78	0.46	0.378	0.437
	HierLM	-0.044	0.339	0.345	0.94	0.81	0.043	0.253
	C-BART	-0.227	0.197	0.391	0.66	0.90	0.232	0.383
	LRC-BART (100)	0.000	0.189	0.202	0.92	1.00	0.005	0.168
	LRC-BART (0.9)	-0.000	0.189	0.202	0.93	1.00	0.006	0.170
	CAHB	-0.333	0.422	0.453	0.95	0.25	0.309	0.398
Sc5, $\delta_2 = 2$	LM-PP	-0.042	0.347	0.339	0.94	0.80	0.042	0.246
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.001	0.185
	BART-CP	-1.390	0.152	1.398	0.00	0.72	1.395	1.399
	BART-PP	-0.012	0.208	0.205	0.95	1.00	0.017	0.171
	PSCL	-1.002	0.269	1.035	0.04	0.04	1.002	1.017
	MAP (100)	-0.395	0.348	0.525	0.78	0.41	0.395	0.465
	HierLM	-0.038	0.338	0.344	0.94	0.82	0.037	0.251
	C-BART	-0.316	0.194	0.650	0.76	0.93	0.322	0.648
	LRC-BART (100)	-0.001	0.190	0.203	0.92	1.00	0.005	0.171
	LRC-BART (0.9)	-0.001	0.190	0.205	0.92	1.00	0.006	0.171
	CAHB	-0.067	0.479	0.400	0.95	0.52	0.041	0.326

Before revision: Regional and global shifts, Gaussian outcomes, $R = 500$: ATE bias and coverage; RMSE of the posterior mean control surface over the RCT profiles inside the compatible region $\mathcal{R}$ and outside it; the borrowing map $\mathcal{B}_c(x) = P(|g(x)| < 0.5 \mid \text{data})$ averaged over the RCT control profiles in each region; the posterior mean of $g(x)$ averaged over each region (true value 0 in $\mathcal{R}$ and $-\delta_2$ in $\mathcal{R}^c$); and the realized ESS. In Sc5 the compatible region is empty. Sc4' places the shift on the null covariate X_7 . LRC-BART ($w = 0$) has no spike, so its map reflects the slab alone.

Scenario	Method	ATE		Surface RMSE		$\tilde{\mathcal{B}}_c$		Mean $\hat{g}$		ESS
		Bias	Cov	$\mathcal{R}$	$\mathcal{R}^c$	$\mathcal{R}$	$\mathcal{R}^c$	$\mathcal{R}$	$\mathcal{R}^c$	
Sc4 (0.5)	BART-PP	0.000	0.95	0.758	0.795	—	—	—	—	—
	C-BART	-0.109	0.90	0.736	0.799	0.98	0.98	-0.06	-0.06	79.4
	LRC-BART ($w = 0$)	0.007	0.93	0.768	0.795	0.71	0.69	-0.21	-0.27	0.1
	LRC-BART (100)	-0.043	0.92	0.751	0.796	0.84	0.83	-0.15	-0.18	42.9
Sc4 (1)	BART-PP	-0.002	0.95	0.784	0.820	—	—	—	—	—
	C-BART	-0.207	0.73	0.749	0.915	0.91	0.90	-0.14	-0.16	73.6
	LRC-BART ($w = 0$)	0.006	0.93	0.801	0.827	0.51	0.40	-0.35	-0.63	0.1
	LRC-BART (100)	-0.041	0.89	0.783	0.843	0.62	0.52	-0.32	-0.52	20.8
Sc4 (2)	BART-PP	-0.006	0.96	0.785	0.822	—	—	—	—	—
	C-BART	-0.227	0.66	0.801	1.087	0.66	0.39	-0.30	-0.97	45.7
	LRC-BART ($w = 0$)	0.004	0.92	0.836	0.858	0.57	0.05	-0.25	-1.73	0.1
	LRC-BART (100)	0.000	0.92	0.824	0.857	0.64	0.05	-0.26	-1.71	0.8
Sc4' (1)	BART-PP	-0.001	0.96	0.807	0.810	—	—	—	—	—
	C-BART	-0.206	0.73	0.767	0.899	0.91	0.90	-0.15	-0.16	73.3
	LRC-BART ($w = 0$)	0.009	0.92	0.820	0.817	0.52	0.41	-0.37	-0.63	0.1
	LRC-BART (100)	-0.038	0.90	0.805	0.833	0.62	0.52	-0.33	-0.52	20.5
Sc4' (2)	BART-PP	-0.002	0.95	0.805	0.806	—	—	—	—	—
	C-BART	-0.218	0.69	0.828	1.075	0.64	0.38	-0.33	-0.97	44.7
	LRC-BART ($w = 0$)	0.007	0.92	0.859	0.848	0.57	0.05	-0.25	-1.74	0.1
	LRC-BART (100)	0.001	0.93	0.850	0.848	0.64	0.05	-0.26	-1.71	0.8
Sc5 (1)	BART-PP	-0.004	0.95	—	0.769	—	—	—	—	—
	C-BART	-0.244	0.68	—	0.821	—	0.43	—	-0.60	41.5
	LRC-BART ($w = 0$)	0.007	0.92	—	0.769	—	0.10	—	-1.00	0.1
	LRC-BART (100)	-0.005	0.91	—	0.768	—	0.10	—	-0.98	1.2
Sc5 (2)	BART-PP	-0.012	0.95	—	0.771	—	—	—	—	—
	C-BART	-0.316	0.76	—	0.932	—	0.19	—	-1.54	45.4
	LRC-BART ($w = 0$)	0.006	0.93	—	0.771	—	0.01	—	-2.00	0.1
	LRC-BART (100)	-0.001	0.92	—	0.771	—	0.01	—	-1.99	0.2

Table 2: Regional and global shifts, Gaussian outcomes, $R = 500$: ATE bias and coverage; RMSE of the posterior mean control surface over all N RCT profiles (treated and control) inside the compatible region $\mathcal{R}$ and outside it; the borrowing map $\mathcal{B}_c(x) = P(|g(x)| < 0.5 \mid \text{data})$ averaged over the RCT control profiles in each region; the posterior mean of $g(x)$ averaged over each region (true value 0 in $\mathcal{R}$ and $-\delta_2$ in $\mathcal{R}^c$); and the realized ESS. In Sc5 the compatible region is empty. Sc4' places the shift on the null covariate X_7 . LRC-BART ($w = 0$) has no spike, so its map reflects the slab alone. CAHB rows use $R = 200$; all other rows use $R = 500$. CAHB precision shares and effective sizes have different definitions and are reported separately in Web Appendix G.2. For Sc1, the two columns use the Sc4 partition although both regions are compatible.

Scenario	Method	ATE		Surface RMSE		$\tilde{\mathcal{B}}_c$		Mean $\hat{g}$		ESS
		Bias	Cov	$\mathcal{R}$	$\mathcal{R}^c$	$\mathcal{R}$	$\mathcal{R}^c$	$\mathcal{R}$	$\mathcal{R}^c$	
Sc1	BART-PP	0.006	0.94	0.750	0.780	—	—	—	—	—
	C-BART	0.007	0.95	0.728	0.756	1.00	1.00	0.00	0.00	81.0
	LRC-BART ($w = 0$)	0.009	0.92	0.755	0.783	0.79	0.79	-0.00	0.01	0.1
	LRC-BART (100)	0.009	0.94	0.736	0.763	0.92	0.92	-0.00	0.00	52.5
	CAHB	-0.003	1.00	2.071	2.273	—	—	—	—	—
Sc4 (0.5)	BART-PP	0.000	0.95	0.758	0.795	—	—	—	—	—
	C-BART	-0.109	0.90	0.736	0.799	0.98	0.98	-0.06	-0.06	79.4
	LRC-BART ($w = 0$)	0.007	0.93	0.768	0.795	0.71	0.69	-0.21	-0.27	0.1
	LRC-BART (100)	-0.043	0.92	0.751	0.796	0.84	0.83	-0.15	-0.18	42.9
Sc4 (1)	BART-PP	-0.002	0.95	0.784	0.820	—	—	—	—	—
	C-BART	-0.207	0.73	0.749	0.915	0.91	0.90	-0.14	-0.16	73.6
	LRC-BART ($w = 0$)	0.006	0.93	0.801	0.827	0.51	0.40	-0.35	-0.63	0.1
	LRC-BART (100)	-0.041	0.89	0.783	0.843	0.62	0.52	-0.32	-0.52	20.8
	CAHB	-0.208	0.98	2.133	2.207	—	—	—	—	—
Sc4 (2)	BART-PP	-0.006	0.96	0.785	0.822	—	—	—	—	—
	C-BART	-0.227	0.66	0.801	1.087	0.66	0.39	-0.30	-0.97	45.7
	LRC-BART ($w = 0$)	0.004	0.92	0.836	0.858	0.57	0.05	-0.25	-1.73	0.1
	LRC-BART (100)	0.000	0.92	0.824	0.857	0.64	0.05	-0.26	-1.71	0.8
	CAHB	-0.333	0.95	2.172	2.157	—	—	—	—	—
Sc4' (1)	BART-PP	-0.001	0.96	0.807	0.810	—	—	—	—	—
	C-BART	-0.206	0.73	0.767	0.899	0.91	0.90	-0.15	-0.16	73.3
	LRC-BART ($w = 0$)	0.009	0.92	0.820	0.817	0.52	0.41	-0.37	-0.63	0.1
	LRC-BART (100)	-0.038	0.90	0.805	0.833	0.62	0.52	-0.33	-0.52	20.5
Sc4' (2)	BART-PP	-0.002	0.95	0.805	0.806	—	—	—	—	—
	C-BART	-0.218	0.69	0.828	1.075	0.64	0.38	-0.33	-0.97	44.7
	LRC-BART ($w = 0$)	0.007	0.92	0.859	0.848	0.57	0.05	-0.25	-1.74	0.1
	LRC-BART (100)	0.001	0.93	0.850	0.848	0.64	0.05	-0.26	-1.71	0.8
Sc5 (1)	BART-PP	-0.004	0.95	—	0.769	—	—	—	—	—
	C-BART	-0.244	0.68	—	0.821	—	0.43	—	-0.60	41.5
	LRC-BART ($w = 0$)	0.007	0.92	—	0.769	—	0.10	—	-1.00	0.1
	LRC-BART (100)	-0.005	0.91	—	0.768	—	0.10	—	-0.98	1.2
	CAHB	-0.437	0.91	—	2.203	—	—	—	—	—
Sc5 (2)	BART-PP	-0.012	0.95	—	0.771	—	—	—	—	—
	C-BART	-0.316	0.76	—	0.932	—	0.19	—	-1.54	45.4
	LRC-BART ($w = 0$)	0.006	0.93	—	0.771	—	0.01	—	-2.00	0.1
	LRC-BART (100)	-0.001	0.92	—	0.771	—	0.01	—	-1.99	0.2
	CAHB	-0.067	0.95	—	1.983	—	—	—	—	—

Scenario	Regional ESS, $X_5 > 2$	Regional ESS, $X_5 \leq 2$
Sc1	32.66	31.86
Sc4 (1)	15.14	12.74
Sc4 (2)	9.51	0.44
Sc5 (2)	0.24	0.24

Regional realized ESS for LRC-BART (100), averaged over the first 200 paired replicates. Each estimand is the control mean standardized to all RCT profiles in that region. The partition is held fixed for Sc1 and Sc5; both regions are compatible in Sc1 and neither is compatible in Sc5. Regional effective sizes are not additive.

3.3 Survival outcomes

Table 3 reports the survival study for the median ratio and the RMST ratio side by side, and Web Table S2 the full set. With 75 RCT controls and about a quarter of the times censored, the median ratio has a wide and skewed posterior under every method, and its posterior mean carries an upward bias that grows with the posterior spread: BART-NP has bias 0.26 and RMSE 0.51 in Sc1, where its RMST-ratio bias is 0.13, and AFT-NP is wider still. Borrowing ~~is what makes the estimand usable at this~~ reduces this uncertainty at the studied sample size.

In Sc1 LRC-BART (100) has median-ratio bias 0.045 and RMSE 0.312 against 0.032 and 0.345 for BART-PP, with power 0.34 against 0.25, and its RMST-ratio RMSE is 0.097 against 0.106; in Sc2, where the ceiling falls to 40 and the absolute targets are capped, the two are within 0.03 of each other on both estimands.

Under unmeasured confounding ~~;~~ (Sc3), the tree models with a fixed commensurate component ~~fail~~ have large biases, as in the Gaussian study~~;~~ BART-CP with median-ratio bias 2.3 and C-BART with 0.58 at $\rho = 0$, and LRC-BART abstains with a realized ESS of 2.3. Its median-ratio bias is then 0.15 against 0.04 for BART-PP and 0.19 for BART-NP, with RMSE 0.558 against 0.451 and coverage 0.94; the paired difference in bias from BART-PP is 0.11 with standard error 0.005, the largest loss to BART-PP in the study. The control log-median is biased by only 0.02, against -0.006 for BART-PP, so most of the difference on the ratio scale comes from the wider posterior LRC-BART has once it has switched borrowing off with 75 controls, ~~a posterior SD of~~ with posterior SD 0.52 against 0.39, ~~passed through the convex ratio together with the nonlinear transformation to the ratio scale~~. On the RMST ratio the biases are 0.031 and -0.005 and the RMSEs 0.146 and 0.137, a paired difference in mean squared error of 0.003 with standard error 0.001. We report both estimands because the median ratio amplifies a small control bias and a wide posterior in a way the RMST ratio does not, and the RMST ratio is the estimand of the application.

Under the regional shift, Sc4 at $\delta_2 = 1$ on the log scale with a residual SD of 1.4, the map separates weakly, 0.64 in $\mathcal{R}$ against 0.58 outside, ~~yet the ATE stays unbiased~~ (and the treatment-effect biases are -0.011 on the median ratio ~~;~~ and -0.037 on the RMST ratio ~~)~~ where C-BART and BART-CP have biases of -0.20 and -0.37 ; under the global shift of Sc5 the map is 0.16, the median-ratio bias 0.03, and C-BART's -0.23 . Web Table S4 gives the map and surface summaries for the survival

study, and the fractional ladder behaves as in the Gaussian study.

Table 3: Survival outcomes, two-arm design, $R = 500$: bias, Monte Carlo RMSE, coverage and power (95% interval excludes one) for the ratio of standardized population medians (true value 1.4) and the ratio of standardized restricted mean survival times at $t^* = 3$. Sc4 raises the RWD log-time surface by $\delta_2 = 1$ outside $\mathcal{R} = \{X_5 > 2\}$ and Sc5 by 1 everywhere. Point estimates are posterior means. The full table is Web Table S2.

Scenario	Method	Median ratio				RMST ratio, $t^* = 3$			
		Bias	RMSE	Cov	Pow	Bias	RMSE	Cov	Pow
Sc1	AFT-PP	0.261	0.737	0.95	0.12	-0.009	0.142	0.94	0.10
	BART-NP	0.260	0.511	0.92	0.46	0.131	0.206	0.87	0.47
	BART-CP	0.064	0.232	0.96	0.63	-0.011	0.072	0.95	0.35
	BART-PP	0.032	0.345	0.95	0.25	-0.023	0.106	0.94	0.17
	HierAFT	0.246	0.733	0.94	0.12	-0.008	0.152	0.92	0.11
	C-BART	0.034	0.248	0.97	0.42	0.003	0.079	0.95	0.32
	LRC-BART (100)	0.045	0.312	0.95	0.34	0.003	0.097	0.94	0.28
	LRC-BART (0.9)	0.042	0.304	0.96	0.38	0.002	0.095	0.95	0.30
Sc2	AFT-PP	0.389	0.818	0.95	0.19	0.050	0.154	0.95	0.20
	BART-NP	0.251	0.495	0.91	0.43	0.110	0.177	0.88	0.45
	BART-CP	0.194	0.383	0.93	0.56	0.037	0.100	0.94	0.43
	BART-PP	0.036	0.335	0.95	0.24	-0.002	0.097	0.95	0.21
	HierAFT	0.233	0.695	0.97	0.14	0.001	0.134	0.95	0.12
	C-BART	0.138	0.331	0.93	0.55	0.043	0.104	0.92	0.49
	LRC-BART (100)	0.061	0.322	0.94	0.34	0.018	0.100	0.93	0.31
	LRC-BART (0.9)	0.058	0.325	0.94	0.33	0.017	0.100	0.94	0.30
Sc3, $\rho = 0$	AFT-PP	0.440	0.919	0.94	0.19	0.054	0.158	0.95	0.19
	BART-NP	0.186	0.571	0.90	0.30	0.058	0.173	0.92	0.21
	BART-CP	2.308	2.412	0.00	1.00	0.588	0.607	0.01	1.00
	BART-PP	0.041	0.451	0.91	0.20	-0.005	0.137	0.92	0.15
	HierAFT	0.337	0.890	0.95	0.13	0.006	0.147	0.95	0.11
	C-BART	0.576	0.909	0.88	0.40	0.139	0.222	0.85	0.38
	LRC-BART (100)	0.155	0.558	0.94	0.17	0.031	0.146	0.94	0.18
	LRC-BART (0.9)	0.148	0.554	0.94	0.17	0.029	0.144	0.93	0.18
Sc4, $X_5, \delta_2 = 1$	AFT-PP	0.191	0.691	0.95	0.10	-0.039	0.139	0.92	0.07
	BART-NP	0.260	0.511	0.92	0.46	0.131	0.206	0.87	0.47
	BART-CP	-0.367	0.401	0.50	0.04	-0.156	0.168	0.32	0.07
	BART-PP	-0.012	0.339	0.97	0.17	-0.059	0.113	0.91	0.07
	HierAFT	0.193	0.700	0.94	0.10	-0.023	0.149	0.91	0.09
	C-BART	-0.202	0.305	0.84	0.08	-0.093	0.120	0.79	0.04
	LRC-BART (100)	-0.011	0.340	0.93	0.24	-0.037	0.108	0.90	0.16
	LRC-BART (0.9)	-0.018	0.342	0.93	0.24	-0.040	0.110	0.90	0.16
Sc5, $\delta_2 = 1$	AFT-PP	0.193	0.695	0.94	0.09	-0.029	0.141	0.93	0.08
	BART-NP	0.260	0.511	0.92	0.46	0.131	0.206	0.87	0.47
	BART-CP	-0.703	0.712	0.00	0.64	-0.256	0.262	0.01	0.65
	BART-PP	-0.029	0.333	0.95	0.17	-0.041	0.110	0.93	0.11
	HierAFT	0.204	0.701	0.94	0.10	-0.018	0.150	0.92	0.09
	C-BART	-0.233	0.405	0.77	0.13	-0.085	0.143	0.77	0.11
	LRC-BART (100)	0.032	0.353	0.93	0.30	0.001	0.112	0.93	0.26
	LRC-BART (0.9)	0.029	0.350	0.94	0.31	0.001	0.111	0.94	0.26

3.4 The single-arm design

~~The application~~ To assess the single-arm setting of Section 4 ~~has no RCT control arm, and we study that design with the mechanisms above.~~ we use the same mechanisms with no RCT control arm:

the trial contributes n_T treated subjects only, 200 for Gaussian outcomes and 30 or 200 for survival outcomes, the 300 RWD controls are the sole control source, and LRC-BART runs in the single-arm mode of Section 2.6 with $w = 1$ against the complete-pooling models, the hierarchical models and

BART-CP; Sc3 is not identifiable without an internal control and is omitted. Table 4 reports the results and Web Appendix G.7 discusses them in detail. Under Sc1 the tree models have Gaussian RMSE 0.16 with coverage 0.94 to 1.00.

Under Sc2 ~~no method does well, because without RCT controls nothing in the data can reveal a discrepancy at trial profiles the RWD does not cover~~, all methods have substantial bias because there are no internal controls to identify a discrepancy and RWD overlap is limited: on the Gaussian ATE, BART-CP has bias 0.41 with coverage 0.75 and LRC-BART 0.42. ~~In every design LRC-BART (100) gives the point estimate of the pooled model within 0.005, with a wider posterior where the target is not capped, and the~~ Adding a zero-mean discrepancy preserves the expected Gaussian control mean, but nonlinear survival-ratio means can change. The $w = 0.9$ sensitivity ~~only adds prior noise to the counterfactual control, which is why we report~~ substantially increases their uncertainty and bias (Web Appendix G.7), supporting $w = 1$ ~~in~~ for the primary single-arm ~~model~~ analysis.

Table 4: Single-arm design, $R = 500$: the trial contributes n_T treated subjects only and the $n_2 = 300$ RWD controls are the sole control source, under Sc1 (no confounding) and Sc2 (measured confounding). LRC-BART (pooled) is $w = 1$ with the spike scale at the grid minimum, LRC-BART (0.9) is $w = 1$ at 0.9 of the ESS ceiling, and LRC-BART ($w = 0.9$) is the slab sensitivity at the absolute target 100. HierLM and HierAFT use the stated hyperprior $\tau_j^2 \sim \text{Inv-Gamma}(1.5, 0.75)$ with no RCT controls, under which the control surface is drawn from the hierarchy; values above 10^3 are printed as such. Control columns are on the outcome scale (Gaussian) or the log-median scale (survival). Web Table S6 adds the absolute-target rows and the median ratio at $n_T = 200$.

Design	Method	Treatment effect					Control			
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Prior	ESS
Gaussian, $n_T = 200$ ATE Sc1	LM-CP	0.007	0.259	0.254	0.95	0.97	-0.003	0.233	—	
	HierLM	-0.014	8.239	0.497	1.00	0.00	0.018	0.487	—	
	BART-CP	0.010	0.163	0.162	0.94	1.00	-0.003	0.127	—	
	LRC-BART (pooled)	0.007	0.165	0.162	0.94	1.00	0.001	0.127	172.3	
	LRC-BART (0.9)	0.007	0.202	0.163	0.98	1.00	0.000	0.128	115.8	
	LRC-BART ($w = 0.9$)	0.010	1.383	0.169	1.00	0.00	-0.003	0.136	81.9	
Gaussian, $n_T = 200$ ATE Sc2	LM-CP	1.682	0.417	1.734	0.01	1.00	-1.676	1.726	—	
	HierLM	1.642	8.317	1.754	1.00	0.00	-1.636	1.747	—	
	BART-CP	0.408	0.310	0.506	0.75	1.00	-0.399	0.492	—	
	LRC-BART (pooled)	0.420	0.316	0.515	0.75	1.00	-0.412	0.501	34.8	
	LRC-BART (0.9)	0.417	0.416	0.511	0.89	0.98	-0.409	0.498	23.1	
	LRC-BART ($w = 0.9$)	0.416	1.329	0.511	1.00	0.00	-0.408	0.498	34.8	
Survival, $n_T = 30$ median ratio Sc1	AFT-CP	1.540	4.678	4.444	0.92	0.15	-0.006	0.204	—	
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	$> 10^3$	$> 10^3$	—	
	BART-CP	-0.013	0.580	0.482	0.97	0.05	-0.026	0.103	—	
	LRC-BART (pooled)	0.182	0.918	0.686	0.98	0.02	-0.017	0.102	80.2	
	LRC-BART (0.9)	0.222	1.174	0.711	0.99	0.01	-0.005	0.133	51.8	
	LRC-BART ($w = 0.9$)	1.423	11.199	1.959	1.00	0.00	0.318	0.433	79.3	

Table 4, continued

Design	Method	Treatment effect					Control			
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Prior	ESS
Survival, $n_T = 30$ median ratio Sc2	AFT-CP	20.944	46.014	51.645	0.19	0.89	-0.441	0.526	—	
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	$> 10^3$	$> 10^3$	—	
	BART-CP	0.814	1.109	1.231	0.92	0.25	-0.199	0.277	—	
	LRC-BART (pooled)	1.324	1.859	1.864	0.95	0.21	-0.207	0.284	29.2	
	LRC-BART (0.9)	1.511	3.461	2.055	0.97	0.12	-0.180	0.279	19.3	
	LRC-BART ($w = 0.9$)	4.243	28.669	5.305	1.00	0.00	0.179	0.537	28.9	
Survival, $n_T = 30$ RMST ratio Sc1	AFT-CP	0.036	0.178	0.202	0.90	0.16	0.079	0.187	—	
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	0.311	0.355	—	
	BART-CP	-0.111	0.176	0.186	0.92	0.02	-0.021	0.089	—	
	LRC-BART (pooled)	-0.046	0.192	0.157	0.97	0.02	0.002	0.085	80.2	
	LRC-BART (0.9)	-0.040	0.218	0.156	0.98	0.01	0.002	0.085	51.8	
	LRC-BART ($w = 0.9$)	0.080	0.998	0.200	1.00	0.00	0.020	0.089	79.3	
Survival, $n_T = 30$ RMST ratio Sc2	AFT-CP	1.216	0.522	1.336	0.13	0.94	-0.518	0.559	—	
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	0.017	0.201	—	
	BART-CP	0.149	0.271	0.286	0.95	0.14	-0.219	0.260	—	
	LRC-BART (pooled)	0.234	0.297	0.339	0.94	0.19	-0.210	0.251	29.2	
	LRC-BART (0.9)	0.256	0.452	0.358	0.97	0.13	-0.207	0.248	19.3	
	LRC-BART ($w = 0.9$)	0.527	2.689	0.643	1.00	0.00	-0.185	0.228	28.9	
Survival, $n_T = 200$ RMST ratio Sc1	AFT-CP	-0.023	0.101	0.098	0.94	0.15	0.074	0.114	—	
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	0.305	0.317	—	
	BART-CP	-0.001	0.084	0.082	0.97	0.32	-0.021	0.061	—	
	LRC-BART (pooled)	-0.024	0.078	0.076	0.95	0.25	0.001	0.054	169.5	
	LRC-BART (0.9)	-0.022	0.094	0.076	0.99	0.17	0.002	0.054	114.0	
	LRC-BART ($w = 0.9$)	0.109	1.179	0.275	1.00	0.00	0.021	0.057	81.8	
Survival, $n_T = 200$ RMST ratio Sc2	AFT-CP	1.164	0.428	1.244	0.01	1.00	-0.535	0.550	—	
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	0.001	0.101	—	
	BART-CP	0.305	0.206	0.361	0.61	0.77	-0.233	0.261	—	
	LRC-BART (pooled)	0.284	0.206	0.336	0.70	0.74	-0.224	0.250	35.9	
	LRC-BART (0.9)	0.301	0.350	0.353	0.85	0.50	-0.222	0.248	23.9	
	LRC-BART ($w = 0.9$)	0.562	2.419	0.633	1.00	0.00	-0.200	0.228	36.2	

4 Application: a single-arm myeloma trial with a registry control

4.1 Data and harmonization

EloKRd is a single-arm phase II trial of elotuzumab with carfilzomib, lenalidomide and dexamethasone in 30 patients with newly diagnosed multiple myeloma, with progression-free survival (PFS) and overall survival (OS) measured from the first day of treatment; 13 patients progressed or died and 10 died over a maximum follow-up of 6.75 years. The external control is the University of Chicago Multiple Myeloma (UCMM) registry, an institutional database of 797 patients from which we retain, in order, those without negative survival times, with a recorded induction ~~regimen, positive survival times, start~~, no transplant on or before induction, and a regimen other than elotuzumab with lenalidomide and dexamethasone, leaving 253 patients on standard induction regimens, mostly VRd, Vd, KRd and CyBorD, with survival measured from the start of induction. ~~Six-Five~~ baseline

covariates are complete in both sources: age, sex, race, ethnicity ~~;~~ and high-risk cytogenetics, ~~and receipt of.~~ We also adjust for recorded autologous stem cell transplantation (ASCT) after induction (Web Table S5). The trial cohort is enriched for high-risk cytogenetics (50% against 23%) and male sex (70% against 51%), and far fewer trial patients received ASCT (30% against 80%). The unadjusted Kaplan–Meier curves give a 5-year RMST ratio of 1.015 for PFS and 0.90 for OS, with log-rank p -values of 0.51 and 0.048.

~~The merged dataset reproduces every count of the baseline table, but the covariates need harmonizing before they enter a model. The non-Hispanic level of ethnicity is labelled differently in the two sources, so a dummy coding taken as recorded produces two non-Hispanic indicators, one per source, and every control model trained on UCMM and evaluated at the trial profiles then predicts the 29 non-Hispanic trial patients from the 13-patient Hispanic stratum of the registry; race as recorded has four levels with Asian as the reference level, although the table pools it into Other/Unknown. We harmonize both, to seven columns (age, male, Black, Other/Unknown, Hispanic, high-risk cytogenetics, ASCT), fit every model of Section 4.3 under the harmonized coding, and keep~~ We harmonize race and ethnicity before fitting the models and retain the original coding as a sensitivity analysis (Web Appendix G.5). The ~~calibration of Section 2.4 flags the problem on its own: ESS ceilings~~ under the original coding ~~the ESS ceiling at the trial profiles is~~ are 12.6 for PFS and 14.6 for OS, ~~so compared with 63.7 and 36.7 after harmonization;~~ a target of 30 is capped ~~;~~ whereas under the harmonized coding it is 63.7 and 36.7 under the original coding.

4.2 Estimand and calibration

Neither trial median is reached, and the registry median PFS of 5.1 years falls near the end of the trial follow-up, so we take the standardized 5-year RMST ratio as the primary estimand and the median ratio as a secondary summary. Writing $\bar{S}_a(t)$ for the arm- a log-normal survival function averaged over the 30 trial profiles, $\text{RMST}_a(5) = \int_0^5 \bar{S}_a(t) dt$ has a closed form under the AFT model and is evaluated at every posterior draw. The treated arm is AFT-BART with 50 trees fitted to the 30 trial patients, and the control arm is LRC-BART in single-arm mode with f an AFT-BART of $H_f = 10$ trees on the registry, $H_g = 5$, the split prior $(0.5, 3)$, $\nu_0 = 3$ and $w = 1$, with $w = 0.9$ and $H_f = 50$ as sensitivity analyses. Four chains of 2,000 burn-in and 2,000 retained iterations give a split- $\hat{R}$ of 1.000 and bulk effective sizes above 6,800 of 8,000 draws for the log RMST ratio.

The ~~calibration uses the registry alone, with~~ application is a descriptive model-based comparison ~~standardized to the recorded trial profiles, including post-induction ASCT. This adjustment does not identify a total causal treatment effect, and the single-arm data cannot verify compatibility with an unobserved trial control surface. Calibration uses the registry alone; $\sigma_1 = \sigma_2$ since the trial has no controls. Its first output is the ceiling, the information the registry posterior carries about the standardized control RMST is also assumed for counterfactual survival prediction. The resulting variance-based ceiling concerns the standardized control log-time mean~~ at the 30 trial profiles: 63.7 RCT-equivalent controls for PFS and 36.7 for OS, far below the 253 registry patients, because the trial profiles, half of them high-risk and 70% without ASCT, lie where ~~the registry is thin~~ ~~registry support is limited~~. A target of 30 is therefore about half the ~~usable registry ceiling~~ for PFS and most of ~~it the ceiling~~ for OS, and the Pr-rule returns $s_0^2 = 0.0056$ and 0.0050 with ESS_0 of 28.1 and 24.4. We report the absolute ladder $N_{\text{target}} \in \{30, 22.5, 15\}$ and the fractional ladder at 0.9, 0.5 and 0.25 of the ceiling; the realized ESS equals ESS_0 within one unit in every row, ~~as it must when consistent with g is remaining~~ a prior draw.

4.3 Results

Table 5 reports the standardized 5-year RMST ratio for both endpoints. At $N_{\text{target}} = 30$ LRC-BART gives 0.97 (0.77, 1.23) for PFS and 0.90 (0.77, 1.04) for OS, with $P(\text{ratio} > 1)$ of 0.37 and 0.08; lowering the target to 15 widens the intervals to (0.75, 1.40) and (0.76, 1.11) and moves the means by at most 0.02, and the fractional ladder does the same. ~~The interval widens as the target falls, so the trade-off between borrowing and uncertainty that the calibration exposes is intact, on a scale of tenths. The point estimates agree with the~~

~~The~~ harmonized reference analyses ~~give similar estimates~~: AFT-BART on the registry with no discrepancy gives 0.97 and 0.90, and the complete-pooling AFT gives 0.87 (0.69, 1.06) and 0.80 (0.65, 0.94), the ~~one-only~~ interval in the table that excludes one, while the hierarchical AFT stays near 1.2 for both endpoints (Web Appendix G.5). Under the harmonized coding ~~no nonparametric analysis supports a marginal benefit of EloKRd on either endpoint, and the OS estimate favours the registry, which is the direction of the unadjusted log-rank test, the nonparametric 95% intervals include one for both endpoints; the OS point estimates are below one. These are adjusted descriptive comparisons conditional on the recorded profiles and discrepancy prior.~~

The ~~lower block of Table 5 carries the earlier analysis, whose earlier~~ MAP-AFT-BART rows ~~in Table 5~~ had 97.5% limits of 16 to 33 for PFS and 4.8 to 7.2 for OS; ~~those tails do not recur under the calibrated five-tree discrepancy, and the current fits have shorter observed tails~~ (Web Appendix G.5 ~~traces the change in the intervals to the prior and the change in the point estimates to the coding and the model~~). The $w = 0.9$ sensitivity widens the PFS interval to (0.74, 1.48) ~~with the mean unchanged, the slab adding prior noise as Section 2.6 predicts and changes its posterior mean from 0.97 to 1.00, illustrating the nonlinear effect of discrepancy uncertainty~~, and $H_f = 50$ changes the means by less than 0.015.

~~For this single-arm analysis,~~ Figure 2 ~~is the covariate-resolved display for a single-arm analysis, the local ESS map reports the local prior ESS~~ $\text{ESS}(x)$ at the 30 trial profiles. It ranges from 5 to 17 for PFS (median 9.5) and 6 to 15 for OS, highest for White patients with ASCT and standard-risk cytogenetics, where ~~the registry is dense~~ registry support is strongest, and lowest for the one Hispanic patient, the Other/Unknown-race patients, and the high-risk profiles without ASCT. The ~~borrowing map compatibility probability~~ $\mathcal{B}_c$ ~~has no~~ is constant across profiles in this single-arm counterpart model, since g is a prior draw here, mean zero with SD 0.26 on the log-time scale at $N_{\text{target}} = 30$ at every profile (Web Figure S1); ~~what the analysis can report per patient is how much of the registry reaches them, and the map shows that a target~~ the patient-specific summary therefore describes registry precision under the stated model. It does not allocate the target across patients. A target of 30 is spread thinly over the profiles the trial enrolled does not imply equally strong external support at every trial profile.

5 Discussion

~~The simulations place the method between two familiar ones. The free-offset model, BART with the source as a covariate, is safe everywhere and never gains from agreement, and complete pooling gains from agreement and fails under any disagreement. LRC-BART gains~~ combines an efficiency gain under agreement with a covariate-resolved assessment of compatibility. In the Gaussian simulations, its RMSE is about 16% in RMSE over the free-offset model when the sources agree, abstains at a bias of lower than BART-PP's under Sc1. Under global disagreement, its residual bias is about 0.04 when they disagree everywhere, and finds a regional disagreement. For a regional shift of 1.3 residual SDs,

Table 5: EloKRd versus UCMM: standardized 5-year RMST ratio, posterior mean [posterior median] and 95% credible interval, for progression-free survival (PFS) and overall survival (OS) at the 30 EloKRd profiles. The ESS ceilings are 63.7 (PFS) and 36.7 (OS). Reference rows are rerun under the harmonized covariate coding of Section 4.1; the Kaplan–Meier row is unadjusted and reports the estimate with its large-sample interval from the survRM2 package. The published rows are those of the earlier analysis under the original coding, reported there as posterior medians.

Method	PFS	OS	ESS ₀ (PFS / OS)
<i>Reference methods, harmonized coding</i>			
Kaplan–Meier, unadjusted	1.01 (0.84, 1.22)	0.90 (0.79, 1.03)	
AFT, complete pooling	0.87 [0.87] (0.69, 1.06)	0.80 [0.80] (0.65, 0.94)	
AFT, hierarchical ($s^2 = 0.5$)	1.23 [1.21] (0.74, 1.90)	1.21 [1.19] (0.77, 1.70)	
AFT-BART on UCMM, no discrepancy	0.97 [0.97] (0.79, 1.16)	0.90 [0.91] (0.78, 1.02)	
<i>LRC-BART, single-arm mode, harmonized coding, $H_f = 10$, $w = 1$</i>			
$N_{\text{target}} = 30$	0.97 [0.96] (0.77, 1.23)	0.90 [0.90] (0.77, 1.04)	28.1 / 24.4
$N_{\text{target}} = 22.5$	0.98 [0.96] (0.76, 1.29)	0.91 [0.90] (0.76, 1.06)	20.6 / 19.0
$N_{\text{target}} = 15$	0.99 [0.96] (0.75, 1.40)	0.91 [0.90] (0.76, 1.11)	14.2 / 12.7
0.9 $\times$ ceiling	0.97 [0.96] (0.79, 1.15)	0.90 [0.90] (0.77, 1.04)	52.4 / 26.8
0.5 $\times$ ceiling	0.97 [0.96] (0.78, 1.22)	0.91 [0.90] (0.76, 1.09)	29.6 / 15.3
0.25 $\times$ ceiling	0.99 [0.96] (0.75, 1.38)	0.92 [0.91] (0.75, 1.16)	15.4 / 8.2
<i>Sensitivity analyses at $N_{\text{target}} = 30$</i>			
$w = 0.9$	1.00 [0.97] (0.74, 1.48)	0.91 [0.90] (0.76, 1.10)	28.1 / 24.4
$H_f = 50$	0.98 [0.97] (0.78, 1.23)	0.91 [0.91] (0.77, 1.05)	27.1 / 25.3
Original covariate coding	1.04 [1.01] (0.79, 1.43)	0.91 [0.90] (0.76, 1.07)	12.9 / 15.0
<i>Published rows (original coding, posterior medians)</i>			
AFT, complete pooling	1.03 (0.71, 1.61)	0.90 (0.71, 1.17)	
AFT, hierarchical ($s^2 = 0.5$)	1.21 (0.73, 2.68)	1.18 (0.77, 1.89)	
Standard BART	1.06 (0.79, 1.51)	0.90 (0.76, 1.08)	
MAP-AFT-BART, $N_{\text{target}} = 30$	1.08 (0.66, 16.40)	0.97 (0.74, 4.77)	
MAP-AFT-BART, $N_{\text{target}} = 15$	1.10 (0.65, 32.80)	1.00 (0.73, 7.22)	

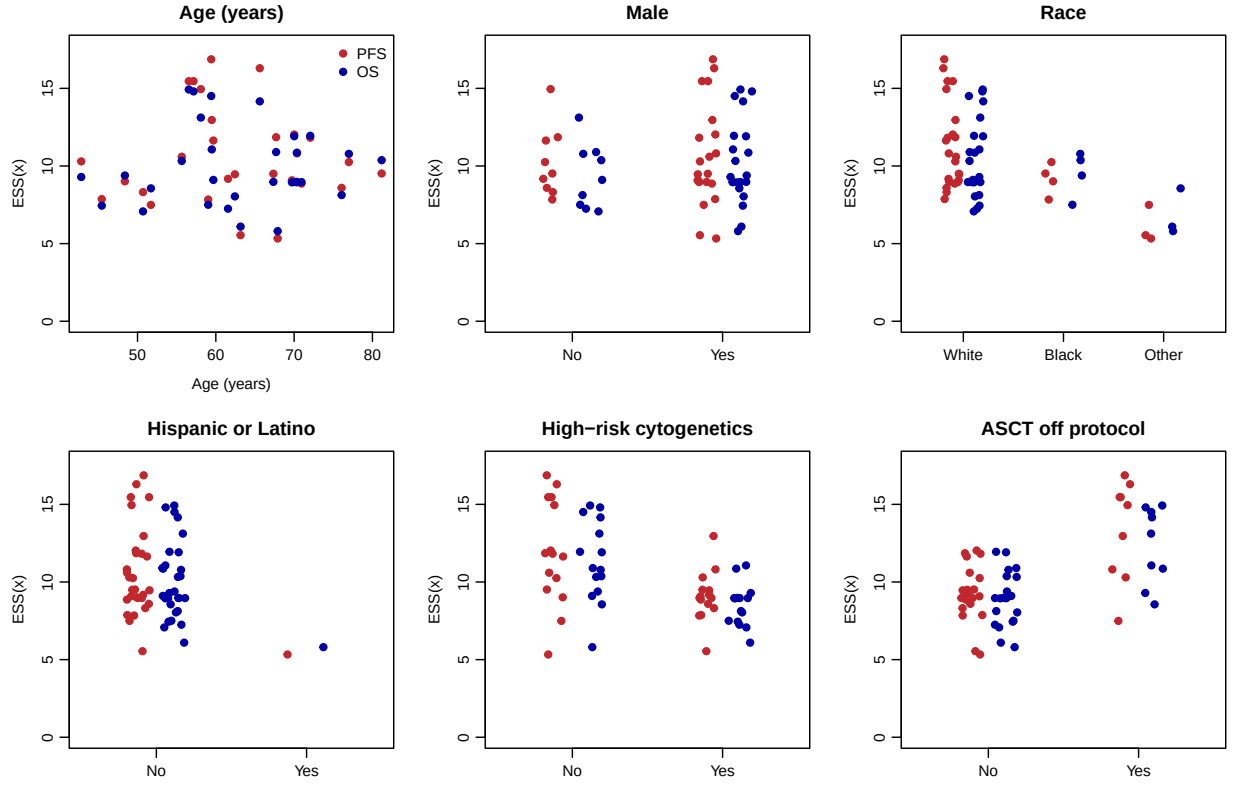


Figure 2: Local prior effective sample size $ESS(x) = \sigma_1^2 / \{V_f(x) + H_g \tau_0^2\}$ at the 30 EloKRd profiles for PFS (red) and OS (blue) at $N_{\text{target}} = 30$, plotted against each harmonized covariate; $V_f(x)$ is the UCMM posterior variance of the log-time surface at x .

with a transition of the compatibility map and realized regional ESS distinguish the two regions, although the transition spans about one unit of the splitting covariate around the boundary that we could not remove by tuning and that comes from learning the boundary from about a hundred RCT controls. The control-surface RMSE remains higher than BART-PP's, including away from the boundary. At a regional shift of one residual SD it borrows across the boundary and pays a two thirds of a residual SD, excessive borrowing produces bias of 0.04 and a coverage of 0.89, which is the resolution of. The leaf calculation in Theorem 2(a) and argues for reporting the map with the estimate rather than for a smaller spike, which would forgo the gain under agreement describes the dependence of detection on the spike scale and local sample size; it does not establish the performance of the full fitted ensemble.

The CAHB comparison illustrates the difficulty of kernel-local borrowing in our nonlinear ten-covariate design. It also depends on our choices for smoothing, precision scaling and joint posterior draws. We therefore interpret it as an assessment of this transfer of CAHB, with the variants reported alongside it. The survival study adds that the median ratio is a poor estimand shows a further limitation: with 75 controls and censoring, and that the median ratio can amplify control-surface bias and posterior uncertainty. We report the RMST ratio should be reported with it. In the a single-arm study, measured confounding without an internal control biases every method, since nothing in the data can reveal a discrepancy at profiles the external source does not cover; what the calibration contributes there is the ceiling and the local ESS map, which report how thin the external support is at the trial profiles and, in the application, exposed a coding error in the data trial data cannot identify a control discrepancy. The ceiling and local prior ESS then describe external support at trial profiles, but cannot demonstrate compatibility with an unobserved trial control surface.

Several limitations remain. The map results are conditional on the partitions of the discrepancy ensemble and on the true discrepancy being representable on. The theoretical results condition on features that are learned in the fitted model. Map consistency requires fixed discrepancy partitions and a true discrepancy in their additive span, and we have no statement about. We have no corresponding result for the posterior over partitions; the leaf results treat the pooled surface as fixed, and with. The leaf-level influence bound also treats f estimated the RWD posterior error at the RCT profiles enters a as fixed; uncertainty in the fitted pooled surface perturbs the residuals

~~used to estimate g_{leaf} as a perturbation of the discrepancy. Both are open, and the second is where the boundary leak lives.~~ These limitations prevent a direct extension of the conditional results to the full posterior. The observed surface loss under Sc4 remains an empirical limitation even when profiles near the boundary are excluded.

The spike scale is global~~and the locality is carried~~, with locality determined by the indicators;~~a leaf-specific scale would let compatible regions differ in how compatible they are. Leaf-specific scales could accommodate different degrees of compatibility,~~ at the cost of the $\text{Inv-}\chi^2$ conjugacy. ~~The model handles one external source; with several, a discrepancy ensemble per source with a common w is the natural extension, and the ceiling would then be a sum over sources of the information each carries at the RCT profiles.~~ Multiple external sources would require source-specific discrepancies and a joint calibration that accounts for their information at the trial profiles; additivity of the ceilings has not been established. Binary and count outcomes ~~need a latent-variable likelihood in place of the Gaussian one, and the AFT model can be replaced by~~ require an alternative likelihood, and a non-parametric survival ~~BART at the cost of the closed-form RMST.~~ Finally, the calibration sets a prior effective sample size, not an operating characteristic; embedding it in a design that fixes the model would require a different RMST calculation. Prior-ESS calibration alone does not control Type I error ~~rate and the power~~under an agreed mechanism, with the map as a pre-specified diagnostic, is the step that would take the method into a confirmatory setting or power. A confirmatory design would need to evaluate those operating characteristics under specified scenarios and pre-specify how the compatibility map would be used.

References

- [1] Ethan M. Alt, Xiuya Chang, Xun Jiang, Qing Liu, May Mo, H. Amy Xia, and Joseph G. Ibrahim. LEAP: the latent exchangeability prior for borrowing information from historical data. *Biometrics*, 80(3):ujae083, 2024. doi: 10.1093/biomtc/ujae083.
- [2] Dehua Bi, Tianjian Zhou, Wei Zhong, and Yuan Ji. SAM-HC: a Bayesian nonparametric construction of hybrid control for randomized clinical trials using external data. *Biostatistics*, 27(1):kxag003, 2026. doi: 10.1093/biostatistics/kxag003. Circulated as PAM-HC, arXiv:2310.20087.

- [3] Noirit Kiran Chandra, Abhra Sarkar, John F. de Groot, Ying Yuan, and Peter Müller. Bayesian nonparametric common atoms regression for generating synthetic controls in clinical trials. *Journal of the American Statistical Association*, 118(544):2301–2314, 2023. doi: 10.1080/01621459.2023.2231581.
- [4] Wei-Chen Chen, Chenguang Wang, Heng Li, Nelson Lu, Ram Tiwari, Yunling Xu, and Lilly Q. Yue. Propensity score-integrated composite likelihood approach for augmenting the control arm of a randomized controlled trial by incorporating real-world data. *Journal of Biopharmaceutical Statistics*, 30(3):508–520, 2020. doi: 10.1080/10543406.2020.1730877.
- [5] Hugh A. Chipman, Edward I. George, and Robert E. McCulloch. BART: Bayesian additive regression trees. *The Annals of Applied Statistics*, 4(1):266–298, 2010. doi: 10.1214/09-AOAS285.
- [6] Sameer K. Deshpande. flexBART: flexible Bayesian regression trees with categorical predictors. *Journal of Computational and Graphical Statistics*, 34(3):1117–1126, 2025. doi: 10.1080/10618600.2024.2431072.
- [7] Yuyan Duan, Keying Ye, and Eric P. Smith. Evaluating water quality using power priors to incorporate historical information. *Environmetrics*, 17(1):95–106, 2006. doi: 10.1002/env.752.
- [8] P. Richard Hahn, Jared S. Murray, and Carlos M. Carvalho. Bayesian regression tree models for causal inference: regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Analysis*, 15(3):965–1056, 2020. doi: 10.1214/19-BA1195.
- [9] Brian P. Hobbs, Bradley P. Carlin, Sumithra J. Mandrekar, and Daniel J. Sargent. Hierarchical commensurate and power prior models for adaptive incorporation of historical information in clinical trials. *Biometrics*, 67(3):1047–1056, 2011. doi: 10.1111/j.1541-0420.2011.01564.x.
- [10] Brian P. Hobbs, Daniel J. Sargent, and Bradley P. Carlin. Commensurate priors for incorporating historical information in clinical trials using general and generalized linear models. *Bayesian Analysis*, 7(3):639–674, 2012. doi: 10.1214/12-BA722.
- [11] Joseph G. Ibrahim and Ming-Hui Chen. Power prior distributions for regression models. *Statistical Science*, 15(1):46–60, 2000. doi: 10.1214/ss/1009212673.

- [12] Liyun Jiang, Lei Nie, and Ying Yuan. Elastic priors to dynamically borrow information from historical data in clinical trials. *Biometrics*, 79(1):49–60, 2023. doi: 10.1111/biom.13551.
- [13] Huaqing Jin, Mi-Ok Kim, Aaron Scheffler, and Fei Jiang. Bayesian adaptive design for covariate-adaptive historical control information borrowing. *Statistics in Medicine*, 42(29):5338–5352, 2023. doi: 10.1002/sim.9913.
- [14] Alexander M. Kaizer, Joseph S. Koopmeiners, and Brian P. Hobbs. Bayesian hierarchical modeling based on multisource exchangeability. *Biostatistics*, 19(2):169–184, 2018. doi: 10.1093/biostatistics/kxx031.
- [15] Ales Kotalik, David M. Vock, Eric C. Donny, Dorothy K. Hatsukami, and Joseph S. Koopmeiners. Dynamic borrowing in the presence of treatment effect heterogeneity. *Biostatistics*, 22(4):789–804, 2021. doi: 10.1093/biostatistics/kxz066.
- [16] Beat Neuenschwander, Michael Branson, and David J. Spiegelhalter. A note on the power prior. *Statistics in Medicine*, 28(28):3562–3566, 2009. doi: 10.1002/sim.3722.
- [17] Beat Neuenschwander, Gorana Capkun-Niggli, Michael Branson, and David J. Spiegelhalter. Summarizing historical information on controls in clinical trials. *Clinical Trials*, 7(1):5–18, 2010. doi: 10.1177/1740774509356002.
- [18] Beat Neuenschwander, Sebastian Weber, Heinz Schmidli, and Anthony O’Hagan. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020. doi: 10.1111/biom.13252.
- [19] Jane Pan, Veronica Bunn, Bradley Hupf, and Jianchang Lin. Bayesian additive regression trees (BART) with covariate adjusted borrowing in subgroup analyses. *Journal of Biopharmaceutical Statistics*, 32(4):613–626, 2022. doi: 10.1080/10543406.2022.2089160.
- [20] Veronika Ročková and Stéphanie van der Pas. Posterior concentration for Bayesian regression trees and forests. *The Annals of Statistics*, 48(4):2108–2131, 2020. doi: 10.1214/19-AOS1879.
- [21] Christian Röver and Tim Friede. Dynamically borrowing strength from another study through shrinkage estimation. *Statistical Methods in Medical Research*, 29(1):293–308, 2020. doi: 10.1177/0962280219833079.

- [22] Christian Röver and Tim Friede. Meta-analytic-predictive priors based on a single study, 2025. arXiv:2505.15502, v2, 10 November 2025.
- [23] H. Schmidli, S. Gsteiger, S. Roychoudhury, A. O’Hagan, P. Guo, and B. Neuenschwander. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014. doi: 10.1111/biom.12209.
- [24] Rodney Sparapani, Charles Spanbauer, and Robert McCulloch. Nonparametric machine learning and efficient computation with Bayesian additive regression trees: the BART R package. *Journal of Statistical Software*, 97(1):1–66, 2021. doi: 10.18637/jss.v097.i01.
- [25] Chenguang Wang, Heng Li, Wei-Chen Chen, Nelson Lu, Ram Tiwari, Yunling Xu, and Lilly Q. Yue. Propensity score-integrated power prior approach for incorporating real-world evidence in single-arm clinical studies. *Journal of Biopharmaceutical Statistics*, 29(5):731–748, 2019. doi: 10.1080/10543406.2019.1657133.
- [26] Bruna Wundervald, Andrew Parnell, and Katarina Domijan. Hierarchical embedded Bayesian additive regression trees, 2022. arXiv:2204.07207, v2, 24 April 2023.
- [27] Tianjian Zhou and Yuan Ji. Incorporating external data into the analysis of clinical trials via Bayesian additive regression trees. *Statistics in Medicine*, 40(30):6782–6795, 2021. doi: 10.1002/sim.9196.

Web Appendices

Web Appendix A: Model, notation, and exact leaf marginals

This appendix collects the proofs for the locally robust commensurate BART (LRC-BART) model of the main text. We fix notation first. The two control sources are indexed by $g = 1$ (RCT control) and $g = 2$ (RWD control), with residual variances σ_1^2 and σ_2^2 . The control outcomes follow

$$y_i = f(x_i) + D_i g(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_{g_i}^2), \quad D_i = \mathbf{1}\{i \text{ is an RCT control}\}, \quad (\text{A11})$$

where f is a standard BART ensemble of H_f trees with leaf prior $\mathcal{N}(0, \lambda_f^2)$ and g is an ensemble of H_g trees whose leaves carry the two-component prior

$$\theta_{h\ell} \mid z_{h\ell} \sim z_{h\ell} \mathcal{N}(0, \tau_0^2) + (1 - z_{h\ell}) \mathcal{N}(0, \tau_1^2), \quad z_{h\ell} \stackrel{\text{iid}}{\sim} \text{Bernoulli}(w), \quad (\text{A12})$$

with $z = 1$ the spike (commensurate component) and $z = 0$ the slab, $w \sim \text{Beta}(a_w, b_w)$, τ_1^2 fixed by the range rule, and

$$\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2) \text{ truncated to } (0, \tau_1^2), \quad (\text{A13})$$

which is the prior the sampler implements (its conditional for τ_0^2 is the scaled-Inv- $\chi^2(\nu_0 + K_0, \cdot)$ law truncated to $(0, \tau_1^2)$, which is conjugate for (A13) and not for the untruncated prior). Throughout we impose

Assumption A1 ((A1)). $0 < \tau_0^2 < \tau_1^2 < \infty$.

Under (A13), Assumption A1 holds almost surely under the prior and along the chain. The ~~prior-ESS calculation of variance-based calibration in~~ Web Appendix D uses the untruncated scaled-Inv- $\chi^2(\nu_0, s_0^2)$ law, ~~and we say there where truncation matters; at the calibrated s_0^2 of order 10^{-3} against τ_1^2 of order one the truncated mass $P(\chi_{\nu_0}^2 < \nu_0 s_0^2 / \tau_1^2)$ is of order 10^{-5} . The mass removed by the fitting prior is $P(\chi_{\nu_0}^2 < \nu_0 s_0^2 / \tau_1^2)$ and depends on the fitted calibration scale; the two laws are distinguished in that appendix.~~

Within a leaf, n_g and $\bar{y}_g$ denote the number and the mean of the source- g partial residuals routed to the leaf, $v_g = \sigma_g^2 / n_g$, and $\phi(\cdot; m, V)$ is the $\mathcal{N}(m, V)$ density; Φ is the standard normal

distribution function and $\bar{\Phi} = 1 - \Phi$. For a g leaf only RCT controls enter, and we write $\bar{y} = \bar{y}_1$, $v = v_1$, $n = n_1$ when no confusion arises. The identity we use repeatedly is the Gaussian product rule

$$\phi(\theta; 0, \tau^2) \phi(\bar{y}; \theta, v) = \phi(\bar{y}; 0, \tau^2 + v) \phi\left(\theta; \frac{\tau^2}{\tau^2 + v} \bar{y}, \frac{\tau^2 v}{\tau^2 + v}\right), \quad (\text{A14})$$

which follows by completing the square in θ .

A.1 Exact leaf marginals

Lemma A1 (Exact leaf marginals). *Fix all trees other than tree h , their leaf parameters, the residual variances, λ_f^2 , and (τ_0^2, τ_1^2, w) .*

(a) (f leaf.) *Let $r_i = y_i - f_{-h}(x_i) - D_i g(x_i)$ be the partial residuals of the control observations routed to a leaf of f -tree h , so that $r_i | \theta \stackrel{\text{indep}}{\sim} \mathcal{N}(\theta, \sigma_{g_i}^2)$ and $\theta \sim \mathcal{N}(0, \lambda_f^2)$. With $A = n_1/\sigma_1^2 + n_2/\sigma_2^2$ and $B = S_1/\sigma_1^2 + S_2/\sigma_2^2$, $S_g = \sum_{i: g_i=g} r_i$,*

$$\int \prod_i \phi(r_i; \theta, \sigma_{g_i}^2) \phi(\theta; 0, \lambda_f^2) d\theta = \left[\prod_i \phi(r_i; 0, \sigma_{g_i}^2) \right] M_f, \quad M_f = \frac{\exp\{\lambda_f^2 B^2 / \{2(1 + \lambda_f^2 A)\}\}}{(1 + \lambda_f^2 A)^{1/2}}. \quad (\text{A15})$$

(b) (g leaf.) *Let $r_i = y_i - f(x_i) - g_{-h}(x_i)$ be the partial residuals of the n_1 RCT controls routed to a leaf of g -tree h , with mean $\bar{y}$. Under the prior (A12),*

$$\sum_{z \in \{0,1\}} P(z) \int \prod_i \phi(r_i; \theta, \sigma_1^2) \phi(\theta; 0, \tau_z^2) d\theta = \left[\prod_i \phi(r_i; 0, \sigma_1^2) \right] M_g, \quad (\text{A16})$$

$$M_g = \frac{w \phi(\bar{y}; 0, \tau_0^2 + v) + (1 - w) \phi(\bar{y}; 0, \tau_1^2 + v)}{\phi(\bar{y}; 0, v)}, \quad (\text{A17})$$

where τ_z^2 denotes τ_0^2 for $z = 1$ and τ_1^2 for $z = 0$. A leaf with $n_1 = 0$ has $M_g = 1$.

(c) (Leaf conditionals.) *In a g leaf,*

$$P(z = 1 | \bar{y}) = \frac{w \phi(\bar{y}; 0, \tau_0^2 + v)}{w \phi(\bar{y}; 0, \tau_0^2 + v) + (1 - w) \phi(\bar{y}; 0, \tau_1^2 + v)}, \quad \theta | z, \bar{y} \sim \mathcal{N}\left(\frac{\tau_z^2}{\tau_z^2 + v} \bar{y}, \frac{\tau_z^2 v}{\tau_z^2 + v}\right), \quad (\text{A18})$$

and in an f leaf $\theta | \cdot \sim \mathcal{N}(\lambda_f^2 B / (1 + \lambda_f^2 A), \lambda_f^2 / (1 + \lambda_f^2 A))$.

Proof. (a) Expanding $-(r_i - \theta)^2/(2\sigma_{g_i}^2) = -r_i^2/(2\sigma_{g_i}^2) + r_i\theta/\sigma_{g_i}^2 - \theta^2/(2\sigma_{g_i}^2)$ and summing over i gives $\prod_i \phi(r_i; \theta, \sigma_{g_i}^2) = \prod_i \phi(r_i; 0, \sigma_{g_i}^2) \exp(\theta B - \theta^2 A/2)$. Multiplying by $\phi(\theta; 0, \lambda_f^2)$ and integrating, the exponent is $-\frac{1}{2}(A + \lambda_f^{-2})\theta^2 + B\theta$, a Gaussian integral equal to $(2\pi)^{1/2}(A + \lambda_f^{-2})^{-1/2} \exp\{B^2/(2(A + \lambda_f^{-2}))\}$; dividing by the prior normalizing constant $(2\pi\lambda_f^2)^{1/2}$ gives M_f . The conditional of θ is read off the same quadratic form.

(b) With a single source, (a) gives $\prod_i \phi(r_i; \theta, \sigma_1^2) = \prod_i \phi(r_i; 0, \sigma_1^2) \exp\{\theta n\bar{y}/\sigma_1^2 - \theta^2 n/(2\sigma_1^2)\}$, and the exponential equals $\phi(\bar{y}; \theta, v)/\phi(\bar{y}; 0, v)$, since $-\{(\bar{y} - \theta)^2 - \bar{y}^2\}/(2v) = (2\bar{y}\theta - \theta^2)n/(2\sigma_1^2)$. Hence the likelihood factors as $\prod_i \phi(r_i; 0, \sigma_1^2) \times \phi(\bar{y}; \theta, v)/\phi(\bar{y}; 0, v)$, the sufficiency of $\bar{y}$. Integrating $\phi(\bar{y}; \theta, v)$ against $\phi(\theta; 0, \tau_z^2)$ by (A14) gives $\phi(\bar{y}; 0, \tau_z^2 + v)$, and the sum over z with weights w and $1 - w$ gives M_g . When $n_1 = 0$ the leaf has no likelihood factor and the integral of the prior is one.

(c) By (A14), the joint density of (z, θ) given $\bar{y}$ is proportional to $P(z)\phi(\bar{y}; 0, \tau_z^2 + v)\phi(\theta; \tau_z^2\bar{y}/(\tau_z^2 + v), \tau_z^2 v/(\tau_z^2 + v))$; summing out θ gives the stated Bernoulli weight and the remaining factor is the stated conditional of θ . $\square$

Proposition A1 (The tree moves target the exact posterior). *Condition on $(\sigma_1^2, \sigma_2^2, \lambda_f^2, \tau_0^2, \tau_1^2, w)$ and on all trees other than h together with their leaf parameters. Let the update of tree h consist of a Metropolis–Hastings step on the partition T_h with grow, prune, and change proposals, acceptance ratio equal to the tree-prior ratio times the proposal ratio times the ratio of products of M_f (for f trees) or M_g (for g trees) over the affected leaves, followed by a draw of the leaf parameters of T_h (and, for a g tree, the indicators) from (A18). Then the composite update leaves the conditional posterior of (T_h, θ_h) (resp. (T_h, θ_h, z_h)) invariant. Consequently the full sweep, with the conjugate updates of $(\sigma_g^2, \tau_0^2, w)$, the update of τ_0^2 being the truncated conditional under the prior (A13), is a partially collapsed Gibbs sampler whose invariant law is the exact joint posterior under (A12) and (A13).*

Proof. Given the conditioning, the marginal conditional of T_h with the leaf parameters and indicators integrated out is $\pi(T_h) \prod_i \phi(r_i; 0, \sigma_{g_i}^2) \prod_{\ell \in T_h} M_\ell$ by Lemma A1, where M_ℓ is M_f or M_g . The partial residuals r_i do not depend on T_h , and the observation-level product is over all observations of the tree, so it is the same for the current and the proposed partition and cancels from the Metropolis–Hastings ratio. The stated acceptance ratio is therefore the exact ratio for the collapsed target; for a grow move that splits leaf P into L and R , the leaf factor ratio is $M_L M_R / M_P$, and the prune and change

ratios follow in the same way. A Metropolis–Hastings kernel with the exact ratio leaves the collapsed conditional invariant. ~~Drawing~~ Following that step by a draw of the leaf parameters (and indicators) from their exact conditional given T_h ~~then yields a draw from~~ gives a kernel that preserves the joint conditional of (T_h, θ_h, z_h) , which is the marginal-then-conditional construction of Chipman et al. [5]. The conjugate updates of σ_g^2 , τ_0^2 , and w are ordinary Gibbs steps, so the sweep is a partially collapsed Gibbs sampler with the joint posterior as invariant law. $\square$

Remark A1 (No surrogate is needed). Every factor entering a tree move is closed form because, conditionally on (τ_0^2, τ_1^2, w) , the leaf prior (A12) is a finite Gaussian mixture, and the Gaussian likelihood integrates against each component in closed form. ~~The previous model placed~~ If a leaf-specific discrepancy variance ~~under~~ instead has a scaled-Inv- χ^2 prior and ~~integrated it is~~ integrated inside the leaf marginal, ~~which produced an the resulting~~ additive scale mixture ~~is~~ integrated inside the leaf marginal, ~~the moment-matched Student- t surrogate and the perturbation bound of the previous Theorem 2 were the response to that integral.~~ In LRC-BART the scale τ_0^2 is a global parameter updated by its own conjugate step, and the local adaptivity is carried by the per-leaf indicator z , so the leaf marginal is exact and the tree kernels target the exact posterior (Proposition A1). ~~The surrogate, its error bound, and the compactness assumption that supported it are therefore no longer part of the method~~ Posterior invariance therefore follows from the exact update construction without a surrogate-likelihood perturbation bound; Web Appendix F ~~lists what is dropped~~ states its scope.

Web Appendix B: Bounded local influence (T1)

B.1 A single g leaf

We fix a g leaf and condition on f , on the other g trees, on the partition, and on $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$, so that $\bar{y} | \theta \sim \mathcal{N}(\theta, v)$ with $v = \sigma_1^2/n$. Write

$$\rho_j = \frac{\tau_j^2}{\tau_j^2 + v}, \quad \hat{\theta}_j = \rho_j \bar{y}, \quad V_j = \rho_j v = \frac{\tau_j^2 v}{\tau_j^2 + v}, \quad j \in \{0, 1\}, \quad (\text{A19})$$

so that $\hat{\theta}_1$ is the slab estimate (close to the RCT-only residual mean $\bar{y}$ when $\tau_1^2 \gg v$) and $\hat{\theta}_0$ the spike-shrunk estimate (close to zero, that is, to the pooled surface f , when $\tau_0^2 \ll v$). The subscript j

of ρ_j , $\hat{\theta}_j$, V_j indexes the scale τ_j^2 , so $j = 1$ is the slab, whereas the indicator convention of (A12) has $z = 1$ for the spike. Under Assumption A1, $0 < \rho_0 < \rho_1 < 1$. Let $p = P(z = 0 | \bar{y})$ be the posterior slab probability and define

$$\kappa = \frac{1}{2} \left(\frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right), \quad a = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2 + v}{\tau_1^2 + v}, \quad C = e^{-a} = \frac{w}{1-w} \sqrt{\frac{\tau_1^2 + v}{\tau_0^2 + v}}. \quad (\text{A20})$$

Under Assumption A1, $\kappa > 0$, and a direct computation gives

$$\rho_1 - \rho_0 = \frac{v(\tau_1^2 - \tau_0^2)}{(\tau_1^2 + v)(\tau_0^2 + v)} = 2v\kappa. \quad (\text{A21})$$

Theorem A1 (Bounded local influence). *Under Assumption A1 and the conditioning above, the following hold for every $\bar{y} \in \mathbb{R}$; parts (b) to (d) assume $0 < w < 1$.*

(a) (Posterior.) $\theta | \bar{y} \sim p\mathcal{N}(\hat{\theta}_1, V_1) + (1-p)\mathcal{N}(\hat{\theta}_0, V_0)$, and hence

$$\mathbb{E}[\theta | \bar{y}] = p \frac{\tau_1^2}{\tau_1^2 + v} \bar{y} + (1-p) \frac{\tau_0^2}{\tau_0^2 + v} \bar{y}. \quad (\text{A22})$$

(b) (Slab probability.) $\text{logit} p = \log \frac{p}{1-p} = a + \kappa \bar{y}^2$, with a and κ as in (A20).

(c) (Bounded spike contribution and bounded RWD-induced shift.) *The spike-shrunk part of (A22) satisfies $|(1-p)\hat{\theta}_0| \leq |\bar{y}| \tau_0^2/(\tau_0^2 + v)$. The shift of the posterior mean relative to the slab estimate,*

$$I(\bar{y}) = \mathbb{E}[\theta | \bar{y}] - \hat{\theta}_1 = -(1-p)(\rho_1 - \rho_0) \bar{y}, \quad (\text{A23})$$

satisfies $|I(\bar{y})| \leq (\rho_1 - \rho_0) |\bar{y}| \min\{1, Ce^{-\kappa \bar{y}^2}\}$ and

$$\sup_{\bar{y} \in \mathbb{R}} |I(\bar{y})| \leq 2v \sqrt{\kappa \max\{\log C, \frac{1}{2}\}} \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}. \quad (\text{A24})$$

(d) (Gaussian rate.) $1-p \leq Ce^{-\kappa \bar{y}^2}$, so $p \rightarrow 1$ and $I(\bar{y}) \rightarrow 0$ as $|\bar{y}| \rightarrow \infty$, at a Gaussian rate.

(e) (Nested cases.) *With $w = 1$ (C-BART), $p \equiv 0$ and $I(\bar{y}) = -(\rho_1 - \rho_0)\bar{y}$, linear and unbounded; if in addition $\tau_0^2 \rightarrow 0$ (BART-CP), $\mathbb{E}[\theta | \bar{y}] \rightarrow 0$ and $I(\bar{y}) \rightarrow -\rho_1 \bar{y}$. With $w = 0$, $p \equiv 1$ and $I \equiv 0$. If Assumption A1 fails with $\tau_0^2 > \tau_1^2$, then $\kappa < 0$, the roles of the two components*

are exchanged, and (c), (d) hold with the wide component in the role of the slab; if $\tau_0^2 = \tau_1^2$, $p \equiv 1 - w$ and $I \equiv 0$.

Proof. (a) The posterior density is proportional to $\{w\phi(\theta; 0, \tau_0^2) + (1 - w)\phi(\theta; 0, \tau_1^2)\}\phi(\bar{y}; \theta, v)$. Applying (A14) to each term gives $w\phi(\bar{y}; 0, \tau_0^2 + v)\phi(\theta; \hat{\theta}_0, V_0) + (1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)\phi(\theta; \hat{\theta}_1, V_1)$, a two-component Gaussian mixture with weights proportional to $w\phi(\bar{y}; 0, \tau_0^2 + v)$ and $(1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)$; normalizing gives $1 - p$ and p as in (A18), and (A22) is the mean of the mixture.

(b) The log odds of the two weights are

$$\log \frac{(1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)}{w\phi(\bar{y}; 0, \tau_0^2 + v)} = \log \frac{1 - w}{w} - \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} + \frac{\bar{y}^2}{2} \left(\frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right) = a + \kappa \bar{y}^2.$$

(c) The first bound is immediate from $0 \leq 1 - p \leq 1$. For I , subtract $\hat{\theta}_1$ from (A22). Since $1 - p = (1 + e^{a + \kappa \bar{y}^2})^{-1} \leq \min\{1, e^{-a - \kappa \bar{y}^2}\} = \min\{1, C e^{-\kappa \bar{y}^2}\}$, the pointwise bound follows. For the supremum we show that for $\kappa > 0$ and $C > 0$,

$$\sup_{u \in \mathbb{R}} |u| \min\{1, C e^{-\kappa u^2}\} \leq \sqrt{\max\{\log C, \frac{1}{2}\}/\kappa}. \quad (\text{A25})$$

Put $t = \kappa u^2$. On $\{t \leq \log C\}$ (empty if $C \leq 1$) the left side is at most $|u| \leq \sqrt{\log C/\kappa}$. On $\{t > \max(\log C, 0)\}$ the left side equals $\sqrt{t/\kappa} C e^{-t}$. If $\log C \geq \frac{1}{2}$, the function $\sqrt{t} e^{-t}$ is decreasing on $[\frac{1}{2}, \infty)$, so the value is at most $\sqrt{\log C/\kappa} C e^{-\log C} = \sqrt{\log C/\kappa}$. If $\log C < \frac{1}{2}$, then $\sup_{t > 0} \sqrt{t} e^{-t} = (2e)^{-1/2}$ gives at most $C(2e\kappa)^{-1/2} < \sqrt{1/(2\kappa)}$ because $C < e^{1/2}$. This proves (A25). Combining with (A21), $\sup |I| \leq 2v\kappa \sqrt{\max\{\log C, \frac{1}{2}\}/\kappa} = 2v\sqrt{\kappa \max\{\log C, \frac{1}{2}\}}$, and $\kappa \leq 1/(2(\tau_0^2 + v)) \leq 1/(2v)$ gives the second inequality in (A24).

(d) is the pointwise bound in (c) together with $\kappa > 0$. (e) With $w = 1$ the slab weight is zero for every $\bar{y}$; with $w = 0$ the spike weight is zero; the remaining statements are read off (A20) and (A23). $\square$

Remark A2 (What is and is not bounded). Theorem A1 bounds the influence of the commensurate component relative to the slab estimate $\hat{\theta}_1 = \rho_1 \bar{y}$. The slab estimate itself is shrunk toward zero, that is toward the pooled surface f , by the factor ρ_1 : $\mathbb{E}[\theta | \bar{y}] - \bar{y} = -(1 - \rho_1)\bar{y} + I(\bar{y})$ with $1 - \rho_1 = v/(\tau_1^2 + v)$. This linear term is the range-rule regularization that any BART leaf with prior variance τ_1^2 applies, and it is of the same form as the shrinkage of a BART-NP leaf toward its

ensemble center; at $n = 50$, $\sigma_1 = 1.5$ and $\tau_1^2 = 1.25$ its slope is 0.035. The bounded term I is the part specific to borrowing.

Remark A3 (Numerical illustration). With $\sigma_1 = 1.5$, $n = 50$, $\tau_0^2 = 0.001$ (the scale calibrated for $N_{\text{target}} = 100$ in the Gaussian study), $\tau_1^2 = 1.25$ (the range rule at $H_g = 5$ for an outcome range of 10) and $w = 0.8$ fixed for illustration, $v = 0.045$, $\kappa = 10.5$, $C = 21.2$, $\log C = 3.06$, and $\rho_1 - \rho_0 = 0.94$. The bound (A24) gives $\sup |I| \leq 0.51$; the exact supremum of (A23), a one-dimensional maximization, is 0.31, attained at $\bar{y} = 0.43$. By comparison C-BART's shift is $0.94 |\bar{y}|$, which exceeds 0.51 once $|\bar{y}| > 0.54$.

B.2 The estimand level: fixed partitions of the g ensemble

The estimand is the RCT-standardized control mean $\mu = N^{-1} \sum_{i \leq N} f_1(x_i)$ over the N RCT covariate profiles, and $f_1 = f + g$, so $\mu = \mu_f + \mu_g$ with $\mu_g = N^{-1} \sum_{i \leq N} g(x_i)$. We condition on f , on the partitions $\mathcal{T} = (T_1, \dots, T_{H_g})$ of the g trees, and on $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$; the RWD then enters μ_g only through the shrinkage of g toward zero. Index the L_g leaves of all g trees by j , let $\theta \in \mathbb{R}^{L_g}$ collect the leaf parameters and $z \in \{0, 1\}^{L_g}$ the indicators, and let $r \in \mathbb{R}^{n_1}$ be the residuals $y_i - f(x_i)$ of the RCT controls. Let $X \in \{0, 1\}^{n_1 \times L_g}$ be the incidence matrix ($X_{ij} = 1$ if control i falls in leaf j), so each row of X has exactly H_g ones and $g(x_i) = (X\theta)_i$. Let $a \in \mathbb{R}^{L_g}$ with $a_j = N_j/N$, N_j the number of estimand profiles in leaf j , so $\mu_g = a^\top \theta$, $\|a\|_1 = H_g$, and $\|a\|_2^2 \leq \max_j a_j \sum_j a_j \leq H_g$, so $\|a\|_2 \leq \sqrt{H_g}$. Then

$$r | \theta \sim \mathcal{N}(X\theta, \sigma_1^2 I), \quad \theta | z \sim \mathcal{N}(0, D_z), \quad D_z = \text{diag}(\tau_{z_j}^2), \quad \pi(z) = \prod_j w^{z_j} (1 - w)^{1 - z_j}. \quad (\text{A26})$$

Standard Gaussian algebra gives, with $V_z = XD_zX^\top + \sigma_1^2 I$ and $\Sigma_z = (X^\top X/\sigma_1^2 + D_z^{-1})^{-1}$,

$$P(z | r) \propto \pi(z) \phi_{n_1}(r; 0, V_z), \quad \theta | z, r \sim \mathcal{N}(\hat{\theta}(z), \Sigma_z), \quad \hat{\theta}(z) = D_z X^\top V_z^{-1} r, \quad (\text{A27})$$

where the form of $\hat{\theta}(z)$ follows from $(X^\top X/\sigma^2 + D^{-1})^{-1} X^\top / \sigma^2 = DX^\top (XD X^\top + \sigma^2 I)^{-1}$. We write $\mathbf{0}$ for the all-slab configuration; the $w = 0$ model (the tree analogue of BART-PP, a free source

offset regularized by the range rule) has $\mathbb{E}_{w=0}[\mu_g | r] = a^\top \hat{\theta}(\mathbf{0})$, and we call

$$\Delta(r) = \mathbb{E}[\mu_g | r] - a^\top \hat{\theta}(\mathbf{0}) = \sum_{z \neq \mathbf{0}} P(z | r) a^\top (\hat{\theta}(z) - \hat{\theta}(\mathbf{0})) \quad (\text{A28})$$

the RWD-induced shift of the estimand at the given partitions.

Corollary A1 (Bounded shift of the standardized control mean). *Under Assumption A1, $0 < w < 1$, and the conditioning of this subsection, let $\delta = \tau_1^2 - \tau_0^2$ and $b = w\tau_1/\{(1-w)\tau_0\}$.*

(a) (General partitions.) $\sup_{r \in \mathbb{R}^{n_1}} |\Delta(r)| \leq \sqrt{H_g} \left(1 + \frac{\tau_1}{\tau_0}\right) (2^{L_g} - 1) \sqrt{2\delta \max\{L_g \log b, \frac{1}{2}\}} < \infty.$

(b) (Single tree, $H_g = 1$.) *With leaves ℓ , RCT counts $n_{1\ell}$, $v_\ell = \sigma_1^2/n_{1\ell}$, leaf means $\bar{y}_\ell$, and κ_ℓ, C_ℓ as in (A20) with $v = v_\ell$,*

$$\Delta(r) = \sum_{\ell: n_{1\ell} > 0} a_\ell I_\ell(\bar{y}_\ell), \quad \sup_r |\Delta(r)| \leq \sum_\ell a_\ell \sqrt{2v_\ell \max\{\log C_\ell, \frac{1}{2}\}},$$

and the right side is at most $\max_\ell \sqrt{2v_\ell \max\{\log C_\ell, \frac{1}{2}\}}$.

(c) (H_g trees with a common partition.) *If all g trees share one partition with leaves ℓ , write $\vartheta_\ell = \sum_h \theta_{h\ell}$, $T_k = k\tau_0^2 + (H_g - k)\tau_1^2$, $\rho_\ell^{(k)} = T_k/(T_k + v_\ell)$,*

$$\kappa_{k\ell} = \frac{1}{2} \left(\frac{1}{T_k + v_\ell} - \frac{1}{T_0 + v_\ell} \right), \quad C_{k\ell} = \binom{H_g}{k} \left(\frac{w}{1-w} \right)^k \sqrt{\frac{T_0 + v_\ell}{T_k + v_\ell}}.$$

Then $P(k \text{ spikes in leaf } \ell | \bar{y}_\ell) \propto \binom{H_g}{k} w^k (1-w)^{H_g-k} \phi(\bar{y}_\ell; 0, T_k + v_\ell)$, the leaves are independent a posteriori,

$$\Delta(r) = - \sum_\ell a_\ell \bar{y}_\ell \sum_{k=1}^{H_g} P(k | \bar{y}_\ell) (\rho_\ell^{(0)} - \rho_\ell^{(k)}), \quad (\text{A29})$$

$$\sup_r |\Delta(r)| \leq \max_\ell 2v_\ell \sum_{k=1}^{H_g} \sqrt{\kappa_{k\ell} \max\{\log C_{k\ell}, \frac{1}{2}\}} \leq \max_\ell H_g v_\ell \sqrt{\frac{2 \max_k \max\{\log C_{k\ell}, \frac{1}{2}\}}{H_g \tau_0^2 + v_\ell}}. \quad (\text{A30})$$

(d) (Contrasts.) *Under complete pooling of the control surface ($w = 1$, $\tau_0^2 \rightarrow 0$) the shift is $-a^\top \hat{\theta}(\mathbf{0})$, and under C-BART ($w = 1$) it is $a^\top (\hat{\theta}(\mathbf{1}) - \hat{\theta}(\mathbf{0}))$; both are linear in r , and each is*

unbounded on $\mathbb{R}^{n_1}$ unless the corresponding linear functional vanishes identically. For a single tree the C-BART functional is $-\sum_{\ell} a_{\ell}(\rho_{1\ell} - \rho_{0\ell})\bar{y}_{\ell}$, which is nonzero whenever some leaf has $a_{\ell} > 0$ and $n_{1\ell} > 0$.

Proof. (a) Fix $z \neq \mathbf{0}$, let $S = \{j : z_j = 1\}$ be its spike leaves, E_S the diagonal indicator of S , X_S the corresponding columns of X , and $u = X_S^{\top} V_{\mathbf{0}}^{-1} r \in \mathbb{R}^{|S|}$. Since $D_{\mathbf{0}} - D_z = \delta E_S$,

$$V_z^{-1} - V_{\mathbf{0}}^{-1} = V_z^{-1}(V_{\mathbf{0}} - V_z)V_{\mathbf{0}}^{-1} = \delta V_z^{-1} X_S X_S^{\top} V_{\mathbf{0}}^{-1}, \quad \text{so} \quad V_z^{-1} r = V_{\mathbf{0}}^{-1} r + \delta V_z^{-1} X_S u.$$

Three consequences follow. First, the quadratic form in the likelihood ratio is

$$r^{\top} (V_z^{-1} - V_{\mathbf{0}}^{-1}) r = \delta (V_z^{-1} r)^{\top} X_S u = \delta (u + \delta X_S^{\top} V_z^{-1} X_S u)^{\top} u = \delta \|u\|^2 + \delta^2 u^{\top} X_S^{\top} V_z^{-1} X_S u \geq \delta \|u\|^2.$$

Second, the difference of posterior means is

$$\hat{\theta}(z) - \hat{\theta}(\mathbf{0}) = (D_z - D_{\mathbf{0}}) X^{\top} V_{\mathbf{0}}^{-1} r + D_z X^{\top} (V_z^{-1} - V_{\mathbf{0}}^{-1}) r = -\delta \iota_S u + \delta D_z X^{\top} V_z^{-1} X_S u,$$

where ι_S embeds $\mathbb{R}^{|S|}$ into the S coordinates of $\mathbb{R}^{L_g}$. The matrix $D_z X^{\top} V_z^{-1} X = D_z^{1/2} \{M^{\top} (M M^{\top} + \sigma_1^2 I)^{-1} M\} D_z^{-1/2}$ with $M = X D_z^{1/2}$, and the bracket is symmetric with eigenvalues $\lambda_i / (\lambda_i + \sigma_1^2) \in [0, 1)$, so $\|D_z X^{\top} V_z^{-1} X_S\| \leq \|D_z X^{\top} V_z^{-1} X\| \leq \|D_z^{1/2}\| \|D_z^{-1/2}\| \leq \tau_1 / \tau_0$, and $\|\hat{\theta}(z) - \hat{\theta}(\mathbf{0})\| \leq \delta(1 + \tau_1 / \tau_0) \|u\|$. Third, $|V_z| = \sigma_1^{2n_1} |D_z| |D_z^{-1} + X^{\top} X / \sigma_1^2|$ and $D_z^{-1} \succeq D_{\mathbf{0}}^{-1}$ give $|V_{\mathbf{0}}| / |V_z| \leq |D_{\mathbf{0}}| / |D_z| = (\tau_1^2 / \tau_0^2)^{|S|}$, so

$$\begin{aligned} P(z | r) &\leq \frac{P(z | r)}{P(\mathbf{0} | r)} = \frac{\pi(z)}{\pi(\mathbf{0})} \sqrt{\frac{|V_{\mathbf{0}}|}{|V_z|}} \exp\left\{-\frac{1}{2} r^{\top} (V_z^{-1} - V_{\mathbf{0}}^{-1}) r\right\} \\ &\leq c_z e^{-\delta \|u\|^2 / 2}, \quad c_z = b^{|S|}. \end{aligned}$$

Combining, and using (A25) in the variable $\|u\|$ with $\kappa = \delta/2$,

$$\begin{aligned} |P(z | r) a^{\top} (\hat{\theta}(z) - \hat{\theta}(\mathbf{0}))| &\leq \|a\| \delta \left(1 + \frac{\tau_1}{\tau_0}\right) \|u\| \min\{1, c_z e^{-\delta \|u\|^2 / 2}\} \\ &\leq \sqrt{H_g} \left(1 + \frac{\tau_1}{\tau_0}\right) \sqrt{2\delta \max\{|S| \log b, \frac{1}{2}\}}. \end{aligned}$$

Summing over the $2^{L_g} - 1$ configurations $z \neq \mathbf{0}$ and using $|S| \leq L_g$ gives (a).

(b) For a single tree $X^\top X$ is diagonal, the residual vector splits by leaf, and (A26) factorizes over leaves, so the leaves are independent a posteriori and each is the single-leaf model of Section B.1 with $v = v_\ell$; leaves with $n_{1\ell} = 0$ have posterior equal to prior under every w and contribute zero to Δ . Theorem A1(c) bounds each $|I_\ell|$, and $\sum_\ell a_\ell = 1$ gives the last inequality.

(c) With a common partition, the residual in leaf ℓ is $r_i = \vartheta_\ell + \varepsilon_i$ and, given the indicators, $\vartheta_\ell \sim \mathcal{N}(0, T_{k_\ell})$ with k_ℓ the number of spikes among the H_g leaves stacked on ℓ ; the leaves are independent a posteriori and $\mu_g = \sum_\ell a_\ell \vartheta_\ell$. Summing $\pi(z)\phi(\bar{y}_\ell; 0, T_k + v_\ell)$ over the $\binom{H_g}{k}$ configurations with k spikes gives the stated posterior of k , and (A14) gives $\mathbb{E}[\vartheta_\ell | k, \bar{y}_\ell] = \rho_\ell^{(k)} \bar{y}_\ell$. Under $w = 0$, $k = 0$ almost surely, which gives the expression for Δ . For $k \geq 1$, $P(k | \bar{y}_\ell) \leq P(k | \bar{y}_\ell) / P(0 | \bar{y}_\ell) = C_{k\ell} e^{-\kappa_{k\ell} \bar{y}_\ell^2}$, and $\rho_\ell^{(0)} - \rho_\ell^{(k)} = v_\ell(T_0 - T_k) / \{(T_0 + v_\ell)(T_k + v_\ell)\} = 2v_\ell \kappa_{k\ell}$; the k -th term is therefore at most $2v_\ell \kappa_{k\ell} |\bar{y}_\ell| \min\{1, C_{k\ell} e^{-\kappa_{k\ell} \bar{y}_\ell^2}\} \leq 2v_\ell \sqrt{\kappa_{k\ell} \max\{\log C_{k\ell}, \frac{1}{2}\}}$ by (A25). The final form uses $\kappa_{k\ell} \leq \kappa_{H_g \ell} \leq 1 / \{2(H_g \tau_0^2 + v_\ell)\}$ and $\sum_\ell a_\ell = 1$.

(d) With $w = 1$ the posterior over z is degenerate at $\mathbf{1}$, so $\mathbb{E}[\mu_g | r] = a^\top \hat{\theta}(\mathbf{1})$, linear in r ; as $\tau_0^2 \rightarrow 0$, $\hat{\theta}(\mathbf{1}) \rightarrow 0$. A nonzero linear functional on $\mathbb{R}^{n_1}$ is unbounded. For a single tree, $\hat{\theta}_\ell(\mathbf{1}) - \hat{\theta}_\ell(\mathbf{0}) = (\rho_{0\ell} - \rho_{1\ell}) \bar{y}_\ell$ by (b). $\square$

Remark A4 (On the constants). The constant in (a) is finite for every design and every partition, which is the content of the statement, but it is far from sharp: it treats each of the $2^{L_g} - 1$ non-slab configurations separately and bounds the posterior probability of each by its odds against the all-slab configuration. Parts (b) and (c) are the cases in which the sum collapses, and they display the dependence on H_g , on τ_0 through κ , and on the leaf shrinkage $\rho_\ell^{(H_g)} = H_g \tau_0^2 / (H_g \tau_0^2 + v_\ell)$ of the all-spike state. For the numbers of Remark A3 with $\tau_0^2 = 0.0012$ and the study's configuration $H_g = 5$, the common-partition bound in (c) is 1.15 per leaf, while the exact supremum of (A29) is 0.148, attained at $\bar{y}_\ell = 0.33$ (at $H_g = 10$ the corresponding values are 1.80 and 0.067 at 0.30); the ensemble abstains sooner than a single leaf because there are H_g ways to place one slab, and one slab leaf absorbs almost all of the discrepancy. The exact supremum is a one-dimensional maximization and is what we recommend reporting.

Web Appendix C: Detection and the borrowing map (T2)

C.1 A single g leaf under a true local shift

We keep the conditioning of Section B.1 and add a data-generating assumption: the RCT residuals in the leaf are $r_i = d + \epsilon_i$ with $\epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_1^2)$, where d is the true local discrepancy between the RCT control surface and the pooled surface f (taken as fixed; see the [Gap-paragraph-scope discussion](#) in Section C.3 for the estimated f). Then $\bar{y} \sim \mathcal{N}(d, v)$, $v = \sigma_1^2/n$. Let $P_{\text{spike}}(\bar{y}) = 1 - p = P(z = 1 \mid \bar{y})$.

Theorem A2 (Leaf-level detection). *Under Assumption A1 and the conditioning above:*

- (a) (Spike probability.) $P_{\text{spike}}(\bar{y}) = \{1 + \exp(a + \kappa \bar{y}^2)\}^{-1}$ with a, κ as in (A20); it depends on $(\bar{y}, n)$ only through $\bar{y}^2$ and $v = \sigma_1^2/n$, is decreasing in $|\bar{y}|$, and satisfies $P_{\text{spike}}(0) = \{1 + e^a\}^{-1}$.
- (b) (Half-detection discrepancy.) If $\log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} > 0$, the equation $P_{\text{spike}}(\bar{y}) = \frac{1}{2}$ has the unique positive solution

$$d_{1/2} = \sqrt{\frac{2(\tau_0^2 + v)(\tau_1^2 + v)}{\tau_1^2 - \tau_0^2} \left\{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} \right\}}; \quad (\text{A31})$$

otherwise $P_{\text{spike}} \leq \frac{1}{2}$ for every $\bar{y}$ (with equality only at $\bar{y} = 0$ when the bracket vanishes) and we set $d_{1/2} = 0$. When $\tau_1^2 \gg \tau_0^2 + v$, $d_{1/2}^2 \approx 2(\tau_0^2 + v) \left\{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2}{\tau_0^2 + v} \right\}$, a few multiples of $\tau_0^2 + \sigma_1^2/n$. The probability that the leaf's slab probability exceeds one half, under the true shift d , is

$$P_d(p(\bar{y}) > \frac{1}{2}) = P_d(|\bar{y}| > d_{1/2}) = \bar{\Phi}\left(\frac{d_{1/2} - d}{\sqrt{v}}\right) + \bar{\Phi}\left(\frac{d_{1/2} + d}{\sqrt{v}}\right). \quad (\text{A32})$$

- (c) (Large n , τ_0^2 fixed.) As $n \rightarrow \infty$, almost surely

$$P_{\text{spike}}(\bar{y}) \rightarrow P_{\text{spike}}^\infty(d) = \left\{ 1 + \frac{(1-w)\tau_0}{w\tau_1} \exp\left(\frac{d^2}{2} \left(\frac{1}{\tau_0^2} - \frac{1}{\tau_1^2}\right)\right) \right\}^{-1} \in (0, 1). \quad (\text{A33})$$

In particular, when $d = 0$ (borrowing is warranted), $P_{\text{spike}} \rightarrow w\tau_1 / \{w\tau_1 + (1-w)\tau_0\} > w$; when $d \neq 0$, $P_{\text{spike}}^\infty(d) \leq \frac{w\tau_1}{(1-w)\tau_0} \exp\{-\frac{d^2}{2}(\tau_0^{-2} - \tau_1^{-2})\}$, which tends to zero as $|d|/\tau_0 \rightarrow \infty$ but is strictly positive for every fixed d .

- (d) (Large n with a vanishing spike.) If $\tau_0^2 = \tau_{0,n}^2 \rightarrow 0$ as $n \rightarrow \infty$ (with τ_1^2 , w fixed), then for every $d \neq 0$, $P_{\text{spike}}(\bar{y}) \rightarrow 0$ almost surely.

Proof. (a) is Theorem A1(b) with $\kappa > 0$. (b) Setting $a + \kappa\bar{y}^2 = 0$ gives $\bar{y}^2 = -a/\kappa$, which is (A31) after substituting (A20); the condition is $-a > 0$. The approximation drops v and τ_0^2 against τ_1^2 in $(\tau_1^2 + v)/(\tau_1^2 - \tau_0^2)$. Since $p > \frac{1}{2}$ iff $|\bar{y}| > d_{1/2}$ and $\bar{y} \sim \mathcal{N}(d, v)$, (A32) follows.

(c) By the strong law, $\bar{y} \rightarrow d$ almost surely; $v \rightarrow 0$ gives $a \rightarrow a_\infty = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2}{\tau_1^2}$ and $\kappa \rightarrow \kappa_\infty = \frac{1}{2}(\tau_0^{-2} - \tau_1^{-2})$, both finite. Hence $a + \kappa\bar{y}^2 \rightarrow a_\infty + \kappa_\infty d^2$ almost surely, and (A33) follows by continuity of $t \mapsto (1 + e^t)^{-1}$; the limit is strictly between 0 and 1 because the exponent is finite. The bound on $P_{\text{spike}}^\infty(d)$ is $(1 + e^t)^{-1} \leq e^{-t}$.

(d) Write $\epsilon_n = \tau_{0,n}^2 + v_n \rightarrow 0$. Then $a_n + \kappa_n \bar{y}_n^2 \geq \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\epsilon_n}{\tau_1^2 + v_n} + \bar{y}_n^2 \left(\frac{1}{2\epsilon_n} - \frac{1}{2\tau_1^2} \right)$. Almost surely $\bar{y}_n^2 \rightarrow d^2 > 0$, and $\frac{d^2}{2\epsilon_n} + \frac{1}{2} \log \epsilon_n \rightarrow \infty$, so the log odds of the slab diverge and $P_{\text{spike}} \rightarrow 0$. $\square$

Gap (consistency of abstention) Fixed and vanishing spike scales. ~~The design specification states that $P_{\text{spike}} \rightarrow 0$ as $n \rightarrow \infty$ for any nonzero shift.~~ Theorem A2(c) ~~shows this is false for a fixed~~ distinguishes increasing local sample size from decreasing spike scale: because the spike is a $\mathcal{N}(0, \tau_0^2)$ component and not a point mass, a discrepancy of the order of τ_0 is consistent with it, and the limiting spike probability (A33) is positive for every d . What is true is (i) the limit is exponentially small in $d^2/(2\tau_0^2)$, so abstention is consistent in the regime $|d|/\tau_0 \rightarrow \infty$, and (ii) abstention is consistent for every fixed $d \neq 0$ when the spike scale shrinks with n (part (d)). ~~In the~~ For a fixed-scale illustration, at $\tau_0^2 = 0.001$ a shift of $d = 0.5$ has $d^2/(2\tau_0^2) = 125$. The fitted two-arm sampler ~~learns τ_0^2 is learned~~ from the spike leaves under a prior calibrated to the order 10^{-3} , ~~so on the outcome scale (i) is the operative statement: at $\tau_0^2 = 0.001$ a shift of $d = 0.5$ has $d^2/(2\tau_0^2) = 125$; the fixed-scale limit therefore does not establish its finite-sample detection behavior, which is assessed in Section 3.~~ The consistency statement that does hold for every fixed $d \neq 0$, without a vanishing spike, is the one for the discrepancy map in Theorem A3 below.

Remark A5 (Numerical illustration). With the constants of Remark A3, (A31) gives $d_{1/2} = 0.54$ ~~(the specification's rough figure was 0.7)~~, and (A32) gives leaf abstention probabilities 0.011, 0.43, 0.98 and 1.00 at true shifts $d = 0, 0.5, 1$ and 2. The limit (A33) at $d = 0$ is 0.993.

C.2 The borrowing map $P(|g(x)| < c \mid \text{data})$

The map we display is the posterior probability that the local discrepancy is smaller than a pre-specified outcome-scale threshold $c > 0$, $\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data})$, together with the all-spike map

$B(x) = P(\text{all } H_g \text{ leaves covering } x \text{ are spike} \mid \text{data})$ as a diagnostic. For a single tree and a fixed partition, $g(x) = \theta_\ell$ for x in leaf ℓ , and the two are related by the leaf mixture of Theorem A1(a).

Theorem A3 (The discrepancy map for a single tree). *Under Assumption A1, for a single g tree with a fixed partition, f and $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$ fixed, and x in a leaf with statistics $(\bar{y}, n)$:*

(a) (Exact form.) $\mathcal{B}_c(x) = p \Psi_1(c) + (1 - p) \Psi_0(c)$ with

$$\Psi_j(c) = \Phi\left(\frac{c - \rho_j \bar{y}}{\sqrt{\rho_j v}}\right) - \Phi\left(\frac{-c - \rho_j \bar{y}}{\sqrt{\rho_j v}}\right), \quad j \in \{0, 1\}.$$

(b) (Relation to the spike probability.) $\mathcal{B}_c(x) \leq P_{\text{spike}}(\bar{y}) + p \Psi_1(c)$, and on $\{\rho_0 |\bar{y}| < c\}$,

$$\mathcal{B}_c(x) \geq P_{\text{spike}}(\bar{y}) \left\{ 1 - 2\bar{\Phi}\left(\frac{c - \rho_0 |\bar{y}|}{\tau_0}\right) \right\}.$$

(c) (Consistency.) *If the RCT residuals in the leaf are $d + \epsilon_i$ as in Section C.1 and $|d| \neq c$, then as $n \rightarrow \infty$, almost surely $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|d| < c\}$, whatever the limit of P_{spike} .*

Proof. (a) By Theorem A1(a), the posterior of $g(x) = \theta$ is the two-component mixture with components $\mathcal{N}(\rho_j \bar{y}, \rho_j v)$, and $\Psi_j(c)$ is the probability that component j lies in $(-c, c)$.

(b) The upper bound uses $\Psi_0 \leq 1$. For the lower bound, $\mathcal{B}_c \geq (1 - p)\Psi_0(c)$ and, for a $\mathcal{N}(m, V)$ variable with $|m| < c$, $P(|N| \geq c) \leq P(|N - m| \geq c - |m|) = 2\bar{\Phi}((c - |m|)/\sqrt{V})$; here $|m| = \rho_0 |\bar{y}|$ and $V = \rho_0 v = \tau_0^2 v / (\tau_0^2 + v) \leq \tau_0^2$, and $\bar{\Phi}$ is decreasing.

(c) Almost surely $\bar{y} \rightarrow d$, and $\rho_j \rightarrow 1$, $\rho_j v \rightarrow 0$ for both j , so $\rho_j \bar{y} \rightarrow d$ and each component concentrates at d : $\Psi_j(c) \rightarrow \mathbf{1}\{|d| < c\}$ when $|d| \neq c$. A convex combination of two sequences with the same limit has that limit, regardless of the weights. $\square$

Remark A6. Parts (b) and (c) are the theoretical counterpart of the empirical finding that $\mathcal{B}_c$ separates shifted from compatible regions while $B(x)$ does not. The spike indicator answers whether the leaf discrepancy is of the order of τ_0 , a question whose answer stays uncertain in the limit (Theorem A2(c)); $\mathcal{B}_c$ answers whether it is smaller than c on the outcome scale, a question that the RCT controls settle as n grows. The threshold c must be pre-specified on the outcome scale; the study uses one third of σ_{RCT} .

C.3 The ensemble with fixed partitions, and what is not proved

We return to the fixed-partition linear form (A26). For a profile x , let $e_x \in \{0, 1\}^{L_g}$ select the H_g leaves covering x , so $g(x) = e_x^\top \theta$. The atoms of the common refinement of the H_g partitions are the cells $A_1, \dots, A_m$; e_x is constant on each cell, and if x and RCT control i lie in the same cell then e_x is row i of X . Let n_{1k} be the number of RCT controls in cell A_k .

Proposition A2 (The discrepancy map for H_g trees). *Under Assumption A1 and the conditioning of Section B.2:*

- (a) (Exact form.) $\mathcal{B}_c(x) = \sum_z P(z|r) \Psi_z(c)$ with $\Psi_z(c) = P\{|\mathcal{N}(e_x^\top \hat{\theta}(z), e_x^\top \Sigma_z e_x)| < c\}$, and $B(x) = \sum_{z \in \mathcal{Z}_x} P(z|r)$ where $\mathcal{Z}_x = \{z : z_j = 1 \text{ for all } j \text{ with } (e_x)_j = 1\}$.
- (b) (Union bound.) $1 - B(x) \leq \sum_{h=1}^{H_g} P(z_{h, \ell_h(x)} = 0|r)$ and $B(x) \leq \min_h P(z_{h, \ell_h(x)} = 1|r)$, where $\ell_h(x)$ is the leaf of tree h containing x . For $z \in \mathcal{Z}_x$, $e_x^\top \Sigma_z e_x \leq e_x^\top D_z e_x = H_g \tau_0^2$, so on $\{|e_x^\top \hat{\theta}(z)| < c\}$, $\Psi_z(c) \geq 1 - 2\bar{\Phi}\{(c - |e_x^\top \hat{\theta}(z)|)/(\sqrt{H_g} \tau_0)\}$.
- (c) (Consistency at fixed partitions.) Suppose the true residual means are representable on the partitions, $\mathbb{E}[r|x] = X\theta^*$ for some $\theta^* \in \mathbb{R}^{L_g}$, and let $x \in A_k$ with $m_k^* = e_x^\top \theta^*$ and $|m_k^*| \neq c$. If $n_{1k} \rightarrow \infty$ (the other cell counts arbitrary), then $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|m_k^*| < c\}$ in probability.

Proof. (a) is (A27) and the definition of the two maps. (b) The first two inequalities are the union bound and monotonicity for the events $\{z_{h, \ell_h(x)} = 0\}$, $h = 1, \dots, H_g$. For Gaussians the posterior covariance is dominated by the prior covariance, $\Sigma_z = (X^\top X / \sigma_1^2 + D_z^{-1})^{-1} \preceq D_z$, and for $z \in \mathcal{Z}_x$, $e_x^\top D_z e_x = H_g \tau_0^2$; the tail bound is as in Theorem A3(b).

(c) Fix z ; write $e = e_x$, $G = X^\top X / \sigma_1^2$, $m = e^\top \theta$. *Variance.* The data enter the Gaussian model (A26) through the cell means $\bar{r}_{k'} | \theta \stackrel{\text{indep}}{\sim} \mathcal{N}(e^{(k')\top} \theta, \sigma_1^2 / n_{1k'})$, where $e^{(k')}$ is the common row of X in cell k' . In the joint Gaussian law of $(m, \bar{r}_1, \dots, \bar{r}_m)$ given z , conditional variances are constants and the law of total variance gives $\text{Var}(m | \bar{r}_k) = \mathbb{E}[\text{Var}(m | \bar{r}) | \bar{r}_k] + \text{Var}(\mathbb{E}[m | \bar{r}] | \bar{r}_k) \geq \text{Var}(m | \bar{r})$; and $\text{Var}(m | \bar{r}_k) = \omega^2 (\sigma_1^2 / n_{1k}) / (\omega^2 + \sigma_1^2 / n_{1k}) \leq \sigma_1^2 / n_{1k}$ with $\omega^2 = e^\top D_z e$. Hence $e^\top \Sigma_z e \leq \sigma_1^2 / n_{1k} \rightarrow 0$. *Mean.* With $r = X\theta^* + \epsilon$, (A27) gives $\mathbb{E}[\theta | z, r] = \Sigma_z G \theta^* + \Sigma_z X^\top \epsilon / \sigma_1^2$, and $\Sigma_z G = I - \Sigma_z D_z^{-1}$, so

$$e^\top \mathbb{E}[\theta | z, r] - m_k^* = -e^\top \Sigma_z D_z^{-1} \theta^* + e^\top \Sigma_z X^\top \epsilon / \sigma_1^2.$$

By the Cauchy–Schwarz inequality in the Σ_z inner product,

$$|e^\top \Sigma_z D_z^{-1} \theta^*| \leq \sqrt{e^\top \Sigma_z e} \sqrt{(D_z^{-1} \theta^*)^\top \Sigma_z (D_z^{-1} \theta^*)} \leq \sqrt{\sigma_1^2/n_{1k}} \tau_1 \|\theta^*\|/\tau_0^2,$$

using $\Sigma_z \preceq D_z \preceq \tau_1^2 I$ and $\|D_z^{-1}\| \leq \tau_0^{-2}$. The noise term has mean zero and variance $e^\top \Sigma_z G \Sigma_z e \leq e^\top \Sigma_z e \leq \sigma_1^2/n_{1k}$, because $G \preceq G + D_z^{-1} = \Sigma_z^{-1}$. Both terms vanish in probability. *Conclusion.* For each of the finitely many z , $\Psi_z(c) = P\{|\mathcal{N}(\hat{m}_z, \hat{V}_z)| < c\}$ with $\hat{m}_z \rightarrow m_k^*$ in probability and $\hat{V}_z \rightarrow 0$, so $\Psi_z(c) \rightarrow \mathbf{1}\{|m_k^*| < c\}$ in probability when $|m_k^*| \neq c$, uniformly over z ; the mixture $\sum_z P(z|r)\Psi_z(c)$ then has the same limit whatever the weights. $\square$

~~Gap (the ensemble with~~ Scope under random partitions~~).~~ Everything in this appendix about the map is conditional on the partitions of the g trees. Three things are not proved. First, the representability assumption in Proposition A2(c) is a real restriction: with two trees splitting on different variables the incidence matrix has rank three on four cells, so a discrepancy with an interaction on those cells is not representable, and the fitted cell means then converge to a weighted projection of the truth onto the additive span of the partitions rather than to the truth. Second, we have no statement about the posterior over the partitions themselves, hence none about $\mathcal{B}_c(x)$ or $B(x)$ under the full posterior; posterior concentration for a single standard BART ensemble is known [20], but the two-ensemble decomposition with a spike-and-slab discrepancy prior is outside the available theory, and the empirical separation of regions by $\mathcal{B}_c$ (~~verification report, check 6~~Section 3) is the evidence we have. Third, the leaf-level results treat f as fixed; with f estimated, the residual in a g leaf carries the error of f at the RCT profiles, which is O_p of the RWD posterior standard deviation of f there, and enters d as an additive perturbation. On the estimand, this perturbation is what V_μ^f of Web Appendix D measures.

Web Appendix D: ~~Monotone, well-posed~~ Variance-based ESS and calibration (T3)

The prior effective sample size of the commensurate component is defined at the level of the estimand as

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)} \left[\frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right],$$

We use a normal location experiment with known variance σ_1^2 as the reference information unit. For a smooth scalar prior density $p(\mu)$, the ELIR definition of Neuenschwander et al. [18] is

$$\text{ESS}_{\text{ELIR}} = \sigma_1^2 \mathbb{E}_p[-\partial_\mu^2 \log p(\mu)].$$

For a normal prior of variance V , this equals σ_1^2/V . For a mixture, the curvature is that of the log *marginal* density, not the average of the component curvatures. Accordingly, our calibration is an average of working precision ratios, with σ_1^2 providing normal-reference units. It is not an exact marginal ELIR calculation and does not inherit ELIR's predictive-consistency property. In the survival analysis these units refer to uncensored log times.

D.1 Leaf aggregation and the calibration reference

At fixed discrepancy partitions $\mathcal{T}$, define $a_{h\ell} = N_{h\ell}/N$ from the profiles used to standardize the estimand. With $\mu_g = \sum_{h,\ell} a_{h\ell} \theta_{h\ell}$, conditional independence of the leaf priors gives

$$V_g(\mathcal{T}, z, t) = \text{Var}(\mu_g | \mathcal{T}, z, \tau_0^2 = t) = \sum_{h,\ell} a_{h\ell}^2 \{z_{h\ell} t + (1 - z_{h\ell}) \tau_1^2\}. \quad (\text{A34})$$

Leaves with $a_{h\ell} = 0$ contribute nothing. Let L_+ count the remaining leaves. For fixed w and a specified scale law P_s , averaging the corresponding working precision over independent leaf states gives

$$\text{ESS}_{\mathcal{T}}(w, s) = \sigma_1^2 \sum_{z \in \{0,1\}^{L_+}} w^{|z|} (1-w)^{L_+ - |z|} \int \frac{P_s(dt)}{V_\mu^f + V_g(\mathcal{T}, z, t)}.$$

An additional expectation over $\mathcal{T}$ averages over the partition prior. When one leaf per tree has positive weight, its weight is one, $L_+ = H_g$, and the sum reduces to a binomial average with variance $kt + (H_g - k) \tau_1^2$. This includes a pointwise profile and stump-only ensembles. With several occupied leaves per tree, their squared proportions and independent indicators must be retained.

Merely counting the total number of spikes is insufficient when their weights differ.

For the informative calibration reference we set every discrepancy leaf to the spike. Then $V_g = tC(\mathcal{T})$, where $V_\mu^f = \text{Var}(N^{-1} \sum_{i \leq N} f(x_i) | \text{RWD}) > 0$ is the Stage-1 posterior variance of the standardized RWD surface at the RCT profiles, $c_g \in (0, 1]$ is the leaf-sharing constant of the g -tree prior, $C(\mathcal{T}) = \sum_{h, \ell} a_{h\ell}^2$ and the prior-averaged version is $\text{ESS}(w, s_0^2) = \sum_{k=0}^{H_g} \binom{H_g}{k} w^k (1-w)^{H_g-k} \sigma_1^2 \mathbb{E}[\{V_\mu^f + c_g(k\tau_0^2 + (H_g-k)\tau_1^2)\}^{-1}]$. In the calibration, V_μ^f is replaced by a block statistic $\mathbb{E}C(\mathcal{T}) = c_g H_g$. The implemented convention replaces $C(\mathcal{T})$ by this mean and uses an untruncated scale law:

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)} \left[\frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right]. \quad (\text{A35})$$

Here $V_\mu^f > 0$ is the variance of the standardized RWD-only BART surface. Its use represents that marginal distribution by its variance. By convexity, substituting $\mathbb{E}C$ before inversion gives no larger a value than averaging $1/(V_\mu^f + tC)$ over partitions at fixed t . Averaging these conditional precisions over t also differs from taking the inverse of the variance after mixing over t . These are distinct operations.

For block variances $V^{(b)}$ from Stage 1, and the selection is the Pr-rule

$$\hat{s}_{0\star}^2 = \inf\{s_0^2 \in \mathcal{S} : \hat{G}(s_0^2) \geq q\}, \quad \hat{G}(s_0^2) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\text{ESS}_0(V^{(b)}; s_0^2) \leq N_{\text{target}}\},$$

grid rule is

$$\hat{s}_{0\star}^2 = \inf\{s_0^2 \in \mathcal{S} : \hat{G}(s_0^2) \geq q\}, \quad \hat{G}(s_0^2) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\text{ESS}_0(V^{(b)}; s_0^2) \leq N_{\text{target}}\}. \quad (\text{A36})$$

with $\text{ESS}_0(V; s_0^2)$ denoting (A35) at $V_\mu^f = V$. Its population counterpart is

$$G(s_0^2) = P_{V \sim \mathbb{P}}\{\text{ESS}_0(V; s_0^2) \leq N_{\text{target}}\},$$

where $\mathbb{P}$ is the stationary law of the block statistic. Write $G(s) = P\{\text{ESS}_0(V; s) \leq N_{\text{target}}\}$ for a specified positive block-variance law. The theorem concerns exact evaluation of (A35), a fixed coefficient c_g , and the rule (A36) before implementation overrides.

Theorem A4 (Monotone, well-posed calibration). *Fix $\sigma_1^2 > 0$, $V > 0$, $c = c_g H_g > 0$, $\nu_0 > 0$, and $N_{\text{target}} > 0$ and $q \in (0, 1)$, and write $s = s_0^2$.*

- (a) (Monotonicity and limits.) ~~$s \mapsto \text{ESS}_0(V; s)$ is continuous, differentiable, convex and strictly decreasing on $(0, \infty)$, with $\text{ESS}_0(V; s) \uparrow \sigma_1^2/V$, with limits σ_1^2/V and 0 as $s \downarrow 0$ and $\text{ESS}_0(V; s) \downarrow 0$ as $s \uparrow \infty$; it is a bijection of $(0, \infty)$ onto $(0, \sigma_1^2/V)$. If the prior on τ_0^2 is truncated. Under truncation to $(0, \tau_1^2)$ as in (A13), $\text{ESS}_0(V; s)$ it remains continuous and strictly decreasing with limit, with limits σ_1^2/V as $s \downarrow 0$, but its limit as $s \uparrow \infty$ is and $\sigma_1^2/(V + c\tau_1^2) > 0$, so it is a bijection onto $(\sigma_1^2/(V + c\tau_1^2), \sigma_1^2/V)$; convexity and differentiability are stated. Convexity is asserted only for the untruncated case only. $\text{ESS}(w, s)$ is strictly decreasing in s and in τ_1^2 (for $w < 1$), and, when the prior on τ_0^2 is truncated to $(0, \tau_1^2)$, strictly increasing in w .~~
- (b) (Well-posed selection.) ~~For each V the sublevel set $\{s : \text{ESS}_0(V; s) \leq N_{\text{target}}\}$ is the half-line $[s^\dagger(V), \infty)$, where $s^\dagger(V)$ law.~~
- (c) (Population selection.) ~~Under the untruncated law, extend $\text{ESS}_0(V; 0) = \sigma_1^2/V$. Then $\{s \geq 0 : \text{ESS}_0(V; s) \leq N_{\text{target}}\} = [s^\dagger(V), \infty)$, where $s^\dagger$ is the unique root of $\text{ESS}_0(V; s) = N_{\text{target}}$ if positive root when $N_{\text{target}} < \sigma_1^2/V$ and $s^\dagger(V) = 0$ otherwise; zero otherwise. The function $s^\dagger$ is continuous and strictly decreasing in V on $\{V < \sigma_1^2/N_{\text{target}}\}$. Hence $G(s) = P_{\mathbb{P}}\{s^\dagger(V) \leq s\}$ is the distribution function of $s^\dagger(V)$, non-decreasing and right-continuous, and $\{s : G(s) \geq q\} = [s_\star, \infty)$ is a nonempty closed half-line whose infimum $s_\star = \inf\{s \geq 0 : G(s) \geq q\}$ is attained and unique. The selected scale is strictly positive, $s_\star > 0$, if and only if~~

$$P_{\mathbb{P}}(N_{\text{target}} < \sigma_1^2/V) > 1 - q,$$

~~where positive. Thus $G(s) = P\{s^\dagger(V) \leq s\}$ is a distribution function, and its q -quantile $s_\star$ is~~

the unique attained infimum of $\{G \geq q\}$. Moreover,

$$\underline{s_\star > 0 \iff P\{N_{\text{target}} < \sigma_1^2/V\} > 1 - q.} \quad (\text{A37})$$

~~that is, the target lies below the ceiling $\sup_s \text{ESS}_0(V; s) = \sigma_1^2/V$ for more than a $1 - q$ fraction of the Stage-1 blocks; when (A37) fails, $s_\star = 0$, the point-mass limit~~ Under truncation, a positive finite root requires the target to lie strictly between the lower limit in (BART-CP), which the implementation reports as capped. Under the truncated prior (A13) a finite root $s^\dagger(V)$ exists only if in addition $N_{\text{target}} > \sigma_1^2/(V + c\tau_1^2)$, the floor of part (a) ; below the floor the sublevel set is empty and no scale meets and the ceiling. At or below the lower limit, no finite scale satisfies the target.

- (d) (Monte Carlo ~~consistency limits.~~) Let the full Stage-1 draws form state process be a stationary, aperiodic , Harris-ergodic Markov chain, and let $V_{\text{raw}}^{(b)}$ Harris-ergodic Markov chain. Suppose the standardized surface is a square-integrable measurable function of that state, with variance $V_\star > 0$. Let U_b be the sample variance of $N^{-1} \sum_i f(x_i)$ over the in its b -th block of m consecutive draws, $b = 1, \dots, B$, with $\mathbb{P}_m$ the stationary law of $V_{\text{raw}}^{(b)}$ and G_m the corresponding population function. The implementation uses $V^{(b)} = \gamma_B V_{\text{raw}}^{(b)}$ with $\gamma_B = \hat{V}_{\text{whole}}/(B^{-1} \sum_b V_{\text{raw}}^{(b)})$, a finite- m mean-matching correction that sets the block mean equal to the whole-chain variance and nothing more. Two regimes apply.
- (e) ~~($m \rightarrow \infty$ at fixed B , the regime of the code, which fixes $B = 20$ blocks of length $m = n_{\text{draw}}/20$.) Under geometric ergodicity every within-block variance converges almost surely to the stationary variance V_μ^f , $\gamma_B \rightarrow 1$ th block of length $m \geq 2$, and set $V^{(b)} = \gamma_B U_b$ with $\gamma_B = \hat{V}_{\text{whole}}/(B^{-1} \sum_b U_b)$. For fixed B and $m \rightarrow \infty$, every $V^{(b)} \rightarrow V_\mu^f$, and $\hat{G}(s) \rightarrow \mathbf{1}\{s^\dagger(V_\mu^f) \leq s\}$ for $s \neq s^\dagger(V_\mu^f)$; the Pr-rule then reduces to the root of $\text{ESS}_0(V_\mu^f; s) = N_{\text{target}}$, $V^{(b)} \rightarrow V_\star$ almost surely. On a finite grid avoiding equality $\text{ESS}_0(V_\star; s) = N_{\text{target}}$ and containing an admissible point, selection by (A36) eventually equals the smallest admissible grid value. For fixed m and $\hat{s}_{0\star}^2$ converges to the smallest grid value at or above that root.~~
- (f) ~~($B \rightarrow \infty$ at fixed m .) For each $B \rightarrow \infty$, suppose $U_b > 0$ almost surely and $u_m = \mathbb{E}U_b > 0$. Put~~

$\gamma = V_*/u_m$ and

$$G_m(s) = P\{\text{ESS}_0(\gamma U_b; s) \leq N_{\text{target}}\}.$$

At any s for which $P\{\text{ESS}_0(\gamma U_b; s) = N_{\text{target}}\} = 0$, $\hat{G}(s) \rightarrow G_m(s)$ almost surely; ~~if G_m is continuous, the convergence is uniform in s . On a fixed finite grid $\mathcal{S}$ finite grid satisfying this condition,~~ with $G_m(s) \neq q$ at every ~~grid point~~, $\hat{s}_{0*}^2 = \inf\{s \in \mathcal{S} : G_m(s) \geq q\}$ for all B large enough, ~~almost surely. Under geometric ergodicity, $\sqrt{B}\{\hat{G}(s) - G_m(s)\}$ is asymptotically normal, with long-run variance given by the Markov-chain central limit theorem of Theorem 3(c) of the paper for the unscaled indicator plus a delta-method term for the random γ_B , and point and at least one $G_m(s) > q$, the $O_p(B^{-1/2})$ rate of that theorem follows. empirical and population grid selections eventually agree.~~

(g) (~~Anchor~~Single-tree reference.) Condition on ~~a single one fixed f tree ($H_f = 1$) with a fixed partition into leaves ℓ , RWD counts $n_{2\ell}$, and σ_2^2 fixed, and let $a_\ell = N_\ell/N$ be the RCT profile proportions and $\pi_\ell = n_{2\ell}/n_2$ the RWD ones. Then~~

$$V_\mu^f = \sum_\ell a_\ell^2 \frac{\lambda_f^2 \sigma_2^2 / n_{2\ell}}{\lambda_f^2 + \sigma_2^2 / n_{2\ell}} \leq \frac{\sigma_2^2}{n_2} \sum_\ell \frac{a_\ell^2}{\pi_\ell}, \quad \lim_{s \downarrow 0} \text{ESS}_0(s) = \frac{\sigma_1^2}{V_\mu^f} \geq \frac{n_2 \sigma_1^2 / \sigma_2^2}{\sum_\ell a_\ell^2 / \pi_\ell},$$

~~. Let a_ℓ and $\pi_\ell = n_{2\ell}/n_2$ be target and RWD leaf proportions, with $\pi_\ell > 0$ whenever $a_\ell > 0$. Terms with $a_\ell = \pi_\ell = 0$ contribute zero to the sums below. Then~~

$$V_\mu^f = \sum_\ell a_\ell^2 (\lambda_f^{-2} + n_{2\ell} / \sigma_2^2)^{-1} \leq \frac{\sigma_2^2}{n_2} \sum_\ell \frac{a_\ell^2}{\pi_\ell}, \quad \frac{\sigma_1^2}{V_\mu^f} \geq \frac{n_2 \sigma_1^2 / \sigma_2^2}{\sum_\ell a_\ell^2 / \pi_\ell}.$$

~~with equality in the first inequality as $\lambda_f^2 \rightarrow \infty$. Since $\sum_\ell a_\ell^2 / \pi_\ell \geq 1$ with equality if and only if $a = \pi$, matched leaf proportions give $\lim_{s \downarrow 0} \text{ESS}_0(s) = n_2 \sigma_1^2 / \sigma_2^2$ in the Equality is attained in the flat-prior limit, and any covariate shift ($a \neq \pi$) lowers the ceiling limit. In that limit the denominator is at least one, with equality exactly when $a = \pi$. At finite λ_f^2 the prior contributes information, and covariate shift need not lower the ceiling relative to its value at $a = \pi$.~~

Proof. (a) Write $\tau_0^2 = \nu_0 s / Q$ ~~with $Q \sim \chi_{\nu_0}^2$, so that $\text{ESS}_0(V; s) = \sigma_1^2 \mathbb{E}[\psi(s, Q)]$ with~~

$\psi(s, Q) = \{V + c\nu_0 s/Q\}^{-1}$. For every $Q > 0$, $s \mapsto \psi(s, Q)Q \sim \chi_{\nu_0}^2$. The integrand $(V + c\nu_0 s/Q)^{-1}$ is continuous, strictly decreasing, ~~convex~~, ~~and convex in s~~ , bounded by $1/V$; dominated convergence (dominating constant $1/V$) gives continuity of the expectation, and strict ~~dominated convergence gives continuity and the limits; integration preserves monotonicity and convexity~~ are preserved by integration against a probability measure. Differentiability: $\partial_s \psi = -c\nu_0 Q^{-1} \{V + c\nu_0 s/Q\}^{-2}$, and $|\partial_s \psi| \leq 1/(4Vs)$ because $(V+x)^2 \geq 4Vx$ for $x \geq 0$, so differentiation under the integral is justified. Its derivative is bounded in absolute value by $1/(4Vs)$, which permits differentiation on compact subsets of $(0, \infty)$. As $s \downarrow 0$, $\psi \uparrow 1/V$ pointwise, and as $s \uparrow \infty$, $\psi \downarrow 0$ pointwise; dominated convergence gives the limits, and a continuous strictly decreasing function with these limits is a bijection onto $(0, \sigma_1^2/V)$. For the truncated prior, the scaled-Inv- $\chi^2(\nu_0, s)$ family has monotone likelihood ratio in s (the density ratio $p_{s'}(t)/p_s(t) \propto \exp\{-\nu_0(s' - s)/(2t)\}$ is increasing in t for Under truncation, the density ratio at $s' > s$), ~~truncation to is proportional to $\exp\{-\nu_0(s' - s)/(2t)\}$, strictly increasing in t . The truncated family is therefore stochastically increasing, giving strict decrease of the expected reciprocal.~~ Continuity follows from integration on a fixed interval ~~preserves monotone likelihood ratio, monotone likelihood ratio implies stochastic ordering, and $t \mapsto (V + ct)^{-1}$ is strictly decreasing and bounded, so the expectation is strictly decreasing and continuous in s by the same dominated convergence argument.~~ For the limits under truncation, the truncated density is proportional to $t^{-\nu_0/2-1} \exp\{-\nu_0 s/(2t)\}$ on $(0, \tau_1^2)$: as $s \downarrow 0$ the truncated law converges weakly to the point mass at 0 (the untruncated law does, and the As $s \downarrow 0$, the untruncated law concentrates at zero and its truncation probability tends to one), ~~and as $s \uparrow \infty$ it converges weakly to the point mass at τ_1^2 , because for any $\epsilon > 0$ the ratio of the density at $t < \tau_1^2 - \epsilon$ to the density at $t \in (\tau_1^2 - \epsilon/2, \tau_1^2)$ is, for $s \geq 1$, at most $K_\epsilon \exp\{-\nu_0 s \epsilon/(8\tau_1^4)\} \rightarrow 0$ with K_ϵ independent of s (split the exponent in two halves; one half absorbs the power factor uniformly in t , the other is at least $\nu_0 s \epsilon/(8\tau_1^4)$); bounded continuity of $t \mapsto (V + ct)^{-1}$ gives the limits σ_1^2/V and $\sigma_1^2/(V + c\tau_1^2)$.~~ The scale representation $\tau_0^2 = \nu_0 s/Q$ is not available under truncation, and we do not claim convexity or differentiability there. For $\text{ESS}(w, s)$: each summand with $k \geq 1$ is strictly decreasing in s by the argument above and the $k=0$ term is constant, so the sum is strictly decreasing when $w > 0$; each summand with $k < H_g$ is strictly decreasing in compare any interval below $\tau_1^2 - \epsilon$ with one inside $(\tau_1^2 - \epsilon/2, \tau_1^2)$: the exponential density

factor makes the former probability tend to zero. The truncated law concentrates at τ_1^2 ; and with $\tau_0^2 < \tau_1^2$ almost surely, $\varphi(k) = \sigma_1^2 \mathbb{E}[\{V + c(k\tau_0^2 + (H_g - k)\tau_1^2)\}^{-1}]$ is strictly increasing in k , so $\text{ESS}(w, s) = \mathbb{E}_{K \sim \text{Bin}(H_g, w)}[\varphi(K)]$ is strictly increasing in w because the binomial family is stochastically increasing in w , giving the stated lower limit.

(b) By Continuity, strict decrease and the limits in (a), for $N_{\text{target}} < \sigma_1^2/V$ the equation $\text{ESS}_0(V; s) = N_{\text{target}}$ has exactly one root $s^\dagger(V) \in (0, \infty)$ and the sublevel set is $[s^\dagger(V), \infty)$; for $N_{\text{target}} \geq \sigma_1^2/V$ every $s > 0$ satisfies $\text{ESS}_0(V; s) < \sigma_1^2/V \leq N_{\text{target}}$, and we set $s^\dagger(V) = 0$. The map $(V, s) \mapsto \text{ESS}_0(V; s)$ is continuous and strictly decreasing in each argument, so $s^\dagger$ is continuous and strictly decreasing on $\{V < \sigma_1^2/N_{\text{target}}\}$ (implicit function theorem for a strictly monotone continuous function, or directly: $V < V'$ gives $\text{ESS}_0(V'; s^\dagger(V)) < N_{\text{target}}$, hence $s^\dagger(V') < s^\dagger(V)$). Then $\{s : \text{ESS}_0(V; s) \leq N_{\text{target}}\} = \{s : s^\dagger(V) \leq s\}$, so G is the distribution function of the random variable $s^\dagger(V)$, $V \sim \mathbb{P}$, hence non-decreasing and right-continuous with $G(s) \rightarrow 1$ as $s \rightarrow \infty$ (since $s^\dagger(V) < \infty$ for every sublevel set. Joint continuity and strict decrease in V). Thus $\{G \geq q\}$ is a nonempty closed half-line and its infimum is attained and unique. Finally give continuity and strict decrease of a positive root. Since $s^\dagger(V)$ is finite almost surely, G is a distribution function with an attained finite q -quantile. Right-continuity gives $s_\star > 0$ if and only if exactly when $G(0) < q$; $G(0) = P\{s^\dagger(V) = 0\} = P\{N_{\text{target}} \geq \sigma_1^2/V\}$, which is (A37). This is the point at which the argument differs from Theorem 3(b) of the paper: there the leaf ELIR diverges as $s \rightarrow 0$, so feasibility was automatic, while here the ceiling σ_1^2/V is finite and feasibility is a condition on the data, equivalent to (A37). The truncated range in (a) gives its feasibility restriction.

(c) Regime $m \rightarrow \infty$. A geometrically ergodic chain with a stationary start satisfies a strong law for the bounded-moment functional $N^{-1} \sum_i f^{(l)}(x_i)$. At fixed B , ergodic laws for the draw and its square, so the sample variance over a block of length m converges almost surely to the stationary variance V_μ^f as $m \rightarrow \infty$; the whole-chain variance converges to imply convergence of every growing block variance to $V_\star$. A block sum is the difference of two partial sums, so the same limit, hence same conclusion holds for later blocks whose starting index also increases. The whole-chain and mean block variances have that limit, giving $\gamma_B \rightarrow 1$ and $V^{(b)} \rightarrow V_\mu^f$ for each of the B blocks. By continuity of $s^\dagger$ (part (b)), $s^\dagger(V^{(b)}) \rightarrow s^\dagger(V_\mu^f)$, so for $s \neq s^\dagger(V_\mu^f)$ every indicator converges and $\hat{G}(s) \rightarrow \mathbf{1}\{s^\dagger(V_\mu^f) \leq s\}$; on a grid avoiding $s^\dagger(V_\mu^f)$ the selected value is eventually the smallest grid

point above the root.—

Regime $B \rightarrow \infty$. The chain being aperiodic and Harris ergodic, the Continuity and strict grid separation yield the selection result. At fixed m -step chain is Harris ergodic (aperiodicity is what excludes the periodic case, in which the m -block sequence can fail to be ergodic); and, the nonoverlapping block sequence is stationary and ergodic. Thus $B^{-1} \sum_b U_b \rightarrow u_m$ and $\gamma_B \rightarrow \gamma$ almost surely. For each small $\epsilon > 0$, eventually $(\gamma - \epsilon)U_b \leq \gamma_B U_b \leq (\gamma + \epsilon)U_b$ for every block. Monotonicity in V bounds the empirical indicators between those at the two fixed scale factors. Apply the ergodic theorem to each bound and let $\epsilon \downarrow 0$; absence of mass at the raw block statistics $(V_{\text{raw}}^{(b)})_{b \geq 1}$, a fixed measurable function of the non-overlapping m -blocks, form a stationary ergodic sequence with marginal law $\mathbb{P}_m$. The indicator $\mathbf{1}\{s^\dagger(V_{\text{raw}}^{(b)}) \leq s\}$ is bounded and measurable, so the ergodic theorem gives pointwise almost-sure convergence of the unsealed empirical function to threshold makes both limits $G_m(s)$. The rescaling factor γ_B is a ratio of two ergodic averages and converges almost surely to a constant γ (equal to the ratio of the stationary variance to the mean within-block variance); when the distribution function of γV_{raw} is continuous, the empirical distribution function of $\gamma_B V_{\text{raw}}^{(b)}$ converges to it uniformly, by the Glivenko–Cantelli argument (monotonicity and pointwise convergence on a countable dense set) applied at γ_B and the continuity of the limit in the scale. This gives $\hat{G}(s) \rightarrow G_m(s)$ pointwise, uniformly when G_m is continuous. On a finite grid A finite set of grid points with $G_m(s) \neq q$ at every grid point, $\hat{G}(s)$ lies almost surely on the same side of eventually has the correct sign relative to q as $G_m(s)$ at every grid point for B large, so the empirical and population selections coincide. For the proving selection. No central limit theorem, write $\hat{G}(s) = \hat{F}_B(s; \gamma_B)$ with $\hat{F}_B(s; \gamma) = B^{-1} \sum_b \mathbf{1}\{s^\dagger(\gamma V_{\text{raw}}^{(b)}) \leq s\}$. At fixed γ , $\sqrt{B}\{\hat{F}_B(s; \gamma) - F(s; \gamma)\}$ satisfies the Markov-chain central limit theorem of Theorem 3(c) of the paper, whose proof uses only boundedness and measurability of the indicator; $\sqrt{B}(\gamma_B - \gamma)$ is asymptotically normal by the same theorem applied to the two averages and the delta method for the ratio; and when $\gamma \mapsto F(s; \gamma)$ is differentiable at γ , the delta method applied to $F(s; \gamma_B) - F(s; \gamma)$ adds the term $\partial_\gamma F(s; \gamma) \sqrt{B}(\gamma_B - \gamma)$, so $\sqrt{B}\{\hat{G}(s) - G_m(s)\}$ is asymptotically normal with the sum of the two contributions and their covariance as long-run variance. The $O_p(B^{-1/2})$ rate of the selection then follows as in Theorem 3(c). The inner Monte Carlo over τ_0^2 in (A35) (common random numbers across the grid) adds an error that vanishes as the number of τ_0^2 draws grows, by the strong law applied to the bounded integrand.—

~~(d) Given the partition and σ_2^2 , the leaf means of the single f tree are independent a posteriori with $\text{Var}(\theta_\ell | \text{RWD}) = \lambda_f^2 v_{2\ell} / (\lambda_f^2 + v_{2\ell})$, $v_{2\ell} = \sigma_2^2 / n_{2\ell}$ (Lemma A1(e) with one source), and $N^{-1} \sum_i f(x_i) = \sum_\ell a_\ell \theta_\ell$, which gives the or convergence rate is asserted.~~

~~(d) The fixed-tree posterior leaf variances are $(\lambda_f^{-2} + n_{2\ell} / \sigma_2^2)^{-1}$ and the leaves are independent, giving the expression for V_μ^f ; the inequality uses $\lambda_f^2 v / (\lambda_f^2 + v) \leq v$ with equality in the limit, and $v_{2\ell} = \sigma_2^2 / (n_{2\ell} \pi_\ell)$. By the V_μ^f . Drop λ_f^{-2} for the upper bound. Cauchy-Schwarz inequality, $1 = (\sum_\ell a_\ell)^2 = (\sum_\ell (a_\ell / \sqrt{\pi_\ell}) \sqrt{\pi_\ell})^2 \leq \sum_\ell a_\ell^2 / \pi_\ell \cdot \sum_\ell \pi_\ell = \sum_\ell a_\ell^2 / \pi_\ell$, with equality iff $a_\ell / \sqrt{\pi_\ell} \propto \sqrt{\pi_\ell}$, that is $a = \pi$. The limit of ESS_0 is (a). gives $1 = (\sum_\ell a_\ell)^2 \leq \sum_\ell a_\ell^2 / \pi_\ell$, with equality exactly at $a = \pi$. If a target leaf has no RWD observations, its finite-prior variance remains λ_f^2 ; the flat-prior reference for that target is unavailable.~~ $\square$

Remark A7 (Units and regularization). ~~The ceiling σ_1^2 / V_μ^f is~~ The fitted ceiling σ_1^2 / V_μ^f is a property of the working precision summary. Both external observations and the BART prior determine V_μ^f . The single-tree reference in part (d) does not justify subtracting a universal prior contribution from a fitted ensemble. Under the range rule, $H_f \lambda_f^2 = (\max y - \min y)^2 / (4k^2)$, so increasing the ELIR of the RWD-informed prior for μ before any discrepancy is added, and it is in RCT units only if number of trees alone does not remove prior information.

Remark 4 (Calibration implementation). Stage 1 fits BART to the RWD and evaluates the standardized surface at the target profiles. In two-arm analyses, σ_1^2 is the RCT-residual-variance; posterior mean residual variance from a separate BART fitted to the RCT controls; a design value or the RWD estimate is used when controls are unavailable. The code fixes $B = 20$, estimates c_g with 2,000 prior trees, and evaluates the untruncated scale expectation with 4,000 common chi-squared draws across the grid. These finite approximations are separate from the exact-integral theorem. For a fixed positive variance and finite grid, their errors vanish as the inner sample sizes grow; conclusions about selection require strict separation at the limiting grid values.

A requested target at or above the whole-chain ceiling triggers an override: the smallest grid scale is returned, without requiring the block rule to pass. Otherwise the first grid value satisfying the rule is selected. If none passes, the verification report found the calibration using the RWD variance, which inflates the ceiling by σ_2^2 / σ_1^2 . Under covariate shift the ceiling can be far below the RWD sample size (the report gives 38 against $n_2 = 300$ in Scenario 2), and then (A37) fails

~~for $N_{\text{target}} \in \{75, 100\}$; Theorem A4(b) says this is a property of the target, not of the rule. The regularization of largest value is returned with an upper-boundary flag; it does not thereby meet the target. A small scale approximates complete pooling of f_1 with f adds $\sigma_1^2/(\Lambda_f^2 e_f)$ to the ceiling in the notation of the specification; this is negligible at $H_f = 50$ and is not subtracted. only when $w = 1$. With $w < 1$, slab discrepancies remain possible. Equation (A35) uses an untruncated calibration law, while posterior fitting uses (A13); replacing either law or the boundary rule would change the fitted analyses.~~

The realized diagnostic averages $\sigma_1^2/\{V_\mu^f + V_g(\mathcal{T}, z, \tau_0^2)\}$ over posterior states, using (A34). This evaluates prior variance at learned states; it is not the posterior variance of the discrepancy or a subtraction of posterior precisions. The application map uses $\hat{\sigma}_1^2/\{V_f(x) + H_g \bar{\tau}_0^2\}$, with the fitted mean scale $\bar{\tau}_0^2$. In single-arm mode this estimates the mean under the calibrated truncated prior. The supplementary pointwise prior summaries instead average $\hat{\sigma}_1^2/\{V_f(x) + H_g \tau_0^2\}$ over the untruncated calibration prior. The inversion and expectation occur in different orders. All regional summaries use their own standardization profiles and are not additive.

Web Appendix E: Identifiability of f and g

Proposition A3 (Population identification). *Let $\mathcal{X}_1$ and $\mathcal{X}_2$ be the supports of the covariate distributions of the RCT controls and of the RWD, and suppose the regression functions $x \mapsto \mathbb{E}[y | x, D = 0]$ on $\mathcal{X}_2$ and $x \mapsto \mathbb{E}[y | x, D = 1]$ on $\mathcal{X}_1$ are known (the infinite-data limit). Under model (A11):*

- (a) *f is identified on $\mathcal{X}_2$ and $f + g$ is identified on $\mathcal{X}_1$.*
- (b) *g is identified on $\mathcal{X}_1 \cap \mathcal{X}_2$, as $g = (f + g) - f$.*
- (c) *On $\mathcal{X}_1 \setminus \mathcal{X}_2$ only the sum $f + g$ is identified; the split between f and g is determined by the prior, with f the BART extrapolation of the RWD surface and g the remainder.*
- (d) *On $\mathcal{X}_2 \setminus \mathcal{X}_1$, f is identified ~~and but~~ g is not ~~constrained by the data at all: its law there is the prior (A12) propagated through the tree prior, with hyperparameters (τ_0^2, w) learned from the leaves in $\mathcal{X}_1$. This region does not enter the RCT-standardized estimand μ , whose profiles lie in the RCT support~~ identified from the source-specific outcome distributions. Its posterior*

predictions depend on extrapolation through tree partitions and on the prior. This population statement does not imply that its fitted posterior law at those profiles equals the prior.

~~The~~ These statements concern unrestricted population regression functions. Fixed tree partitions impose additional restrictions that can transmit information across these regions. In the single-arm case $\mathcal{X}_1 = \emptyset$: f is the RWD surface, the likelihood does not involve g is a prior draw everywhere, and $f_1 = f + g$ is the RWD surface plus a zero-mean discrepancy process of pointwise variance anywhere. Its entire posterior law equals its calibrated prior, and, conditional on the scales and w , its pointwise variance is $H_g\{w\tau_0^2 + (1-w)\tau_1^2\}$ (at a profile covered by H_g distinct leaves).

Proof. (a) is the model: $\mathbb{E}[y | x, D = 0] = f(x)$ on $\mathcal{X}_2$ and $\mathbb{E}[y | x, D = 1] = f(x) + g(x)$ on $\mathcal{X}_1$. (b), (c), (d) are immediate from (a) and from the fact that no other moment of the data involves f or g . For the single-arm variance, condition on the scales and w . Then $g(x) = \sum_h \theta_{h, \ell_h(x)}$ is a sum of H_g independent ~~draws from (A12)~~ leaf contributions, each with variance $w\tau_0^2 + (1-w)\tau_1^2$. $\square$

Remark A8 (Finite samples). In finite samples the statement is leaf-wise. A g leaf with no RCT controls has $M_g = 1$ (Lemma A1(b)) and its (z, θ) is drawn from the prior at the current (τ_0^2, w) ; a g leaf that straddles the RCT support extends the RCT-informed discrepancy as a constant into the region without RCT controls, which is a piecewise-constant extrapolation and not a prior draw. On $\mathcal{X}_1 \cap \mathcal{X}_2$ the separation of f and g rests on the two priors: f is pinned by the RWD, whose observations do not involve g , and the spike shrinks g toward zero, so a shift common to both sources is attributed to f and a shift of the RCT controls alone to g . The mixture (A12) is identifiable as a two-component scale mixture as long as $\tau_0^2 \neq \tau_1^2$, and Assumption A1 fixes which component is the spike, so the label of z is well defined.

Web Appendix F: ~~What is dropped from~~ Scope of the previous version posterior-invariance result

~~The previous supplement contained, besides the results replaced above, four items that supported the moment-matched Student- t surrogate for the tree moves: Lemma S2.1 (the exact node marginal as a one-dimensional τ^2 integral with no closed form), Definition S2.2 (the moment-matched Student- t node marginal), Assumption S2.4 (finite tree space and bounded leaf statistics), and~~

~~Theorem 2 (the bound on the surrogate’s log-error and the total-variation perturbation of the stationary law). All four are dropped, for one reason: the leaf marginals of the present model are exact (leaf marginals in Lemma A1), so the tree kernels target the exact posterior (and the marginal-then-conditional update in Proposition A1) and there is no approximation whose error would need to be bounded. The compactness assumption S2.4 was needed only to make the surrogate’s worst-case error finite and to invoke uniform ergodicity on a finite tree space; neither is needed for a kernel with the exact acceptance ratio, whose invariance holds on the full state space. The old Lemma S2.1 explains why the previous model needed a surrogate (the leaf-specific τ^2 enters additively in $v_g + \tau^2$ and is integrated inside the leaf), and Remark A1 records why the present one does not (the scale is global and conditioned on; locality is carried by the indicator). Theorem 1 of the paper (location invariance of the exchangeable prior) is replaced by Theorem A1, whose Remark S1.2 survives as the $w = 1$ contrast in Theorem A1(e) and Corollary A1(d). Theorem 3 of the paper is replaced by Theorem A4; parts (b) and (c) of its proof carry over with the changes stated there, and Lemma S3.x (monotonicity of the data-informed predictive ELIR) is no longer needed because the calibration target (A35) has the closed monotone form of Theorem A4(a) establish invariance for the specified posterior kernel. This construction uses the global discrepancy scale as a conditioned-on parameter during a tree move and updates it separately from its truncated conditional. Thus no moment-matched likelihood or approximation-error bound enters the tree acceptance ratio. Invariance does not imply independent posterior draws, rapid mixing, or adequate exploration by a finite run. Those properties require separate analysis and diagnostics. Likewise, the conditional influence and map results in Appendices B and C and the working calibration result in Appendix D have their own assumptions; the posterior-invariance proposition does not establish them for the full fitted ensemble.~~

Web Appendix G: Related work, methodological remarks, and simulation and application details

G.0 Related work: covariate-adaptive borrowing

~~Section 1.1 positions LRC-BART against~~ In addition to the study-level priors and ~~the tree methods~~ . A second group of methods lets the amount of borrowing vary with tree methods discussed in Section 1.1, several approaches allow borrowing to depend on covariates. The propensity-score-integrated power prior [25] and composite likelihood [4] stratify the pooled patients on the estimated probability of belonging to the trial and give each stratum its own discount, so borrowing is constant within strata fixed by a one-dimensional summary before the outcomes are seen. The covariate-adaptive ~~hybrid-historical control~~ borrowing method of Jin et al. [13] places a commensurate prior on the ~~historical-control-meanfunction-current control mean, centered on the historical mean function,~~ with a covariate-dependent precision fitted by kernel local regression and defines a covariate-conditional effective sample size; it is the closest ~~parametric~~ kernel-local relative of our borrowing map, with locality from a kernel bandwidth and a single covariate-adjusted mean model. Multisource exchangeability models [14] average over the subsets of sources deemed exchangeable with the current study, and Kotalik et al. [15] extend the averaging to treatment effects that vary with covariates. The latent exchangeability prior [1] attaches to each historical patient a latent class, so that a subset of patients is borrowed from, with membership driven by the outcome model. Elastic priors [12] set the prior precision through a global congruence statistic and are not local. SAM-HC [2] and Chandra et al. [3] cluster trial and external patients on the covariates with shared- or common-atoms mixtures and borrow only within clusters both sources occupy; ~~they find locality before their borrowing regions are determined before fitting~~ the outcome model ~~is fitted, whereas~~. In LRC-BART finds it through the partition that fits the outcome, these regions are learned through the outcome-model partition.

G.1 Data-generating constants

~~The mean function (10) carries an~~ We choose the intercept α ~~that holds in (10) to maintain~~ the outcome location ~~fixed against~~ after including the bounded bump term: $\alpha = 0.42$ under Sc1,

Sc4 and Sc5 and $\alpha = -0.08$ under Sc2 and Sc3 for Gaussian outcomes, and $\alpha = -0.20$ and -0.70 on the log-time scale for survival outcomes. Under Sc2 and Sc3 the source is assigned by $P(D_i = 1 | \mathbf{W}_i) = \text{logit}^{-1}\{(\ell_i - q_{0.5}(\ell))/0.5\}$ with $\ell_i = -1.20(W_{i5} + W_{i6})$, where $\mathbf{W}_i = (X_{i5}, X_{i6})$ under Sc2 and (U_{i1}, U_{i2}) under Sc3; the median centring and the temperature 0.5 give the two sources equal size on average. Under Sc3 the unmeasured confounders are $U_j \sim \mathcal{N}(\mu_{j+4}, 1)$ and the observed proxies are $X_{j+4} = \rho\tilde{U}_j + \sqrt{1 - \rho^2}\epsilon_j$ with $\tilde{U}_j$ the standardized U_j and $\epsilon_j \sim \mathcal{N}(0, 1)$, so that $\mathbf{U}$ drives both the assignment and the outcome and (X_5, X_6) carry information about it only through ρ . Under Sc4 and Sc5 the RWD mean is $f_0(\mathbf{X}) + \delta_2 \mathbf{1}\{\mathbf{X} \notin \mathcal{R}\}$ with the RCT mean unchanged, so the true local discrepancy $g = f_1 - f$ is $-\delta_2$ outside $\mathcal{R}$ and 0 inside it; the random-number stream is arranged so that Sc4 with $\delta_2 = 0$ reproduces Sc1 replicate by replicate. Each replicate draws 300 RCT subjects and 300 RWD controls afresh. ~~The truths are standardized to the~~ We compute the true estimands by standardizing to each replicate’s RCT profiles: the Gaussian ATE is $\delta = 1$ exactly, and for survival outcomes the population median of the standardized log-normal mixture is ~~root-found~~ obtained by numerical root-finding for each arm and the RMST at $t^* = 3$ is the average over profiles of the closed-form log-normal RMST, both with σ_1 in the control arm.

G.2 Implementation of the comparators

~~The comparators of~~ We implement the comparators in Section 3.1 ~~fall into three groups as follows.~~ LM-NP, LM-CP and LM-PP are Bayesian linear regressions fitted to the RCT controls alone, to the pooled controls, and to the pooled controls with the source indicator as a covariate, and AFT-NP, AFT-CP and AFT-PP are their log-normal AFT counterparts; BART-NP, BART-CP and BART-PP are standard BART and AFT-BART [5, 24] in the same three configurations, BART-PP being the implicit borrowing of Zhou and Ji [27]. PSCL is the propensity-score composite likelihood of Chen et al. [4] with a borrowed size of 100; MAP is the meta-analytic predictive prior for the control mean, without covariate adjustment, with its heterogeneity scale calibrated to the prior ESS target; HierLM and HierAFT give the two sources distinct coefficients drawn from common means with $\tau^2 \sim \text{Inv-Gamma}(1.5, 0.75)$. ~~Two nested versions of the proposed model complete the set~~ We also include two nested LRC-BART models: C-BART fixes $w = 1$, the commensurate model with a single learned scale and no slab, and LRC-BART ($w = 0$) fixes every leaf in the slab, the free-offset model. We implement the analysis component of covariate-adaptive historical control borrowing (CAHB)

from Jin et al. [13]. Its kernel-local control mean has a commensurate prior centred on a historical mean function, with covariate-dependent precision. Local precision estimates are thresholded and projected onto an L_1 ball. We retain the fixed allocation of our simulation; the sequential allocation component of CAHB is not used.

~~The linear~~ Five transfer choices define our primary CAHB analysis. We estimate the historical mean by Nadaraya–Watson regression on the RWD with the same kernel used for the current control mean, treating that estimate as known. We select four kernel covariates by their variable-inclusion counts in an RWD-only BART fit. We enlarge Scott’s bandwidth until the median local control sample size is 20. We rescale the precision constants to $\gamma^2 = 3 \times 0.11/\hat{\phi}^2$ and $\lambda_2 = 300 \log(n) \times 0.11/\hat{\phi}^2$, where $\hat{\phi}^2$ is the residual variance of the current-control no-borrowing smoother; the precision threshold uses its 10% quantile. Finally, we preserve the profile-specific normal marginals and induce dependence across posterior draws through the smoother weights before standardizing to the trial profiles. This last choice affects posterior SD, coverage and power, but not the point estimates or fitted surfaces. The treated mean is fitted to trial treated subjects with a flat prior. Precision estimation and projection use all trial profiles.

CAHB-p10 uses all ten covariates with the other primary choices unchanged. CAHB-lit uses historical linear regression, all ten covariates, Scott’s bandwidth without enlargement, $\gamma^2 = 3$, $\lambda_2 = 300 \log(n)$ and independent profile draws. The primary choices were assessed on diagnostic seeds 2027, 2050 and 2101, which overlap the study replicates. All variants use 1,000 draws, $R = 200$ and data seed $2026 + r$ for replicate r ; code is in `01-code/R/cahb.R` and the run configuration in `02-validation/cahb/run_cahb.R`. The comparison therefore assesses our transfer of the analysis, including its interval construction.

For CAHB, the local summary is the historical precision share $1 - 1/R(x)$, where $R(x)$ is its information ratio, and its effective size is the sum of $R(x) - 1$ over RCT controls. These are distinct from $\mathcal{B}_c$ and the estimand-level ESS of LRC-BART. The literal variant has Sc1 posterior SD 12.226 and RMSE 0.950, compared with 0.542 and 0.262 for CAHB-p10 and 0.410 and 0.258 for primary CAHB. The local kernel fits are unstable or oversmoothed in this design; these results support examining tree partitions for locality, subject to the limitations of the transfer.

Scenario	Transfer	Bias	SD	RMSE	Share, $\mathcal{R}$	Share, $\mathcal{R}^c$
Sc1	CAHB	-0.003	0.410	0.258	0.52	0.51
Sc1	CAHB-p10	-0.018	0.542	0.262	0.50	0.50
Sc1	CAHB-lit	-0.087	12.226	0.950	0.01	0.01
Sc4 (1)	CAHB	-0.208	0.413	0.333	0.53	0.48
Sc4 (1)	CAHB-p10	-0.260	0.545	0.380	0.50	0.49
Sc4 (1)	CAHB-lit	-0.088	12.207	0.948	0.01	0.01
Sc4 (2)	CAHB	-0.333	0.422	0.453	0.49	0.41
Sc4 (2)	CAHB-p10	-0.265	0.585	0.533	0.31	0.31
Sc4 (2)	CAHB-lit	-0.089	12.187	0.947	0.00	0.00
Sc5 (1)	CAHB	-0.437	0.419	0.546	—	0.47
Sc5 (1)	CAHB-p10	-0.290	0.581	0.549	—	0.32
Sc5 (1)	CAHB-lit	-0.089	12.198	0.948	—	0.01
Sc5 (2)	CAHB	-0.067	0.479	0.400	—	0.03
Sc5 (2)	CAHB-p10	-0.028	0.648	0.340	—	0.00
Sc5 (2)	CAHB-lit	-0.090	12.165	0.947	—	0.00

The linear models use $\alpha \sim \mathcal{N}(0, 5^2)$ and $\beta \sim \mathcal{N}(0, 2^2 I)$, BART the default prior with $H = 50$ trees, and the residual variance of ~~every method these non-CAHB methods~~ the BART-style prior with $\nu = 3$ and $q = 0.9$. ~~Our implementation departs from, or specifies where it was silent, the description of the earlier version at the following points. Every sampler runs~~ The implementation details are given below. The non-CAHB samplers run 1,000 burn-in and 1,000 retained iterations. The linear, AFT and hierarchical models are fitted by exact conjugate Gibbs samplers, with right censoring handled by data augmentation of the log survival times; the HierLM sampler was checked against a Hamiltonian Monte Carlo implementation in Stan on the same data. HierLM and HierAFT use one hypermean per coefficient, $\mu_{\beta_j} \sim \mathcal{N}(0, 10^2)$, and the group residual variances carry the BART-style prior with the scale from a group-specific least-squares or parametric AFT fit. ~~The AFT-NP, AFT-CP and AFT-PP priors, unstated before, are those of~~ use the same coefficient priors as the linear models. The MAP prior is built from the RWD sample mean and, with a fixed Half-Normal(0, 1) heterogeneity scale, gives an expected-local-information-ratio ESS of about 2 with a single external study of 300, ~~which cannot produce the borrowing reported before; we therefore so we~~ calibrate the half-normal scale so that the ESS of the mixture-approximated MAP prior equals the target, falling back to the largest attainable ESS where the target is out of reach, with the mixture, ESS and posterior update from the RBesT package. ~~PSCL follows~~

For PSCL, we follow Chen et al. [4]: a logistic propensity model on X_1 to X_{10} over both

sources, trimming of RWD subjects outside the RCT propensity range, five strata at the RCT propensity quintiles, the total borrowed size of 100 allocated across strata in proportion to the overlap coefficient of the stratum propensity distributions, composite-likelihood stratum means with a sandwich variance, and standardization by the RCT stratum shares; its interval is Wald. BART uses the BART package with $H = 50$, $\alpha = 0.95$, $\beta = 2$, $k = 2$, $\nu = 3$ and $q = 0.9$; the treated arm always has its own ensemble on the RCT treated subjects.

~~We also record how our tables differ from those of the earlier version in definition. The power column there~~ The simulation summaries use the main-text definitions. Comparisons with the previous analysis require several distinctions. Previously, power was computed as the fraction of replicates with $P(\text{ATE} > 0.5 \mid \text{data}) > 0.95$, a one-sided test against 0.5, and for survival as $P(\text{ratio} > 1 \mid \text{data}) > 0.95$, ~~whereas its caption stated although the table caption specified~~ the 95% interval excluding the null; we ~~report the stated definition, under which~~ use that interval-based definition. Under it, every tree model has power 1.00 at $\delta = 1$ for Gaussian outcomes, ~~and note that the~~. The 90%-interval version (`pow90` in the result files) is 1.00 for the tree models and 0.90 to 0.91 for LM-NP, LM-PP and HierLM in Sc1, so neither version discriminates among the tree models. ~~The~~

The previous survival point estimate ~~there~~ was the posterior median of the ratio draws; we use the posterior mean throughout, which moves the BART-NP median-ratio bias in Sc1 by about 0.05 and does not account for the remaining difference from the earlier row, which we could not reproduce from the code and the stated design. ~~The~~

The previous single-arm tables ~~there were produced by a mechanism other than the one stated~~ used a different mechanism, with a selection slope of -0.55 at temperature 1.0 in place of -1.20 at 0.5, different intercepts and residual scales, a dropout rate of 0.09, entry over $(0, 0.5)$, administrative censoring at 5.5 minus entry, an RMST horizon of 5, and a hierarchical hyperprior scale of 0.05 with covariates centred at the trial means; Section 3.4 reports the stated design.

G.3 Configuration of the discrepancy ensemble

The configuration of g was chosen on a development run of $R = 200$ Gaussian replicates paired with the first 200 replicates of the full study. The initial configuration, $H_g = 10$ with $w \sim \text{Beta}(4, 1)$, gave a bias of 0.06 under Sc3 and the same Sc4 map as the final one. A tuning study varied H_g , the

split prior, the slab scale, the leaf minimum, the prior on w , the number of g sweeps per iteration, and a warm start of g from the BART-PP source offset, and ran oracle fits with f fixed at the truth.

~~Two findings fixed the configuration. The under-shoot~~

~~The configuration was informed by two observations. Underestimation of g at $\delta_2 = 2$ persisted under every prior, under long chains, and with f known, and is therefore a property of learning the boundary from the RCT controls; and a moderate shift is decided by a prior trade-off in which the split-and-slab state pays the tree-rule cost of one cut plus the slab's cost $\log\{w/(1-w)\} + \log(\tau_1/\tau_0)$ against a likelihood gain of order $n_\ell \delta_2^2 / (2\sigma_1^2)$, so that fewer trees improve the map at $\delta_2 = 2$ and the Sc1 efficiency but raise~~ was not resolved by those configurations. The observed tradeoffs involved both H_g and the slab range rule, which increases τ_1 through the range rule and with it the when the number of discrepancy trees decreases. These joint changes do not isolate one cause of the moderate-shift bias, while a flat prior on w lowers the slab's cost. Web Appendix G.6 gives a conditional prior-cost comparison, with its limits stated explicitly.

The final configuration, $H_g = 5$, $(\alpha_g, \beta_g) = (0.5, 3)$, the range-rule slab, a leaf minimum of 5, one g sweep per iteration and $w \sim \text{Beta}(1, 1)$, lowered the Sc3 bias to 0.04 and left every other quantity within Monte Carlo error of the initial one; LRC-BART ($H_g = 10$) in Web Table S1 is the initial ensemble size under the final prior on w .

The development run evaluated 25 pre-specified criteria, of which 18 pass at $R = 500$; the seven that fail are the Sc1 power criterion, which is uninformative at $\delta = 1$, the Sc4 $\delta_2 = 1$ RMSE criterion against BART-PP, and five Sc4 $\delta_2 = 2$ criteria: the control surface RMSE in $\mathcal{R}$ below 0.9 times BART-PP's, the all-spike map in $\mathcal{R}$ above 0.5, $\mathcal{B}_c$ in $\mathcal{R}$ above 0.7, and the mean of g within 0.25 of the truth in each region, ~~which as reported in Section 3.2 reports as the boundary leak.~~

A secondary version of Sc4, Sc4' in Table 2 and Web Table S1, places the shift outside $\mathcal{R}' = \{X_7 \leq 2\}$, a covariate with negligible effect, so that a split of g on it is driven by the discrepancy alone. ~~The results are the same on the~~ Results for the null covariate X_7 ~~as on are similar to those for~~ X_5 : at $\delta_2 = 2$ the map $\bar{\mathcal{B}}_c$ is 0.64 over the RCT control profiles in the compatible region and 0.05 outside it, and the ATE is unbiased empirical ATE bias is near zero because the errors on the two sides of the boundary have opposite signs and the estimand averages over both; at $\delta_2 = 1$ LRC-BART (100) borrows too much outside the compatible region, with ATE bias -0.041 , RMSE 0.223 against 0.207 for BART-PP, and coverage 0.89, on both X_5 and X_7 .

Theorem 2 collects Theorem A2(a),(b) and Theorem A3 of Web Appendix C. Part (a) gives a conditional half-detection threshold determined by the spike scale, the RCT standard error of the leaf, the slab scale and the mixing weight. The simulation study includes shifts at and below this scale. Part (b) establishes conditional consistency of the displayed map: as the RCT controls in a region accumulate, $\mathcal{B}_c$ converges to the truth about whether the local discrepancy is below c , irrespective of the limiting spike indicators. Part (c) is the reason we display $\mathcal{B}_c$ and not the all-spike probability: the spike is a Gaussian component of scale τ_0 , not a point mass, so a discrepancy of that order is consistent with it and the spike indicator can remain uncertain in the limit even when the RCT data identify whether the discrepancy is below the threshold used by $\mathcal{B}_c$. At fixed scales and large local counts, the bound in (c) can make the limiting spike probability small when the discrepancy is large relative to the spike scale. In the fitted model τ_0^2 is learned, so this limit does not guarantee finite-sample abstention; Section 3 examines shifts near $d_{1/2}$. Proposition A2 extends (b) to the H_g -tree ensemble with fixed partitions, provided the true discrepancy is representable on those partitions, with the map converging cell by cell as the RCT count in a cell grows. These results are conditional on the partitions of the g trees: we have no statement about the posterior over the partitions themselves, hence none about $\mathcal{B}_c$ under the full posterior, since the two-ensemble decomposition with a spike-and-slab discrepancy prior lies outside the available posterior-concentration theory for BART [20]. The leaf-level results also treat f as fixed, whereas with f estimated the residual in a g leaf carries the RWD posterior error of f at the RCT profiles as an additive perturbation of d . The separation of shifted from compatible regions by $\mathcal{B}_c$ under the full posterior is shown empirically in Section 3.

We evaluate additional surface errors over RCT control profiles and over the subsets with $|X_5 - 2| > 0.5$, using the first 200 paired replicates. These differ from Table 2, which uses all RCT profiles. At Sc4 $\delta_2 = 2$, the control-profile RMSE in $\mathcal{R}$ is 0.751 for LRC-BART and 0.704 for BART-PP, a difference of 0.047 with paired standard error 0.005. Away from the boundary it is 0.747 versus 0.723, with difference 0.024 and standard error 0.005. Boundary exclusion reduces this loss but does not remove it. At Sc4 $\delta_2 = 1$, the far-region difference in $\mathcal{R}$ is close to zero. Control-profile errors also place LRC-BART slightly behind BART-PP in Sc1 and Sc5, whereas the all-profile definition shows an Sc1 advantage and an Sc5 tie.

Scenario	Control profiles	LRC-BART	BART-PP	Difference	Paired SE
Sc1	$X_5 > 2$	0.688	0.682	0.006	0.005
Sc1	$X_5 \leq 2$	0.717	0.711	0.006	0.004
Sc1	$X_5 > 2.5$	0.701	0.696	0.005	0.006
Sc1	$X_5 < 1.5$	0.733	0.726	0.007	0.005
Sc4 (0.5)	$X_5 > 2$	0.695	0.685	0.010	0.004
Sc4 (0.5)	$X_5 \leq 2$	0.740	0.719	0.021	0.004
Sc4 (0.5)	$X_5 > 2.5$	0.713	0.704	0.009	0.005
Sc4 (0.5)	$X_5 < 1.5$	0.746	0.719	0.028	0.005
Sc4 (1)	$X_5 > 2$	0.713	0.705	0.008	0.004
Sc4 (1)	$X_5 \leq 2$	0.772	0.744	0.028	0.005
Sc4 (1)	$X_5 > 2.5$	0.731	0.729	0.002	0.005
Sc4 (1)	$X_5 < 1.5$	0.774	0.748	0.026	0.006
Sc4 (2)	$X_5 > 2$	0.751	0.704	0.047	0.005
Sc4 (2)	$X_5 \leq 2$	0.780	0.746	0.034	0.006
Sc4 (2)	$X_5 > 2.5$	0.747	0.723	0.024	0.005
Sc4 (2)	$X_5 < 1.5$	0.770	0.748	0.022	0.006
Sc5 (2)	$X_5 > 2$	0.709	0.688	0.020	0.004
Sc5 (2)	$X_5 \leq 2$	0.736	0.713	0.023	0.005
Sc5 (2)	$X_5 > 2.5$	0.722	0.699	0.023	0.005
Sc5 (2)	$X_5 < 1.5$	0.749	0.723	0.025	0.006

For the regional ESS in Table 2, we standardize the control mean to all RCT profiles in each of $\{X_5 > 2\}$ and $\{X_5 \leq 2\}$. We use its RWD-only posterior variance and the posterior leaf proportions, spike indicators and scales to evaluate the realized ESS within each region. In Sc1 and Sc5 the partition is retained for comparison; it does not define a compatible region in Sc5. Using an average over the untruncated scale prior, the prior pointwise ESS, about 2.1–2.3 across regions and scenarios, describes external support before the discrepancy is learned and does not distinguish compatible from conflicting regions.

G.4 Full simulation tables

Table S1: Gaussian outcomes, two-arm design, all scenarios and methods, $R = 500$. Columns as in Table 1, plus bias and RMSE of the RCT-standardized treated mean. MAP (N) is the meta-analytic predictive prior calibrated to prior ESS N ; LRC-BART (N) targets prior ESS N and LRC-BART (p) the fraction p of the ESS ceiling; LRC-BART ($w = 0$) fixes every leaf in the slab; LRC-BART ($H_g = 10$) doubles the discrepancy ensemble.

Scenario	Method	ATE					Treated mean		Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Bias	RMSE
Sc1	LM-NP	-0.002	0.341	0.342	0.94	0.83	-0.001	0.145	0.001	0.250
	LM-CP	0.009	0.243	0.236	0.96	0.99	-0.000	0.145	-0.009	0.175
	LM-PP	-0.002	0.347	0.337	0.94	0.85	-0.000	0.145	0.001	0.243
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.005	0.109	0.001	0.185
	BART-CP	0.008	0.145	0.142	0.96	1.00	0.005	0.109	-0.003	0.096
	BART-PP	0.006	0.207	0.205	0.94	1.00	0.005	0.110	-0.001	0.170
	PSCL	0.005	0.269	0.261	0.95	0.97	0.000	0.148	-0.005	0.171
	MAP (100)	0.006	0.292	0.265	0.97	0.95	0.000	0.146	-0.005	0.174
	MAP (75)	0.004	0.301	0.274	0.97	0.94	-0.000	0.146	-0.004	0.178
	MAP (50)	0.003	0.311	0.285	0.97	0.92	-0.000	0.146	-0.004	0.186
	HierLM	-0.003	0.338	0.341	0.93	0.85	-0.001	0.145	0.003	0.249
	C-BART	0.007	0.155	0.150	0.95	1.00	0.005	0.108	-0.002	0.106
	LRC-BART ($w = 0$)	0.009	0.189	0.205	0.92	1.00	0.005	0.110	-0.004	0.169
	LRC-BART (100)	0.009	0.171	0.172	0.94	1.00	0.005	0.109	-0.004	0.134
	LRC-BART (75)	0.008	0.174	0.176	0.94	1.00	0.005	0.110	-0.003	0.136
	LRC-BART (50)	0.009	0.177	0.181	0.94	1.00	0.005	0.110	-0.004	0.142
	LRC-BART (0.9)	0.011	0.171	0.171	0.95	1.00	0.005	0.109	-0.005	0.132
	LRC-BART (0.5)	0.009	0.175	0.178	0.93	1.00	0.005	0.109	-0.004	0.139
	LRC-BART (0.25)	0.008	0.180	0.184	0.94	1.00	0.005	0.109	-0.002	0.147
	LRC-BART ($H_g = 10$)	0.009	0.171	0.169	0.94	1.00	0.005	0.109	-0.004	0.131
Sc2	LM-NP	-0.016	0.320	0.324	0.95	0.87	-0.000	0.140	0.016	0.241
	LM-CP	0.379	0.286	0.487	0.73	0.99	0.000	0.140	-0.379	0.445
	LM-PP	0.010	0.340	0.331	0.95	0.84	0.000	0.140	-0.010	0.248
	BART-NP	-0.008	0.201	0.219	0.93	1.00	0.001	0.115	0.009	0.180
	BART-CP	0.058	0.177	0.195	0.91	1.00	0.002	0.115	-0.056	0.151
	BART-PP	0.007	0.208	0.204	0.96	1.00	0.002	0.115	-0.005	0.163
	PSCL	0.024	0.321	0.292	0.97	0.91	-0.005	0.141	-0.029	0.205
	MAP (100)	0.332	0.360	0.491	0.83	0.97	-0.005	0.149	-0.337	0.424
	MAP (75)	0.250	0.363	0.445	0.89	0.93	-0.006	0.149	-0.256	0.368
	MAP (50)	0.177	0.365	0.412	0.92	0.88	-0.006	0.149	-0.183	0.324
	HierLM	-0.011	0.318	0.321	0.95	0.88	-0.000	0.140	0.011	0.238
	C-BART	0.042	0.168	0.186	0.91	1.00	0.001	0.114	-0.041	0.141
	LRC-BART ($w = 0$)	0.008	0.190	0.204	0.94	1.00	0.002	0.114	-0.005	0.165
	LRC-BART (100)	0.023	0.181	0.192	0.94	1.00	0.002	0.114	-0.021	0.148
	LRC-BART (75)	0.022	0.182	0.192	0.93	1.00	0.002	0.114	-0.021	0.150
	LRC-BART (50)	0.023	0.181	0.193	0.93	1.00	0.002	0.115	-0.021	0.149
	LRC-BART (0.9)	0.019	0.183	0.193	0.93	1.00	0.002	0.114	-0.017	0.151
	LRC-BART (0.5)	0.016	0.186	0.197	0.93	1.00	0.002	0.114	-0.013	0.156
	LRC-BART (0.25)	0.012	0.188	0.200	0.94	1.00	0.002	0.114	-0.010	0.160
	LRC-BART ($H_g = 10$)	0.022	0.181	0.190	0.93	1.00	0.001	0.114	-0.021	0.147
	LM-NP	0.003	0.359	0.352	0.95	0.81	0.015	0.151	0.011	0.255

continued on the next page

Web Table S1, continued

Scenario	Method	ATE					Treated mean		Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Bias	RMSE
	LM-CP	0.888	0.264	0.927	0.10	1.00	0.015	0.150	-0.873	0.893
	LM-PP	0.041	0.353	0.342	0.95	0.85	0.015	0.150	-0.026	0.240
	BART-NP	-0.004	0.284	0.282	0.94	0.94	0.020	0.131	0.024	0.216
	BART-CP	0.846	0.206	0.871	0.01	1.00	0.020	0.131	-0.825	0.839
	BART-PP	0.027	0.271	0.270	0.93	0.96	0.020	0.131	-0.008	0.200
	PSCL	0.675	0.277	0.728	0.31	1.00	0.009	0.148	-0.665	0.690
	MAP (100)	0.363	0.360	0.504	0.82	0.97	0.011	0.148	-0.352	0.431
	MAP (75)	0.280	0.363	0.453	0.88	0.93	0.010	0.149	-0.270	0.371
	MAP (50)	0.207	0.365	0.415	0.92	0.91	0.010	0.149	-0.197	0.323
	HierLM	0.033	0.356	0.348	0.95	0.84	0.015	0.151	-0.018	0.251
	C-BART	0.280	0.292	0.431	0.81	0.99	0.020	0.131	-0.259	0.375
	LRC-BART ($w = 0$)	0.024	0.272	0.271	0.93	0.96	0.020	0.132	-0.003	0.201
	LRC-BART (100)	0.042	0.276	0.278	0.93	0.96	0.021	0.131	-0.021	0.209
	LRC-BART (75)	0.043	0.276	0.278	0.93	0.96	0.021	0.131	-0.022	0.208
	LRC-BART (50)	0.045	0.276	0.278	0.93	0.97	0.020	0.131	-0.025	0.208
	LRC-BART (0.9)	0.039	0.275	0.276	0.93	0.96	0.020	0.132	-0.019	0.206
	LRC-BART (0.5)	0.042	0.277	0.277	0.93	0.96	0.020	0.131	-0.022	0.208
	LRC-BART (0.25)	0.047	0.276	0.279	0.92	0.96	0.020	0.131	-0.026	0.209
	LRC-BART ($H_g = 10$)	0.054	0.276	0.280	0.93	0.96	0.021	0.132	-0.034	0.210
Sc3, $\rho = 0$	LM-NP	0.005	0.366	0.355	0.96	0.80	0.016	0.152	0.011	0.256
	LM-CP	1.037	0.256	1.067	0.03	1.00	0.017	0.152	-1.020	1.035
	LM-PP	0.043	0.356	0.346	0.95	0.85	0.016	0.152	-0.027	0.241
	BART-NP	-0.002	0.292	0.289	0.94	0.92	0.022	0.134	0.024	0.219
	BART-CP	0.984	0.202	1.003	0.00	1.00	0.022	0.134	-0.962	0.971
	BART-PP	0.029	0.275	0.273	0.94	0.96	0.023	0.135	-0.007	0.197
	PSCL	0.704	0.269	0.753	0.26	1.00	0.012	0.149	-0.693	0.713
	MAP (100)	0.363	0.360	0.504	0.82	0.97	0.011	0.148	-0.352	0.431
	MAP (75)	0.280	0.363	0.453	0.88	0.93	0.010	0.149	-0.270	0.371
	MAP (50)	0.207	0.365	0.415	0.92	0.91	0.010	0.149	-0.197	0.323
	HierLM	0.039	0.362	0.354	0.96	0.82	0.016	0.152	-0.023	0.253
	C-BART	0.226	0.300	0.380	0.87	0.98	0.023	0.134	-0.204	0.311
	LRC-BART ($w = 0$)	0.023	0.280	0.275	0.94	0.96	0.023	0.134	-0.000	0.201
	LRC-BART (100)	0.041	0.284	0.280	0.94	0.96	0.023	0.134	-0.018	0.206
	LRC-BART (75)	0.042	0.283	0.281	0.94	0.96	0.023	0.134	-0.019	0.207
	LRC-BART (50)	0.045	0.283	0.282	0.94	0.96	0.023	0.134	-0.022	0.208
	LRC-BART (0.9)	0.037	0.282	0.277	0.94	0.96	0.023	0.134	-0.015	0.203
	LRC-BART (0.5)	0.041	0.282	0.280	0.94	0.96	0.023	0.134	-0.018	0.205
	LRC-BART (0.25)	0.045	0.283	0.279	0.94	0.97	0.023	0.134	-0.022	0.205
	LRC-BART ($H_g = 10$)	0.055	0.283	0.282	0.94	0.97	0.023	0.134	-0.032	0.206
	LM-NP	0.008	0.360	0.350	0.96	0.80	0.017	0.151	0.009	0.252
	LM-CP	0.899	0.265	0.937	0.08	1.00	0.017	0.151	-0.882	0.901
	LM-PP	0.043	0.353	0.343	0.95	0.84	0.017	0.151	-0.027	0.241
	BART-NP	0.001	0.284	0.280	0.95	0.94	0.021	0.132	0.020	0.213
	BART-CP	0.847	0.205	0.872	0.01	1.00	0.021	0.133	-0.826	0.838
	BART-PP	0.029	0.271	0.267	0.94	0.96	0.021	0.132	-0.008	0.197
	PSCL	0.678	0.277	0.731	0.30	1.00	0.010	0.149	-0.668	0.692

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Web Table S1, continued

Scenario	Method	ATE					Treated mean		Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Bias	RMSE
	MAP (100)	0.363	0.360	0.504	0.82	0.97	0.011	0.148	-0.352	0.431
	MAP (75)	0.280	0.363	0.453	0.88	0.93	0.010	0.149	-0.270	0.371
	MAP (50)	0.207	0.365	0.415	0.92	0.91	0.010	0.149	-0.197	0.323
	HierLM	0.038	0.356	0.349	0.96	0.82	0.017	0.151	-0.021	0.250
	C-BART	0.277	0.292	0.430	0.82	0.99	0.021	0.133	-0.256	0.371
	LRC-BART ($w = 0$)	0.021	0.273	0.266	0.95	0.96	0.021	0.133	-0.001	0.196
	LRC-BART (100)	0.040	0.276	0.275	0.94	0.96	0.021	0.132	-0.020	0.204
	LRC-BART (75)	0.042	0.276	0.274	0.94	0.97	0.021	0.132	-0.021	0.203
	LRC-BART (50)	0.044	0.277	0.274	0.95	0.97	0.021	0.132	-0.023	0.204
	LRC-BART (0.9)	0.039	0.276	0.273	0.95	0.97	0.021	0.133	-0.018	0.202
	LRC-BART (0.5)	0.043	0.276	0.274	0.95	0.97	0.021	0.132	-0.021	0.203
	LRC-BART (0.25)	0.046	0.277	0.275	0.94	0.97	0.021	0.132	-0.025	0.205
	LRC-BART ($H_g = 10$)	0.053	0.277	0.276	0.94	0.97	0.021	0.132	-0.032	0.206
Sc4, X_5 , $\delta_2 = 0.5$	LM-NP	-0.002	0.341	0.342	0.94	0.83	-0.001	0.145	0.001	0.250
	LM-CP	-0.176	0.243	0.295	0.90	0.92	-0.000	0.145	0.176	0.249
	LM-PP	-0.007	0.348	0.338	0.94	0.83	-0.000	0.145	0.006	0.243
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.005	0.109	0.001	0.185
	BART-CP	-0.164	0.145	0.217	0.78	1.00	0.005	0.109	0.169	0.195
	BART-PP	0.000	0.208	0.203	0.95	1.00	0.005	0.110	0.005	0.168
	PSCL	-0.120	0.270	0.287	0.95	0.92	0.000	0.148	0.121	0.209
	MAP (100)	-0.115	0.296	0.294	0.95	0.88	0.000	0.146	0.116	0.212
	MAP (75)	-0.101	0.306	0.297	0.95	0.88	-0.000	0.146	0.100	0.209
	MAP (50)	-0.082	0.316	0.303	0.96	0.88	-0.000	0.146	0.082	0.210
	HierLM	-0.014	0.338	0.342	0.94	0.84	-0.001	0.145	0.014	0.249
	C-BART	-0.109	0.159	0.190	0.90	1.00	0.005	0.108	0.115	0.162
	LRC-BART ($w = 0$)	0.007	0.190	0.203	0.93	1.00	0.005	0.110	-0.001	0.169
	LRC-BART (100)	-0.043	0.180	0.200	0.92	1.00	0.005	0.109	0.048	0.168
	LRC-BART (75)	-0.040	0.181	0.197	0.93	1.00	0.005	0.110	0.045	0.165
	LRC-BART (50)	-0.031	0.182	0.194	0.94	1.00	0.005	0.110	0.036	0.160
	LRC-BART (0.9)	-0.043	0.180	0.200	0.93	1.00	0.005	0.109	0.048	0.169
	LRC-BART (0.5)	-0.035	0.182	0.196	0.93	1.00	0.005	0.109	0.040	0.163
	LRC-BART (0.25)	-0.022	0.184	0.195	0.93	1.00	0.005	0.109	0.028	0.161
	LRC-BART ($H_g = 10$)	-0.048	0.179	0.198	0.91	1.00	0.005	0.109	0.053	0.167
Sc4, X_5 , $\delta_2 = 1$	LM-NP	-0.002	0.341	0.342	0.94	0.83	-0.001	0.145	0.001	0.250
	LM-CP	-0.361	0.244	0.433	0.67	0.74	-0.000	0.145	0.361	0.403
	LM-PP	-0.012	0.349	0.339	0.95	0.83	-0.000	0.145	0.011	0.245
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.005	0.109	0.001	0.185
	BART-CP	-0.338	0.146	0.368	0.35	1.00	0.005	0.109	0.343	0.357
	BART-PP	-0.002	0.209	0.207	0.95	1.00	0.005	0.110	0.007	0.173
	PSCL	-0.246	0.271	0.358	0.87	0.80	0.000	0.148	0.247	0.300
	MAP (100)	-0.224	0.306	0.361	0.91	0.75	0.000	0.146	0.224	0.292
	MAP (75)	-0.190	0.317	0.352	0.92	0.75	-0.000	0.146	0.190	0.274
	MAP (50)	-0.152	0.328	0.346	0.94	0.77	-0.000	0.146	0.152	0.260
	HierLM	-0.025	0.338	0.343	0.94	0.83	-0.001	0.145	0.024	0.250
	C-BART	-0.207	0.173	0.273	0.73	1.00	0.005	0.108	0.212	0.258
	LRC-BART ($w = 0$)	0.006	0.191	0.204	0.93	1.00	0.005	0.110	-0.000	0.171

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Web Table S1, continued

Scenario	Method	ATE					Treated mean		Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Bias	RMSE
	LRC-BART (100)	-0.041	0.195	0.223	0.89	1.00	0.005	0.109	0.046	0.194
	LRC-BART (75)	-0.041	0.194	0.221	0.90	1.00	0.005	0.110	0.046	0.191
	LRC-BART (50)	-0.036	0.193	0.213	0.90	1.00	0.005	0.110	0.042	0.184
	LRC-BART (0.9)	-0.041	0.196	0.224	0.90	1.00	0.005	0.109	0.047	0.194
	LRC-BART (0.5)	-0.037	0.194	0.218	0.90	1.00	0.005	0.109	0.042	0.187
	LRC-BART (0.25)	-0.032	0.192	0.208	0.92	1.00	0.005	0.109	0.037	0.178
	LRC-BART ($H_g = 10$)	-0.046	0.195	0.222	0.89	1.00	0.005	0.109	0.051	0.195
Sc4, X_5 , $\delta_2 = 2$	LM-NP	-0.002	0.341	0.342	0.94	0.83	-0.001	0.145	0.001	0.250
	LM-CP	-0.732	0.247	0.771	0.16	0.21	-0.000	0.145	0.732	0.755
	LM-PP	-0.023	0.354	0.343	0.95	0.81	-0.000	0.145	0.022	0.249
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.005	0.109	0.001	0.185
	BART-CP	-0.685	0.149	0.701	0.00	0.56	0.005	0.109	0.690	0.698
	BART-PP	-0.006	0.209	0.205	0.96	1.00	0.005	0.110	0.011	0.170
	PSCL	-0.498	0.273	0.562	0.54	0.44	0.000	0.148	0.498	0.528
	MAP (100)	-0.377	0.333	0.495	0.78	0.46	0.000	0.146	0.378	0.437
	MAP (75)	-0.296	0.344	0.451	0.84	0.54	-0.000	0.146	0.296	0.380
	MAP (50)	-0.219	0.352	0.415	0.92	0.60	-0.000	0.146	0.219	0.333
	HierLM	-0.044	0.339	0.345	0.94	0.81	-0.001	0.145	0.043	0.253
	C-BART	-0.227	0.197	0.391	0.66	0.90	0.005	0.108	0.232	0.383
	LRC-BART ($w = 0$)	0.004	0.192	0.207	0.92	1.00	0.005	0.110	0.001	0.173
	LRC-BART (100)	0.000	0.189	0.202	0.92	1.00	0.005	0.109	0.005	0.168
	LRC-BART (75)	0.001	0.190	0.201	0.93	1.00	0.005	0.110	0.004	0.167
	LRC-BART (50)	-0.002	0.190	0.201	0.92	1.00	0.005	0.110	0.008	0.168
	LRC-BART (0.9)	-0.000	0.189	0.202	0.93	1.00	0.005	0.109	0.006	0.170
	LRC-BART (0.5)	-0.001	0.190	0.202	0.92	1.00	0.005	0.109	0.006	0.168
	LRC-BART (0.25)	-0.001	0.191	0.204	0.93	1.00	0.005	0.109	0.007	0.168
	LRC-BART ($H_g = 10$)	-0.004	0.188	0.199	0.93	1.00	0.005	0.109	0.009	0.166
Sc4, X_7 , $\delta_2 = 1$	LM-NP	-0.002	0.341	0.342	0.94	0.83	-0.001	0.145	0.001	0.250
	LM-CP	-0.363	0.244	0.434	0.66	0.72	-0.000	0.145	0.363	0.404
	LM-PP	-0.012	0.348	0.337	0.94	0.83	-0.000	0.145	0.012	0.244
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.005	0.109	0.001	0.185
	BART-CP	-0.340	0.146	0.369	0.35	1.00	0.005	0.109	0.345	0.359
	BART-PP	-0.001	0.209	0.206	0.96	1.00	0.005	0.110	0.006	0.172
	PSCL	-0.246	0.269	0.358	0.87	0.81	0.000	0.148	0.246	0.301
	MAP (100)	-0.222	0.308	0.360	0.91	0.75	0.000	0.146	0.222	0.291
	MAP (75)	-0.189	0.318	0.352	0.92	0.74	-0.000	0.146	0.189	0.274
	MAP (50)	-0.152	0.328	0.347	0.94	0.77	-0.000	0.146	0.151	0.260
	HierLM	0.001	0.338	0.341	0.93	0.85	-0.001	0.145	-0.001	0.249
	C-BART	-0.206	0.173	0.276	0.73	1.00	0.005	0.108	0.211	0.258
	LRC-BART ($w = 0$)	0.009	0.191	0.206	0.92	1.00	0.005	0.110	-0.004	0.172
	LRC-BART (100)	-0.038	0.195	0.224	0.90	1.00	0.005	0.109	0.043	0.196
	LRC-BART (75)	-0.035	0.194	0.220	0.91	1.00	0.005	0.110	0.041	0.190
	LRC-BART (50)	-0.031	0.193	0.217	0.90	1.00	0.005	0.110	0.037	0.187
	LRC-BART (0.9)	-0.040	0.195	0.223	0.90	1.00	0.005	0.109	0.045	0.194
	LRC-BART (0.5)	-0.034	0.193	0.218	0.91	1.00	0.005	0.109	0.039	0.188
	LRC-BART (0.25)	-0.025	0.191	0.212	0.91	1.00	0.005	0.109	0.031	0.180

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Web Table S1, continued

Scenario	Method	ATE					Treated mean		Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Bias	RMSE
Sc4, X_7 , $\delta_2 = 2$	LRC-BART ($H_g = 10$)	-0.044	0.195	0.221	0.89	1.00	0.005	0.109	0.049	0.192
	LM-NP	-0.002	0.341	0.342	0.94	0.83	-0.001	0.145	0.001	0.250
	LM-CP	-0.735	0.246	0.773	0.16	0.20	-0.000	0.145	0.735	0.758
	LM-PP	-0.023	0.352	0.339	0.95	0.81	-0.000	0.145	0.023	0.247
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.005	0.109	0.001	0.185
	BART-CP	-0.685	0.148	0.701	0.00	0.54	0.005	0.109	0.690	0.698
	BART-PP	-0.002	0.209	0.206	0.95	1.00	0.005	0.110	0.007	0.173
	PSCL	-0.498	0.271	0.562	0.53	0.45	0.000	0.148	0.498	0.528
	MAP (100)	-0.358	0.338	0.486	0.80	0.48	0.000	0.146	0.358	0.424
	MAP (75)	-0.284	0.347	0.446	0.88	0.54	-0.000	0.146	0.283	0.374
	MAP (50)	-0.212	0.353	0.413	0.92	0.59	-0.000	0.146	0.211	0.331
	HierLM	0.005	0.338	0.341	0.93	0.85	-0.001	0.145	-0.005	0.249
	C-BART	-0.218	0.199	0.383	0.69	0.91	0.005	0.108	0.223	0.374
	LRC-BART ($w = 0$)	0.007	0.192	0.209	0.92	1.00	0.005	0.110	-0.002	0.176
	LRC-BART (100)	0.001	0.189	0.203	0.93	1.00	0.005	0.109	0.004	0.171
	LRC-BART (75)	0.004	0.189	0.202	0.93	1.00	0.005	0.110	0.001	0.169
	LRC-BART (50)	0.002	0.190	0.205	0.93	1.00	0.005	0.110	0.003	0.173
	LRC-BART (0.9)	0.001	0.189	0.204	0.93	1.00	0.005	0.109	0.004	0.171
	LRC-BART (0.5)	0.003	0.190	0.203	0.93	1.00	0.005	0.109	0.002	0.169
	LRC-BART (0.25)	0.002	0.190	0.204	0.92	1.00	0.005	0.109	0.004	0.172
	LRC-BART ($H_g = 10$)	-0.002	0.189	0.204	0.92	1.00	0.005	0.109	0.007	0.170
Sc5, $\delta_2 = 1$	LM-NP	-0.002	0.341	0.342	0.94	0.83	-0.001	0.145	0.001	0.250
	LM-CP	-0.734	0.244	0.771	0.15	0.20	-0.000	0.145	0.734	0.755
	LM-PP	-0.022	0.347	0.338	0.95	0.82	-0.000	0.145	0.022	0.244
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.005	0.109	0.001	0.185
	BART-CP	-0.687	0.147	0.702	0.00	0.55	0.005	0.109	0.692	0.699
	BART-PP	-0.004	0.208	0.205	0.95	1.00	0.005	0.110	0.010	0.171
	PSCL	-0.498	0.269	0.562	0.53	0.45	0.000	0.148	0.498	0.527
	MAP (100)	-0.345	0.341	0.480	0.82	0.49	0.000	0.146	0.345	0.417
	MAP (75)	-0.275	0.349	0.444	0.88	0.54	-0.000	0.146	0.275	0.370
	MAP (50)	-0.207	0.355	0.412	0.92	0.60	-0.000	0.146	0.206	0.329
	HierLM	-0.021	0.338	0.342	0.94	0.84	-0.001	0.145	0.021	0.249
	C-BART	-0.244	0.198	0.373	0.68	0.94	0.005	0.108	0.249	0.362
	LRC-BART ($w = 0$)	0.007	0.190	0.205	0.92	1.00	0.005	0.110	-0.002	0.170
	LRC-BART (100)	-0.005	0.192	0.210	0.91	1.00	0.005	0.109	0.010	0.176
	LRC-BART (75)	-0.005	0.192	0.210	0.90	1.00	0.005	0.110	0.010	0.177
	LRC-BART (50)	-0.008	0.192	0.209	0.92	1.00	0.005	0.110	0.013	0.176
	LRC-BART (0.9)	-0.002	0.192	0.208	0.92	1.00	0.005	0.109	0.007	0.176
	LRC-BART (0.5)	-0.006	0.192	0.211	0.92	1.00	0.005	0.109	0.011	0.178
	LRC-BART (0.25)	-0.011	0.192	0.211	0.92	1.00	0.005	0.109	0.016	0.179
	LRC-BART ($H_g = 10$)	-0.009	0.191	0.207	0.92	1.00	0.005	0.109	0.014	0.176
	LM-NP	-0.002	0.341	0.342	0.94	0.83	-0.001	0.145	0.001	0.250
	LM-CP	-1.478	0.247	1.497	0.00	0.51	-0.000	0.145	1.478	1.489
	LM-PP	-0.042	0.347	0.339	0.94	0.80	-0.000	0.145	0.042	0.246
	BART-NP	0.004	0.203	0.220	0.91	1.00	0.005	0.109	0.001	0.185
	BART-CP	-1.390	0.152	1.398	0.00	0.72	0.005	0.109	1.395	1.399

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Web Table S1, continued

Scenario	Method	ATE					Treated mean		Control mean	
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Bias	RMSE
	BART-PP	-0.012	0.208	0.205	0.95	1.00	0.005	0.110	0.017	0.171
	PSCL	-1.002	0.269	1.035	0.04	0.04	0.000	0.148	1.002	1.017
	MAP (100)	-0.395	0.348	0.525	0.78	0.41	0.000	0.146	0.395	0.465
	MAP (75)	-0.279	0.355	0.452	0.89	0.50	-0.000	0.146	0.278	0.377
	MAP (50)	-0.187	0.357	0.406	0.93	0.60	-0.000	0.146	0.186	0.319
	HierLM	-0.038	0.338	0.344	0.94	0.82	-0.001	0.145	0.037	0.251
	C-BART	-0.316	0.194	0.650	0.76	0.93	0.005	0.108	0.322	0.648
	LRC-BART ($w = 0$)	0.006	0.189	0.203	0.93	1.00	0.005	0.110	-0.001	0.169
	LRC-BART (100)	-0.001	0.190	0.203	0.92	1.00	0.005	0.109	0.005	0.171
	LRC-BART (75)	-0.001	0.190	0.205	0.93	1.00	0.005	0.110	0.006	0.172
	LRC-BART (50)	-0.001	0.190	0.203	0.93	1.00	0.005	0.110	0.006	0.169
	LRC-BART (0.9)	-0.001	0.190	0.205	0.92	1.00	0.005	0.109	0.006	0.171
	LRC-BART (0.5)	-0.001	0.190	0.204	0.93	1.00	0.005	0.109	0.006	0.171
	LRC-BART (0.25)	0.000	0.189	0.203	0.92	1.00	0.005	0.109	0.005	0.169
	LRC-BART ($H_g = 10$)	-0.007	0.190	0.201	0.92	1.00	0.005	0.109	0.012	0.169

Table S2: Survival outcomes, two-arm design, all scenarios and methods, $R = 500$. Columns as in Table 3, with the mean posterior SD and the bias of the standardized control median on the log scale (Ctrl bias).

Scenario	Method	Median ratio						RMST ratio, $t^* = 3$				
		Bias	SD	RMSE	Cov	Pow	Ctrl bias	Bias	SD	RMSE	Cov	Pow
Sc1	AFT-NP	0.233	0.713	0.737	0.95	0.12	0.072	-0.020	0.147	0.152	0.92	0.09
	AFT-CP	0.165	0.427	0.428	0.95	0.29	-0.007	-0.012	0.094	0.089	0.95	0.22
	AFT-PP	0.261	0.719	0.737	0.95	0.12	0.052	-0.009	0.144	0.142	0.94	0.10
	BART-NP	0.260	0.417	0.511	0.92	0.46	-0.036	0.131	0.151	0.206	0.87	0.47
	BART-CP	0.064	0.238	0.232	0.96	0.63	-0.019	-0.011	0.074	0.072	0.95	0.35
	BART-PP	0.032	0.350	0.345	0.95	0.25	0.016	-0.023	0.103	0.106	0.94	0.17
	HierAFT	0.246	0.701	0.733	0.94	0.12	0.060	-0.008	0.148	0.152	0.92	0.11
	C-BART	0.034	0.268	0.248	0.97	0.42	-0.007	0.003	0.085	0.079	0.95	0.32
	LRC-BART ($w = 0$)	0.050	0.329	0.351	0.94	0.33	0.006	-0.001	0.101	0.108	0.94	0.25
	LRC-BART (100)	0.045	0.305	0.312	0.95	0.34	-0.001	0.003	0.095	0.097	0.94	0.28
	LRC-BART (75)	0.048	0.308	0.312	0.96	0.36	-0.001	0.003	0.096	0.097	0.95	0.28
	LRC-BART (50)	0.047	0.312	0.318	0.96	0.35	0.001	0.003	0.097	0.099	0.95	0.27
	LRC-BART (0.9)	0.042	0.298	0.304	0.96	0.38	-0.001	0.002	0.093	0.095	0.95	0.30
	LRC-BART (0.5)	0.048	0.305	0.309	0.96	0.35	-0.002	0.004	0.095	0.097	0.94	0.28
	LRC-BART (0.25)	0.047	0.311	0.320	0.95	0.35	0.001	0.003	0.097	0.099	0.94	0.28
Sc2	AFT-NP	0.175	0.672	0.672	0.96	0.12	0.102	-0.024	0.131	0.131	0.93	0.09
	AFT-CP	1.519	0.996	1.831	0.55	0.83	-0.207	0.261	0.153	0.304	0.57	0.82
	AFT-PP	0.389	0.763	0.818	0.95	0.19	0.020	0.050	0.149	0.154	0.95	0.20
	BART-NP	0.251	0.427	0.495	0.91	0.43	-0.042	0.110	0.137	0.177	0.88	0.45
	BART-CP	0.194	0.330	0.383	0.93	0.56	-0.048	0.037	0.092	0.100	0.94	0.43

continued on the next page

Sc2

Web Table S2, continued

Scenario	Method	Median ratio						RMST ratio, $t^* = 3$				
		Bias	SD	RMSE	Cov	Pow	Ctrl bias	Bias	SD	RMSE	Cov	Pow
Sc3, $\rho = -0.5$	BART-PP	0.036	0.360	0.335	0.95	0.24	0.017	-0.002	0.101	0.097	0.95	0.21
	HierAFT	0.233	0.679	0.695	0.97	0.14	0.069	0.001	0.136	0.134	0.95	0.12
	C-BART	0.138	0.300	0.331	0.93	0.55	-0.041	0.043	0.090	0.104	0.92	0.49
	LRC-BART ($w = 0$)	0.028	0.328	0.332	0.93	0.29	0.009	0.004	0.096	0.099	0.94	0.24
	LRC-BART (100)	0.061	0.322	0.322	0.94	0.34	-0.007	0.018	0.096	0.100	0.93	0.31
	LRC-BART (75)	0.065	0.322	0.323	0.94	0.34	-0.008	0.019	0.096	0.100	0.93	0.32
	LRC-BART (50)	0.061	0.322	0.322	0.94	0.34	-0.007	0.018	0.095	0.100	0.94	0.31
	LRC-BART (0.9)	0.058	0.324	0.325	0.94	0.33	-0.006	0.017	0.096	0.100	0.94	0.30
	LRC-BART (0.5)	0.051	0.326	0.322	0.95	0.31	-0.002	0.015	0.096	0.099	0.94	0.29
	LRC-BART (0.25)	0.042	0.327	0.327	0.94	0.31	0.002	0.011	0.096	0.100	0.94	0.28
	AFT-NP	0.222	0.757	0.806	0.95	0.10	0.126	-0.025	0.140	0.142	0.93	0.08
	AFT-CP	3.017	1.311	3.271	0.03	1.00	-0.320	0.483	0.159	0.507	0.04	1.00
	AFT-PP	0.438	0.802	0.909	0.95	0.18	0.009	0.054	0.153	0.157	0.95	0.19
	BART-NP	0.188	0.439	0.555	0.91	0.32	-0.041	0.064	0.150	0.172	0.92	0.24
	BART-CP	1.953	0.596	2.063	0.00	1.00	-0.302	0.499	0.145	0.523	0.02	1.00
Sc3, $\rho = 0$	BART-PP	0.047	0.387	0.441	0.92	0.23	-0.007	-0.000	0.124	0.134	0.92	0.15
	HierAFT	0.321	0.778	0.857	0.95	0.12	0.070	0.005	0.146	0.143	0.95	0.10
	C-BART	0.627	0.668	0.948	0.84	0.45	-0.094	0.156	0.158	0.236	0.81	0.43
	LRC-BART ($w = 0$)	0.094	0.477	0.494	0.95	0.17	0.034	0.015	0.131	0.135	0.94	0.16
	LRC-BART (100)	0.147	0.502	0.539	0.95	0.18	0.018	0.030	0.136	0.144	0.94	0.19
	LRC-BART (75)	0.147	0.501	0.531	0.94	0.19	0.017	0.031	0.136	0.142	0.94	0.19
	LRC-BART (50)	0.150	0.499	0.534	0.95	0.19	0.016	0.032	0.135	0.143	0.94	0.19
	LRC-BART (0.9)	0.145	0.501	0.539	0.94	0.19	0.018	0.030	0.135	0.143	0.94	0.20
	LRC-BART (0.5)	0.147	0.501	0.540	0.95	0.17	0.018	0.031	0.135	0.144	0.94	0.19
	LRC-BART (0.25)	0.151	0.501	0.539	0.95	0.19	0.016	0.032	0.136	0.144	0.94	0.19
	AFT-NP	0.231	0.771	0.832	0.94	0.10	0.128	-0.025	0.141	0.145	0.92	0.09
	AFT-CP	3.527	1.395	3.769	0.01	1.00	-0.342	0.553	0.158	0.573	0.01	1.00
	AFT-PP	0.440	0.806	0.919	0.94	0.19	0.008	0.054	0.153	0.158	0.95	0.19
	BART-NP	0.186	0.443	0.571	0.90	0.30	-0.041	0.058	0.152	0.173	0.92	0.21
	BART-CP	2.308	0.631	2.412	0.00	1.00	-0.323	0.588	0.148	0.607	0.01	1.00
Sc3, $\rho = 0.5$	BART-PP	0.041	0.387	0.451	0.91	0.20	-0.006	-0.005	0.124	0.137	0.92	0.15
	HierAFT	0.337	0.795	0.890	0.95	0.13	0.070	0.006	0.147	0.147	0.95	0.11
	C-BART	0.576	0.670	0.909	0.88	0.40	-0.081	0.139	0.159	0.222	0.85	0.38
	LRC-BART ($w = 0$)	0.096	0.489	0.510	0.95	0.15	0.038	0.014	0.133	0.137	0.94	0.17
	LRC-BART (100)	0.155	0.516	0.558	0.94	0.17	0.020	0.031	0.138	0.146	0.94	0.18
	LRC-BART (75)	0.156	0.516	0.553	0.94	0.17	0.019	0.031	0.138	0.145	0.93	0.18
	LRC-BART (50)	0.156	0.514	0.554	0.94	0.18	0.019	0.031	0.137	0.145	0.93	0.19
	LRC-BART (0.9)	0.148	0.511	0.554	0.94	0.17	0.021	0.029	0.136	0.144	0.93	0.18
	LRC-BART (0.5)	0.151	0.515	0.552	0.95	0.18	0.020	0.030	0.137	0.145	0.94	0.19
	LRC-BART (0.25)	0.156	0.515	0.557	0.94	0.18	0.019	0.031	0.138	0.146	0.93	0.18
	AFT-NP	0.226	0.760	0.819	0.94	0.11	0.127	-0.024	0.140	0.144	0.93	0.09
	AFT-CP	3.038	1.318	3.285	0.03	1.00	-0.321	0.486	0.160	0.509	0.04	1.00
	AFT-PP	0.441	0.803	0.914	0.95	0.18	0.009	0.054	0.153	0.157	0.94	0.19
	BART-NP	0.195	0.440	0.574	0.90	0.31	-0.043	0.066	0.151	0.176	0.92	0.23
	BART-CP	1.957	0.597	2.069	0.01	1.00	-0.302	0.499	0.145	0.522	0.02	1.00
	BART-PP	0.046	0.387	0.445	0.91	0.21	-0.007	-0.000	0.124	0.134	0.92	0.15

continued on the next page

Sc3, $\rho = 0.5$

Web Table S2, continued

Scenario	Method	Median ratio						RMST ratio, $t^* = 3$				
		Bias	SD	RMSE	Cov	Pow	Ctrl bias	Bias	SD	RMSE	Cov	Pow
Sc4, $X_5, \delta_2 = 0.5$	HierAFT	0.327	0.781	0.872	0.95	0.14	0.071	0.006	0.146	0.146	0.94	0.12
	C-BART	0.617	0.662	0.948	0.85	0.44	-0.091	0.154	0.158	0.236	0.82	0.42
	LRC-BART ($w = 0$)	0.096	0.476	0.503	0.94	0.16	0.033	0.016	0.131	0.137	0.93	0.17
	LRC-BART (100)	0.152	0.501	0.552	0.93	0.19	0.016	0.032	0.136	0.146	0.92	0.20
	LRC-BART (75)	0.153	0.502	0.551	0.94	0.19	0.016	0.033	0.136	0.146	0.93	0.19
	LRC-BART (50)	0.156	0.502	0.545	0.94	0.20	0.014	0.034	0.136	0.144	0.93	0.20
	LRC-BART (0.9)	0.146	0.501	0.549	0.94	0.18	0.018	0.031	0.136	0.146	0.94	0.19
	LRC-BART (0.5)	0.150	0.502	0.552	0.93	0.18	0.017	0.031	0.136	0.146	0.94	0.19
	LRC-BART (0.25)	0.156	0.501	0.548	0.94	0.20	0.014	0.034	0.136	0.146	0.93	0.21
	AFT-NP	0.233	0.713	0.737	0.95	0.12	0.072	-0.020	0.147	0.152	0.92	0.09
	AFT-CP	-0.123	0.351	0.347	0.92	0.09	0.083	-0.083	0.086	0.116	0.85	0.06
	AFT-PP	0.228	0.714	0.712	0.95	0.10	0.062	-0.023	0.141	0.139	0.94	0.08
	BART-NP	0.260	0.417	0.511	0.92	0.46	-0.036	0.131	0.151	0.206	0.87	0.47
	BART-CP	-0.168	0.205	0.252	0.86	0.17	0.053	-0.088	0.068	0.109	0.74	0.06
	BART-PP	0.011	0.354	0.339	0.96	0.21	0.023	-0.040	0.102	0.107	0.94	0.12
Sc4, $X_5, \delta_2 = 1$	HierAFT	0.219	0.690	0.713	0.95	0.11	0.068	-0.015	0.147	0.150	0.92	0.10
	C-BART	-0.094	0.253	0.247	0.94	0.21	0.033	-0.048	0.081	0.089	0.92	0.13
	LRC-BART ($w = 0$)	0.047	0.329	0.354	0.94	0.32	0.007	-0.011	0.099	0.107	0.93	0.24
	LRC-BART (100)	0.001	0.308	0.317	0.95	0.29	0.014	-0.022	0.094	0.100	0.93	0.19
	LRC-BART (75)	0.005	0.308	0.317	0.95	0.29	0.014	-0.021	0.094	0.100	0.93	0.20
	LRC-BART (50)	0.014	0.312	0.322	0.95	0.30	0.011	-0.018	0.095	0.099	0.93	0.22
	LRC-BART (0.9)	-0.004	0.305	0.313	0.95	0.28	0.015	-0.023	0.093	0.099	0.92	0.19
	LRC-BART (0.5)	0.004	0.307	0.315	0.95	0.28	0.013	-0.021	0.094	0.099	0.93	0.20
	LRC-BART (0.25)	0.015	0.312	0.324	0.95	0.30	0.011	-0.018	0.095	0.100	0.93	0.20
	AFT-NP	0.233	0.713	0.737	0.95	0.12	0.072	-0.020	0.147	0.152	0.92	0.09
	AFT-CP	-0.357	0.289	0.446	0.75	0.05	0.193	-0.144	0.079	0.163	0.57	0.05
	AFT-PP	0.191	0.703	0.691	0.95	0.10	0.075	-0.039	0.137	0.139	0.92	0.07
	BART-NP	0.260	0.417	0.511	0.92	0.46	-0.036	0.131	0.151	0.206	0.87	0.47
	BART-CP	-0.367	0.176	0.401	0.50	0.04	0.142	-0.156	0.062	0.168	0.32	0.07
	BART-PP	-0.012	0.361	0.339	0.97	0.17	0.033	-0.059	0.101	0.113	0.91	0.07
Sc5, $\delta_2 = 1$	HierAFT	0.193	0.681	0.700	0.94	0.10	0.078	-0.023	0.145	0.149	0.91	0.09
	C-BART	-0.202	0.251	0.305	0.84	0.08	0.076	-0.093	0.080	0.120	0.79	0.04
	LRC-BART ($w = 0$)	0.040	0.331	0.353	0.94	0.31	0.010	-0.023	0.097	0.107	0.92	0.19
	LRC-BART (100)	-0.011	0.321	0.340	0.93	0.24	0.024	-0.037	0.096	0.108	0.90	0.16
	LRC-BART (75)	-0.007	0.322	0.337	0.93	0.24	0.022	-0.036	0.096	0.107	0.90	0.17
	LRC-BART (50)	-0.001	0.322	0.333	0.94	0.24	0.020	-0.034	0.096	0.106	0.91	0.17
	LRC-BART (0.9)	-0.018	0.321	0.342	0.93	0.24	0.027	-0.040	0.096	0.110	0.90	0.16
	LRC-BART (0.5)	-0.013	0.322	0.338	0.94	0.24	0.024	-0.038	0.096	0.108	0.91	0.17
	LRC-BART (0.25)	0.001	0.321	0.338	0.93	0.26	0.020	-0.034	0.095	0.106	0.91	0.17
	AFT-NP	0.233	0.713	0.737	0.95	0.12	0.072	-0.020	0.147	0.152	0.92	0.09
	AFT-CP	-0.720	0.191	0.741	0.18	0.39	0.510	-0.244	0.069	0.253	0.09	0.40
	AFT-PP	0.193	0.702	0.695	0.94	0.09	0.075	-0.029	0.141	0.141	0.93	0.08
	BART-NP	0.260	0.417	0.511	0.92	0.46	-0.036	0.131	0.151	0.206	0.87	0.47
	BART-CP	-0.703	0.119	0.712	0.00	0.64	0.402	-0.256	0.054	0.262	0.01	0.65
	BART-PP	-0.029	0.352	0.333	0.95	0.17	0.037	-0.041	0.107	0.110	0.93	0.11
	HierAFT	0.204	0.683	0.701	0.94	0.10	0.073	-0.018	0.146	0.150	0.92	0.09

Sc5, $\delta_2 = 1$

continued on the next page

Web Table S2, continued

Scenario	Method	Median ratio						RMST ratio, $t^* = 3$				
		Bias	SD	RMSE	Cov	Pow	Ctrl bias	Bias	SD	RMSE	Cov	Pow
	C-BART	-0.233	0.288	0.405	0.77	0.13	0.120	-0.085	0.099	0.143	0.77	0.11
	LRC-BART ($w = 0$)	0.053	0.329	0.353	0.94	0.32	0.005	0.006	0.103	0.110	0.93	0.26
	LRC-BART (100)	0.032	0.328	0.353	0.93	0.30	0.013	0.001	0.104	0.112	0.93	0.26
	LRC-BART (75)	0.030	0.328	0.348	0.94	0.29	0.013	0.000	0.104	0.111	0.93	0.25
	LRC-BART (50)	0.026	0.326	0.350	0.94	0.30	0.014	-0.001	0.104	0.111	0.93	0.24
	LRC-BART (0.9)	0.029	0.327	0.350	0.94	0.31	0.014	0.001	0.104	0.111	0.94	0.26
	LRC-BART (0.5)	0.028	0.326	0.349	0.94	0.30	0.013	0.000	0.104	0.111	0.93	0.25
	LRC-BART (0.25)	0.028	0.325	0.351	0.94	0.30	0.013	0.000	0.103	0.112	0.93	0.25

G.5 Application

We harmonize the covariate labels before constructing the model matrix. The non-Hispanic category is labelled differently in the two sources, so the original coding creates two non-Hispanic dummy columns. For the 29 non-Hispanic trial patients, the registry non-Hispanic indicator takes the value used for the registry’s 13 Hispanic patients, while a trial-only indicator has no variation in registry controls. Predictions then depend on extrapolation and prior regularization in each model as well as the other covariates. Race was recorded with four levels and Asian as reference although the cohort table pools Asian into Other/Unknown. Harmonization gives seven columns: age, male, Black, Other/Unknown, Hispanic, high-risk cytogenetics and ASCT after induction. The original coding remains a sensitivity analysis. Its ESS ceilings are 12.6 for PFS and 14.6 for OS, compared with 63.7 and 36.7 under harmonization; a target of 30 is capped under the original coding.

~~The registry cohort is drawn~~ We construct the registry cohort from 797 patients by excluding 7 with ~~inconsistent~~ ~~negative~~ survival times, those without a recorded induction ~~regimen~~start, 49 transplanted on or before induction, and 1 on elotuzumab with lenalidomide and dexamethasone. Under the original covariate coding the reference analyses of the earlier version ~~are reproduced to rounding by the~~ give the following comparisons under the same scripts (posterior medians: complete-pooling AFT 1.03 and 0.90, hierarchical AFT 1.24 and 1.25, AFT-BART 1.06 and 0.91 for PFS and OS, against 1.03, 0.90, 1.21, 1.18, 1.06 and 0.90), which ~~confirms that the harmonization, and not a change of code, accounts for the movement of the~~ permits a coding sensitivity comparison of the

Table S3: The ESS ladder, Gaussian outcomes, $R = 500$: absolute targets $N_{\text{target}} \in \{50, 75, 100\}$ and fractional targets 0.25, 0.5, 0.9 of the ceiling σ_1^2/V_μ^f . Prior ESS is ESS_0 at the calibrated scale, realized ESS its posterior counterpart, ceiling the mean ceiling, and capped the fraction of replicates in which the target was at or above the ceiling and the calibration returned the point-mass limit. Prior ESS is a block average and can exceed the whole-chain ceiling by a few percent when capped (Section 3.1).

Scenario	Target	Bias	RMSE	Cov	Prior ESS	Realized ESS	Ceiling	Capped
Sc1	LRC-BART (50)	0.009	0.181	0.94	42.7	28.5	128	0.00
	LRC-BART (75)	0.008	0.176	0.94	62.1	40.6	127	0.04
	LRC-BART (100)	0.009	0.172	0.94	82.1	52.5	127	0.20
	LRC-BART (0.25)	0.008	0.184	0.94	28.0	18.7	127	0.00
	LRC-BART (0.5)	0.009	0.178	0.93	53.6	34.8	128	0.00
	LRC-BART (0.9)	0.011	0.171	0.95	90.0	55.7	127	0.00
Sc2	LRC-BART (50)	0.023	0.193	0.93	24.5	13.5	23	1.00
	LRC-BART (75)	0.022	0.192	0.93	24.5	13.5	23	1.00
	LRC-BART (100)	0.023	0.192	0.94	24.5	13.5	23	1.00
	LRC-BART (0.25)	0.012	0.200	0.94	5.1	3.3	23	0.00
	LRC-BART (0.5)	0.016	0.197	0.93	9.7	6.0	23	0.00
	LRC-BART (0.9)	0.019	0.193	0.93	16.5	9.6	23	0.00
Sc3, $\rho = 0$	LRC-BART (50)	0.045	0.282	0.94	44.2	1.9	224	0.00
	LRC-BART (75)	0.042	0.281	0.94	64.7	1.7	224	0.00
	LRC-BART (100)	0.041	0.280	0.94	84.5	1.6	224	0.00
	LRC-BART (0.25)	0.045	0.279	0.94	49.4	1.9	224	0.00
	LRC-BART (0.5)	0.041	0.280	0.94	94.1	1.6	224	0.00
	LRC-BART (0.9)	0.037	0.277	0.94	160.7	1.3	224	0.00
Sc4, $X_5, \delta_2 = 1$	LRC-BART (50)	-0.036	0.213	0.90	42.5	13.1	126	0.00
	LRC-BART (75)	-0.041	0.221	0.90	62.0	17.1	125	0.05
	LRC-BART (100)	-0.041	0.223	0.89	82.0	20.8	125	0.22
	LRC-BART (0.25)	-0.032	0.208	0.92	27.6	9.7	126	0.00
	LRC-BART (0.5)	-0.037	0.218	0.90	52.5	14.8	125	0.00
	LRC-BART (0.9)	-0.041	0.224	0.90	89.4	21.1	126	0.00
Sc4, $X_5, \delta_2 = 2$	LRC-BART (50)	-0.002	0.201	0.92	42.4	0.8	125	0.00
	LRC-BART (75)	0.001	0.201	0.93	61.8	0.8	125	0.04
	LRC-BART (100)	0.000	0.202	0.92	83.1	0.8	124	0.27
	LRC-BART (0.25)	-0.001	0.204	0.93	27.4	0.8	125	0.00
	LRC-BART (0.5)	-0.001	0.202	0.92	52.2	0.8	125	0.00
	LRC-BART (0.9)	-0.000	0.202	0.93	88.5	0.8	124	0.00
Sc5, $\delta_2 = 2$	LRC-BART (50)	-0.001	0.203	0.93	42.7	0.3	128	0.00
	LRC-BART (75)	-0.001	0.205	0.93	62.1	0.3	127	0.04
	LRC-BART (100)	-0.001	0.203	0.92	82.1	0.2	127	0.20
	LRC-BART (0.25)	0.000	0.203	0.92	28.0	0.3	127	0.00
	LRC-BART (0.5)	-0.001	0.204	0.93	53.6	0.3	128	0.00
	LRC-BART (0.9)	-0.001	0.205	0.92	90.0	0.3	127	0.00

Table S4: Regional and global shifts, survival outcomes (median ratio), $R = 500$: columns as in Table 2, with the surface RMSE, the map threshold $c = 0.5$ and g on the log-time scale.

Scenario	Method	ATE		Surface RMSE		$\bar{B}_c$		Mean $\hat{g}$		ESS
		Bias	Cov	$\mathcal{R}$	$\mathcal{R}^c$	$\mathcal{R}$	$\mathcal{R}^c$	$\mathcal{R}$	$\mathcal{R}^c$	
Sc4 (0.5)	BART-PP	0.011	0.96	0.783	0.838	—	—	—	—	—
	C-BART	−0.094	0.94	0.747	0.805	0.96	0.96	−0.08	−0.08	44.6
	LRC-BART ($w = 0$)	0.047	0.94	0.784	0.820	0.65	0.64	−0.18	−0.24	0.2
	LRC-BART (100)	0.001	0.95	0.764	0.809	0.79	0.78	−0.15	−0.18	19.2
Sc4 (1)	BART-PP	−0.012	0.97	0.803	0.888	—	—	—	—	—
	C-BART	−0.202	0.84	0.762	0.920	0.91	0.90	−0.16	−0.17	44.5
	LRC-BART ($w = 0$)	0.040	0.94	0.814	0.861	0.52	0.46	−0.33	−0.53	0.2
	LRC-BART (100)	−0.011	0.93	0.792	0.868	0.64	0.58	−0.30	−0.44	12.8
Sc5 (1)	BART-PP	−0.029	0.95	—	0.827	—	—	—	—	—
	C-BART	−0.233	0.77	—	0.843	—	0.43	—	−0.62	25.9
	LRC-BART ($w = 0$)	0.053	0.94	—	0.810	—	0.15	—	−0.96	0.3
	LRC-BART (100)	0.032	0.93	—	0.807	—	0.16	—	−0.94	2.0

Table S5: ~~Baseline characteristics~~ Characteristics of the merged analytic cohort: median (range) for age, n (%) otherwise. Survival is measured from the first day of treatment (EloKRd) or the start of induction (UCMM); RMST at 5 years is the Kaplan–Meier estimate.

Characteristic	EloKRd ($n = 30$)	UCMM ($n = 253$)
Age, years	62.8 (43–81)	61.3 (32–97)
Male	21 (70%)	130 (51%)
Race: White	23 (77%)	164 (65%)
Race: Black	4 (13%)	76 (30%)
Race: Other/Unknown	3 (10%)	13 (5%)
Hispanic or Latino	1 (3%)	13 (5%)
High-risk cytogenetics	15 (50%)	58 (23%)
ASCT after induction	9 (30%)	203 (80%)
PFS events	13 (43%)	159 (63%)
Median PFS, years	not reached	5.09
PFS RMST at 5 years	3.71	3.65
OS events	10 (33%)	60 (24%)
Median OS, years	not reached	not reached
OS RMST at 5 years	4.12	4.57

reference rows in Table 5; the MAP-AFT-BART rows cannot be rerun because their sampler is not in the repository.

The posterior of the RMST ratio under the three absolute targets (Web Figure S2) is unimodal with a ~~right tail that ends at~~ largest observed draw of 2.0 for OS and ~~reaches one draw reaching~~ 3.1 ~~once in among~~ 8,000 ~~draws~~ for PFS.

The ~~covariate coding behind the model-based analyses of the earlier version does not match the baseline table, in the two respects described in Section 4.1: the non-Hispanic level of ethnicity was labelled differently in the two sources, so the dummy coding produced two non-Hispanic indicators, one per source, and race kept a four-level coding with Asian as the reference level although the table pools it into Other/Unknown. The reference analyses of Table 5 are rerun under the harmonized coding; the hierarchical AFT row keeps the prior scale used in the earlier analysis. The lower block of Table 5 carries the earlier analysis. Its MAP-AFT-BART rows had~~ reports the earlier MAP-AFT-BART analysis, whose sampler is unavailable for a controlled comparison. Its 97.5% limits ~~of were~~ 16 to 33 for PFS and 4.8 to 7.2 for OS, ~~attributed at the time to an identification limit of a 30-patient single-arm trial. They do not recur: the limits here.~~ The corresponding limits under the calibrated discrepancy model are 1.23 to 1.40 and 1.04 to 1.11, ~~and over.~~ Over 8,000 draws at $N_{\text{target}} = 30$, the PFS ratio ~~at $N_{\text{target}} = 30$~~ has a 99.9th percentile of 1.6 and a maximum of 3.1. The ~~tails were a property of the earlier prior, under which a leaf-level discrepancy of unbounded scale could send the control RMST toward zero on rare draws; with the discrepancy carried by a five-tree ensemble of pointwise variance~~ conditional discrepancy variance is $H_g \tau_0^2$, with fitted mean about 0.07 on the log-time scale ~~at $N_{\text{target}} = 30$, that route is closed, and the calibration keeps the point estimate within 0.02 across the ladder. The point estimates move from 1.08 to 1.10 down to~~ . These fitted distributions have shorter observed tails; the comparison does not isolate the effects of prior, model, coding and implementation. Across the absolute target ladder, point estimates are 0.97 to 0.99 for PFS and ~~from 0.97 to 1.00 down to~~ 0.90 to 0.92 for OS; ~~the,~~ compared with 1.08 to 1.10 and 0.97 to 1.00 in the earlier analysis. The original-coding sensitivity ~~row~~ gives 1.04 for PFS, ~~so about 0.06 of the PFS change is the coding and the rest is the model. Within the current model this demonstrates a coding effect, without attributing the remaining cross-model difference to one cause.~~

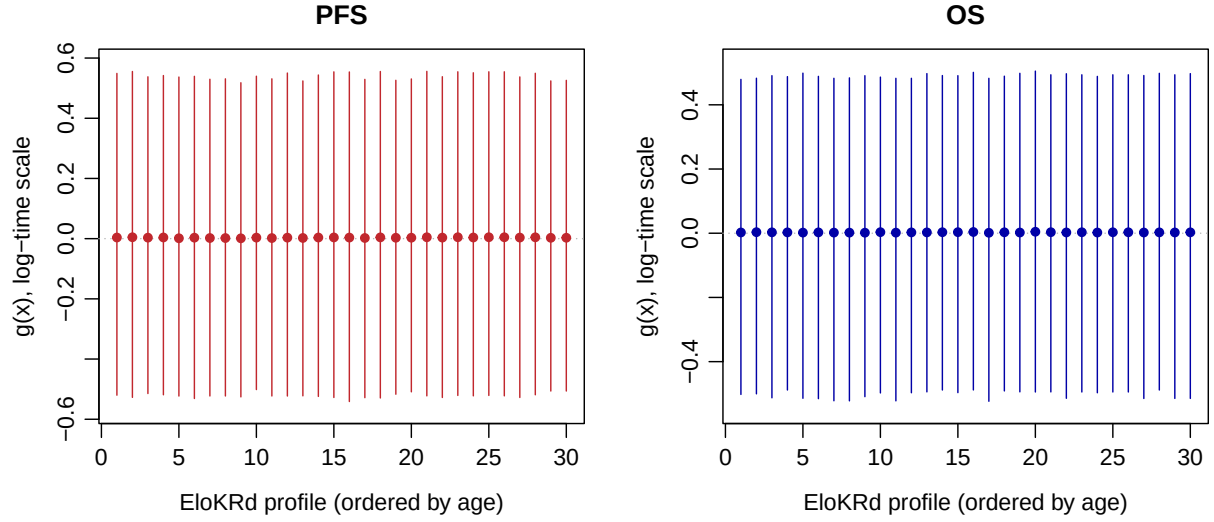


Figure S1: Posterior mean and 95% interval of the discrepancy $g(x)$ on the log-time scale at the 30 EloKRd profiles, ordered by age, at $N_{\text{target}} = 30$ for PFS (red) and OS (blue). In single-arm mode g is a prior draw, so the intervals are centred at zero with SD 0.26 everywhere; discrepancy uncertainty is therefore the same at every profile. Variation in external support is treated differently, which is why summarized by the covariate-resolved display for a single-arm analysis is the local ESS map of in Figure 2.

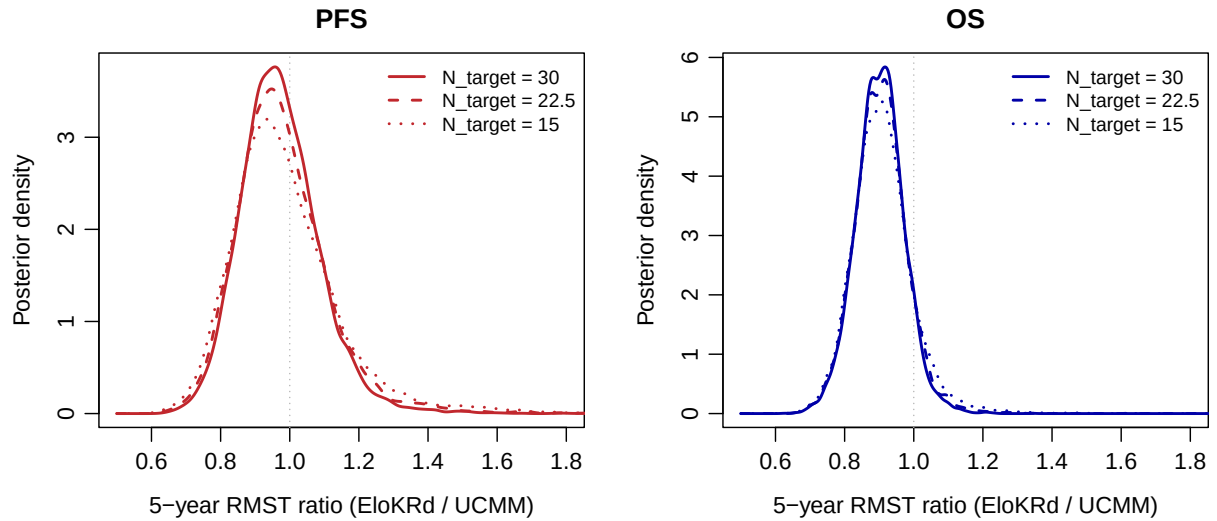


Figure S2: Posterior density of the standardized 5-year RMST ratio for PFS and OS under LRC-BART at $N_{\text{target}} = 30, 22.5$ and 15 (harmonized coding, $H_f = 10$, $w = 1$).

G.6 ~~The prior cost of a regional shift in a shared-leaf design~~A conditional prior-cost comparison

Section 2.2 states that the decomposition into a pooled surface f and a discrepancy ensemble g is what lets a regional shift be detected, and we give the calculation here because the alternative, one ensemble with two means per leaf coupled through a common variance, looks equally natural. Suppose the RWD surface is shifted by δ on a region. In the shared-leaf design with ~~Consider a~~ profile in a fixed region represented by one reached leaf in each of H trees and a commensurate scale τ^2 , the cheapest fit spreads δ/H across every tree under the spike, at prior cost $\delta^2/(2H\tau^2)$; routing the shift through m slab leaves of variance λ^2 instead costs $mc + \delta^2/(2m\lambda^2)$ with c the log prior odds plus the log ratio of the two scales, and the slab route is cheaper only if $\delta > 2H\tau^2\sqrt{2c}/\lambda$. At $H = 50$ and the scales of our simulation study this threshold is between 17 and 86 on an outcome with range 10, so no shift of clinical size flips an indicator and the borrowing map stays flat. In the decomposition (1) the shift lands in trees. Hold the partitions, weight $0 < w < 1$ and variances $0 < \tau^2 < \lambda^2$ fixed. For a specified configuration with m slab leaves, let u_h denote the H discrepancy contributions and require $\sum_h u_h = \delta$. Relative to the all-spike density normalizers, each slab adds $c = \log\{w/(1-w)\} + \log(\lambda/\tau)$ to the negative log joint prior. The constrained minimum is

$$Q_m(\delta) = mc + \min_{\sum_h u_h = \delta} \sum_h \frac{u_h^2}{2v_h} = mc + \frac{\delta^2}{2S_m}, \quad S_m = (H-m)\tau^2 + m\lambda^2, \quad m \in \{0, \dots, H\}.$$

Here v_h is the variance of contribution h . A Lagrange multiplier gives $u_h = \delta v_h / S_m$, so the remaining spike contributions participate in the optimum. If $c > 0$, a specified configuration with $m \geq 1$ has a smaller optimized cost than the all-spike configuration exactly when

$$\delta^2 > \frac{2mc}{(H\tau^2)^{-1} - S_m^{-1}}.$$

The allowed values of m are integers; minimizing over an unconstrained continuous value can give an infeasible allocation. If $c \leq 0$, the normalizing and weight penalty no longer favors the all-spike configuration even at zero discrepancy.

This calculation compares joint prior densities at constrained modes. Posterior indicator

probabilities integrate over leaf parameters, sum over configurations and account for the likelihood and partition probabilities. They therefore do not follow from the ordering of Q_m . Nor does the calculation imply that a learned regional shift is assigned to one or two leaves of one g tree after a single split on the region boundary, which the tree prior allows at ordinary cost, and the slab wins as soon as $\frac{\delta^2}{2} \{(H_g \tau_0^2)^{-1} - \tau_1^{-2}\}$ exceeds the slab's prior cost $\log \frac{w}{1-w} + \log \frac{\tau_1}{\tau_0}$. The design keeps H_g small on purpose: with few trees and a small τ_0 . In a Gaussian common-mean comparison with regional sample sizes n_1, n_2 and a difference δ between source means, optimizing the pooled mean gives a negative log-likelihood loss of $\delta^2/[2(\sigma_1^2/n_1 + \sigma_2^2/n_2)]$; its information is limited by the less precise source.

The separate ensemble in (1) permits different partition and shrinkage priors for the external surface and its discrepancy. With fixed per-leaf spike variance, fewer discrepancy trees reduce the total spike variance at a profile. The slab range rule and the spike cannot hide a shift by spreading it. Letting f absorb part of the shift misfits both sources and pays in the likelihood in proportion to $n_1 + n_2$. partitions also change with the model configuration, so this observation alone does not establish superior detection. The simulations assess the specified $H_g = 5$ model, including the remaining regional surface loss.

G.7 The single-arm simulation in detail

Section 3.4 summarizes. We provide further details of the single-arm study ; this section gives the full account in Section 3.4. The trial contributes n_T treated subjects only, 200 for Gaussian outcomes and 30 or 200 for survival outcomes, the 300 RWD controls are the sole control source, and the estimands are standardized to the trial profiles. The comparators reduce to the complete-pooling models, the hierarchical models, BART-CP, and LRC-BART in the single-arm mode of Section 2.6 with $w = 1$, at the absolute target 100, at 0.9 of the ceiling, and with the spike scale at the grid minimum, which is near-complete pooling of the leaf means; $w = 0.9$ is the slab sensitivity. Sc3 is not identifiable without an internal control and is omitted. Table 4 and Web Table S6 report the results.

Under Sc1 the tree models behave as expected: BART-CP, the pooled LRC-BART and the target-100 LRC-BART have Gaussian RMSE 0.16 with coverage 0.94 to 1.00, and at $n_T = 30$ the survival RMST ratio has RMSE 0.16 to 0.19 with coverage 0.92 to 0.97.

Under Sc2~~no method does well~~, all methods have substantial bias. With the source assignment of Sc2 and no RCT controls, the RWD is the only information about the control surface at trial profiles that lie where ~~the RWD is thin~~RWD support is limited, and every method is far more biased than in the two-arm study: on the Gaussian ATE, LM-CP has bias 1.68, BART-CP 0.41 with coverage 0.75, and LRC-BART 0.42, and the survival median ratio at $n_T = 30$ has biases of 0.8 to 1.3 for the tree models. Single-arm borrowing under measured confounding is a harder problem than the two-arm study, because ~~nothing in the data can~~no internal controls are available to reveal a discrepancy~~at profiles the RWD does not cover~~, and limited RWD overlap increases extrapolation; the rows reported for this scenario in the earlier version came from a different mechanism (Web Appendix G.2). ~~Two further points concern~~

For LRC-BART. ~~First, LRC-BART (100) gives the same point estimate as the pooled model in every design, within 0.005, and the same posterior wherever the target is capped: the~~, uncertainty depends on the target relative to the ceiling. The single-arm ceiling is 27 to 34 in Sc2 and about 75 at $n_T = 30$ in Sc1, so ~~the absolute target is capped there, and where it is not, about 160 in an absolute target of 100 is usually capped there. With $w = 1$, the resulting small discrepancy scale approximates pooling. In Sc1 at $n_T = 200$ the ceiling is about 160, and target 100 allows appreciable discrepancy uncertainty. The prior mean of a Gaussian control outcome or control log time remains centered on the external surface. Nonlinear survival-ratio means need not remain unchanged: under Sc1 with $n_T = 200$, the~~ prior draw of g widens the posterior without moving it (Web Table S6); the fractional rung median-ratio bias is 0.076 for the pooled LRC-BART and 0.108 for target 100. The fractional target at 0.9 of the ceiling ~~gives the same point estimates with wider intervals and widens intervals, with coverage 0.82 to 0.99 , which is what a single-arm calibration can offer. Second, the~~ across the reported designs (Web Table S6).

The $w = 0.9$ sensitivity has ~~a posterior SD of~~ posterior SD 1.4 ~~on for~~ the Gaussian ATE and ~~of~~ 10 to 34 ~~on for~~ the median ratio, with power 0 and coverage 1, ~~the point estimate unchanged, because with no RCT controls the slab draws add prior noise of variance~~. Conditional on w and the scales, the slab contributes $(1 - w)H_g\tau_1^2$ to the ~~counterfactual control and nothing else, which is why we pointwise prior variance of the counterfactual log-time mean. It can also change nonlinear ratio means substantially: under Sc1, the median-ratio bias at $n_T = 200$ is 1.492 with $w = 0.9$, compared with 0.108 with $w = 1$. We therefore report $w = 1$ in for the primary~~ single-arm ~~mode~~analysis.

HierLM and HierAFT with the stated hyperprior and no RCT controls draw the control surface from the hierarchy, and their posterior SDs are 8 on for the Gaussian ATE and astronomically large on extremely large for the median ratio ; we report the stated model as unusable instead of tuning it (Web Appendix G.2).

Table S6: Single-arm design, all methods, $R = 500$. Columns as in Table 4; LRC-BART (100) and LRC-BART ($w = 0.9$) use the absolute target 100, capped at the ceiling where infeasible, and LRC-BART (0.9, $w = 0.9$) the fractional target with the slab.

Design	Method	Treatment effect					Control		
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Prior ESS
Gaussian, $n_T = 200$ ATE Sc1	LM-CP	0.007	0.259	0.254	0.95	0.97	-0.003	0.233	—
	HierLM	-0.014	8.239	0.497	1.00	0.00	0.018	0.487	—
	BART-CP	0.010	0.163	0.162	0.94	1.00	-0.003	0.127	—
	LRC-BART (pooled)	0.007	0.165	0.162	0.94	1.00	0.001	0.127	172.3
	LRC-BART (100)	0.006	0.252	0.161	1.00	1.00	0.001	0.127	81.9
	LRC-BART (0.9)	0.007	0.202	0.163	0.98	1.00	0.000	0.128	115.8
	LRC-BART ($w = 0.9$)	0.010	1.383	0.169	1.00	0.00	-0.003	0.136	81.9
	LRC-BART (0.9, $w = 0.9$)	0.006	1.372	0.169	1.00	0.00	0.001	0.133	116.4
Gaussian, $n_T = 200$ ATE Sc2	LM-CP	1.682	0.417	1.734	0.01	1.00	-1.676	1.726	—
	HierLM	1.642	8.317	1.754	1.00	0.00	-1.636	1.747	—
	BART-CP	0.408	0.310	0.506	0.75	1.00	-0.399	0.492	—
	LRC-BART (pooled)	0.420	0.316	0.515	0.75	1.00	-0.412	0.501	34.8
	LRC-BART (100)	0.421	0.316	0.515	0.76	1.00	-0.413	0.501	34.7
	LRC-BART (0.9)	0.417	0.416	0.511	0.89	0.98	-0.409	0.498	23.1
	LRC-BART ($w = 0.9$)	0.416	1.329	0.511	1.00	0.00	-0.408	0.498	34.8
	LRC-BART (0.9, $w = 0.9$)	0.420	1.345	0.514	1.00	0.00	-0.411	0.500	23.0
Survival, $n_T = 30$ median ratio Sc1	AFT-CP	1.540	4.678	4.444	0.92	0.15	-0.006	0.204	—
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	$> 10^3$	$> 10^3$	—
	BART-CP	-0.013	0.580	0.482	0.97	0.05	-0.026	0.103	—
	LRC-BART (pooled)	0.182	0.918	0.686	0.98	0.02	-0.017	0.102	80.2
	LRC-BART (100)	0.180	0.917	0.686	0.98	0.03	-0.017	0.099	79.0
	LRC-BART (0.9)	0.222	1.174	0.711	0.99	0.01	-0.005	0.133	51.8
	LRC-BART ($w = 0.9$)	1.423	11.199	1.959	1.00	0.00	0.318	0.433	79.3
	LRC-BART (0.9, $w = 0.9$)	1.555	13.493	2.257	1.00	0.00	0.335	0.469	52.2
Survival, $n_T = 30$ median ratio Sc2	AFT-CP	20.944	46.014	51.645	0.19	0.89	-0.441	0.526	—
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	$> 10^3$	$> 10^3$	—
	BART-CP	0.814	1.109	1.231	0.92	0.25	-0.199	0.277	—
	LRC-BART (pooled)	1.324	1.859	1.864	0.95	0.21	-0.207	0.284	29.2
	LRC-BART (100)	1.329	1.865	1.875	0.95	0.20	-0.208	0.284	29.1
	LRC-BART (0.9)	1.511	3.461	2.055	0.97	0.12	-0.180	0.279	19.3
	LRC-BART ($w = 0.9$)	4.243	28.669	5.305	1.00	0.00	0.179	0.537	28.9
	LRC-BART (0.9, $w = 0.9$)	4.544	33.228	6.324	1.00	0.00	0.193	0.446	19.0
Survival, $n_T = 30$ RMST ratio Sc1	AFT-CP	0.036	0.178	0.202	0.90	0.16	0.079	0.187	—
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	0.311	0.355	—

continued on the next page

Web Table S6, continued

Design	Method	Treatment effect					Control		
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Prior ESS
	BART-CP	-0.111	0.176	0.186	0.92	0.02	-0.021	0.089	—
	LRC-BART (pooled)	-0.046	0.192	0.157	0.97	0.02	0.002	0.085	80.2
	LRC-BART (100)	-0.046	0.192	0.157	0.97	0.02	0.002	0.084	79.0
	LRC-BART (0.9)	-0.040	0.218	0.156	0.98	0.01	0.002	0.085	51.8
	LRC-BART ($w = 0.9$)	0.080	0.998	0.200	1.00	0.00	0.020	0.089	79.3
	LRC-BART (0.9, $w = 0.9$)	0.086	1.077	0.204	1.00	0.00	0.020	0.090	52.2
Survival, $n_T = 30$ RMST ratio Sc2	AFT-CP	1.216	0.522	1.336	0.13	0.94	-0.518	0.559	—
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	0.017	0.201	—
	BART-CP	0.149	0.271	0.286	0.95	0.14	-0.219	0.260	—
	LRC-BART (pooled)	0.234	0.297	0.339	0.94	0.19	-0.210	0.251	29.2
	LRC-BART (100)	0.235	0.297	0.340	0.94	0.19	-0.211	0.251	29.1
	LRC-BART (0.9)	0.256	0.452	0.358	0.97	0.13	-0.207	0.248	19.3
	LRC-BART ($w = 0.9$)	0.527	2.689	0.643	1.00	0.00	-0.185	0.228	28.9
	LRC-BART (0.9, $w = 0.9$)	0.560	3.085	0.731	1.00	0.00	-0.184	0.228	19.0
Survival, $n_T = 200$ median ratio Sc1	AFT-CP	0.168	0.470	0.480	0.95	0.23	0.002	0.098	—
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	$> 10^3$	$> 10^3$	—
	BART-CP	0.108	0.273	0.286	0.95	0.57	-0.026	0.058	—
	LRC-BART (pooled)	0.076	0.272	0.262	0.97	0.49	-0.018	0.055	169.5
	LRC-BART (100)	0.108	0.565	0.284	1.00	0.17	-0.008	0.055	81.7
	LRC-BART (0.9)	0.087	0.350	0.268	0.99	0.34	-0.014	0.055	114.0
	LRC-BART ($w = 0.9$)	1.492	15.408	4.227	1.00	0.00	0.318	0.402	81.8
	LRC-BART (0.9, $w = 0.9$)	1.302	10.341	1.607	1.00	0.00	0.312	0.428	113.7
Survival, $n_T = 200$ median ratio Sc2	AFT-CP	9.478	5.162	10.850	0.01	1.00	-0.407	0.419	—
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	$> 10^3$	$> 10^3$	—
	BART-CP	1.136	0.778	1.359	0.60	0.82	-0.189	0.214	—
	LRC-BART (pooled)	1.195	0.851	1.409	0.64	0.82	-0.196	0.219	35.9
	LRC-BART (100)	1.193	0.856	1.410	0.65	0.81	-0.196	0.219	35.9
	LRC-BART (0.9)	1.336	2.306	1.562	0.82	0.57	-0.181	0.207	23.9
	LRC-BART ($w = 0.9$)	3.935	26.373	4.611	1.00	0.00	0.142	0.396	36.2
	LRC-BART (0.9, $w = 0.9$)	4.380	34.059	7.322	1.00	0.00	0.158	0.379	23.8
Survival, $n_T = 200$ RMST ratio Sc1	AFT-CP	-0.023	0.101	0.098	0.94	0.15	0.074	0.114	—
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	0.305	0.317	—
	BART-CP	-0.001	0.084	0.082	0.97	0.32	-0.021	0.061	—
	LRC-BART (pooled)	-0.024	0.078	0.076	0.95	0.25	0.001	0.054	169.5
	LRC-BART (100)	-0.020	0.121	0.076	0.99	0.09	0.003	0.054	81.7
	LRC-BART (0.9)	-0.022	0.094	0.076	0.99	0.17	0.002	0.054	114.0
	LRC-BART ($w = 0.9$)	0.109	1.179	0.275	1.00	0.00	0.021	0.057	81.8
	LRC-BART (0.9, $w = 0.9$)	0.097	0.860	0.138	1.00	0.00	0.019	0.057	113.7
Survival, $n_T = 200$ RMST ratio Sc2	AFT-CP	1.164	0.428	1.244	0.01	1.00	-0.535	0.550	—
	HierAFT	$> 10^3$	$> 10^3$	$> 10^3$	1.00	0.00	0.001	0.101	—
	BART-CP	0.305	0.206	0.361	0.61	0.77	-0.233	0.261	—
	LRC-BART (pooled)	0.284	0.206	0.336	0.70	0.74	-0.224	0.250	35.9
	LRC-BART (100)	0.284	0.207	0.336	0.69	0.73	-0.224	0.250	35.9
	LRC-BART (0.9)	0.301	0.350	0.353	0.85	0.50	-0.222	0.248	23.9
	LRC-BART ($w = 0.9$)	0.562	2.419	0.633	1.00	0.00	-0.200	0.228	36.2

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Web Table S6, continued

Design	Method	Treatment effect					Control		
		Bias	SD	RMSE	Cov	Pow	Bias	RMSE	Prior ESS
	LRC-BART (0.9, $w = 0.9$)	0.585	2.496	0.662	1.00	0.00	-0.198	0.226	23.8